

GENERAL OVERVIEW AND CLASSIFICATION OF MATERIALS, CRYSTALLINE STRUCTURES, DEFECTS

The aim of this course:

- The aim of this course is to provide Mechanical and Automotive Engineering students basic knowledge on different classes of materials
- Relationship between structure of materials and their properties (Materials Science)
- **Structure of materials → chemical and physical characteristics (e.g. chemical bonds, composition, ...)**
- *Materials Engineering → to design/choose materials*
- **Choice of materials: production process, structure-properties, performances, cost, life cycle, sustainability etc.**

MATERIALS OF INTEREST IN ENGINEERING

- metals and alloys
- polymers
- glasses
- ceramics
- composites

(fuels, lubricants, paints,)

The periodic table

METALS

NON-METALS

**“NOBLE”
GASES**

1 IA 2 IIA 3 IIIB 4 IVB 5 VB 6 VIB 7 VII 8 VIII 9 VIII 10 VIII 11 IB 12 IIB 13 IIIA 14 IVA 15 VA 16 VIA 17 VIIA 18 VIIIA																	
1 1,0079 H Idrogeno	2 4,0026 He Elio											3 10,81 B Boro	4 12,011 C Carbonio	5 14,0067 N Azoto	6 15,9994 O Ossigeno	7 18,998 F Fluoro	8 20,179 Ne Neon
9 6,941 Li Litio	10 9,0122 Be Berillio	11 22,9898 Na Sodio	12 24,305 Mg Magnesio	13 26,9815 Al Alluminio	14 28,0855 Si Silicio	15 30,9738 P Fosforo	16 32,06 S Zolfo	17 35,45 Cl Cloro	18 39,948 Ar Argon								
19 39,0983 K Potassio	20 40,08 Ca Calcio	21 44,9559 Sc Scandio	22 47,9 Ti Titanio	23 50,9415 V Vanadio	24 51,996 Cr Cromo	25 54,938 Mn Manganese	26 55,847 Fe Ferro	27 58,9332 Co Cobalto	28 58,7 Ni Nikel	29 63,546 Cu Rame	30 65,38 Zn Zinco	31 69,72 Ga Gallio	32 72,59 Ge Germanio	33 74,9216 As Arsenico	34 78,96 Se Selenio	35 79,90 Br Bromo	36 83,8 Kr Kriptone
37 85,4678 Rb Rubidio	38 87,62 Sr Stronzio	39 88,9059 Y Ittrio	40 91,22 Zr Zirconio	41 92,9064 Nb Niobio	42 95,94 Mo Molibdeno	43 98 Tc Tecnezio	44 101,07 Ru Rutenio	45 102,9055 Rh Rodio	46 106,4 Pd Palladio	47 107,868 Ag Argento	48 112,41 Cd Cadmio	49 114,82 In Indio	50 118,69 Sn Stagno	51 121,75 Sb Antimonio	52 127,6 Te Tellurio	53 126,9045 I Iodio	54 131,3 Xe Xenone
55 132,9054 Cs Cesio	56 137,33 Ba Bario	57 138,9055 La Lantanio	58 178,49 Ce Cerio	59 180,9479 Pr Praseodimio	60 183,85 Nd Neodimio	61 186,207 Pm Promezio	62 190,2 Sm Samario	63 192,22 Eu Europio	64 195,09 Gd Gadolonio	65 196,9665 Tb Terbio	66 200,59 Dy Disprosio	67 204,37 Ho Olmio	68 207,2 Er Erbio	69 208,9804 Tm Tullio	70 210 Yb Iterbio	71 210 Lu Lutezio	72 210 Hf Hafnio
87 223 Fr Francio	88 226,025 Ra Radio	89 227,028 Ac Attinio	90 232,0381 Th Torio	91 232,0381 Pa Protoattinio	92 238,029 U Uranio	93 238,029 Np Nettunio	94 238,029 Pu Plutonio	95 238,029 Am Americio	96 238,029 Cm Curio	97 238,029 Bk Berchelio	98 238,029 Cf Californio	99 238,029 Es Einsteinio	100 238,029 Fm Fermio	101 238,029 Md Mendelevio	102 238,029 No Nobelio	103 238,029 Lr Laurenzio	104 238,029 Rf Rutherfordio
101 102,0718 Nb Niobio	102 102,9055 Mo Molibdeno	103 102,9055 Tc Tecnezio	104 102,9055 Ru Rutenio	105 102,9055 Rh Rodio	106 102,9055 Pd Palladio	107 102,9055 Ag Argento	108 102,9055 Cd Cadmio	109 102,9055 In Indio	110 102,9055 Sn Stagno	111 102,9055 Sb Antimonio	112 102,9055 Te Tellurio	113 102,9055 I Iodio	114 102,9055 Xe Xenone	115 102,9055 Kr Kriptone	116 102,9055 Ar Argon	117 102,9055 Ne Neon	118 102,9055 He Elio

Chemical bond



↓

**PURE
METALS and
METAL
ALLOYS**



↓

Me + O
OXIDES
(with high
electronegativity)

Me + non-Me
(e.g. NaCl)



↓

C (DIAMOND)

metal + C, N
CARBIDES, NITRIDES

Me + O
OXIDES
(with low to mid
electronegativity)



↓

C + H +
(Cl, F, O...)
POLYMERS

Electronegativity: Grade of attraction of one atom's electrons by another atom.

OVERVIEW ON METALS AND ALLOYS



- **ELEMENTS** (Fe, Ti, Al, Mg, Cu,...)
- **Conductors for heat and electricity**
- Prepared by **melting**
- Can be **machined**
- (almost) All are **solid** at room temp.
- Can form **alloys**
- **Corrosion and oxidation** issues
- **Density** ranging (typically) between **2,7 and 8 g/cm³**



Metallic bond

Pure metals and metal alloys consist of **atoms** and are characterized by **metallic bonding**: the valence electrons of each atom, i.e. the electron available for the bond, are delocalized and shared among all the atoms.

Most of the elements in the Periodic Table are metals.

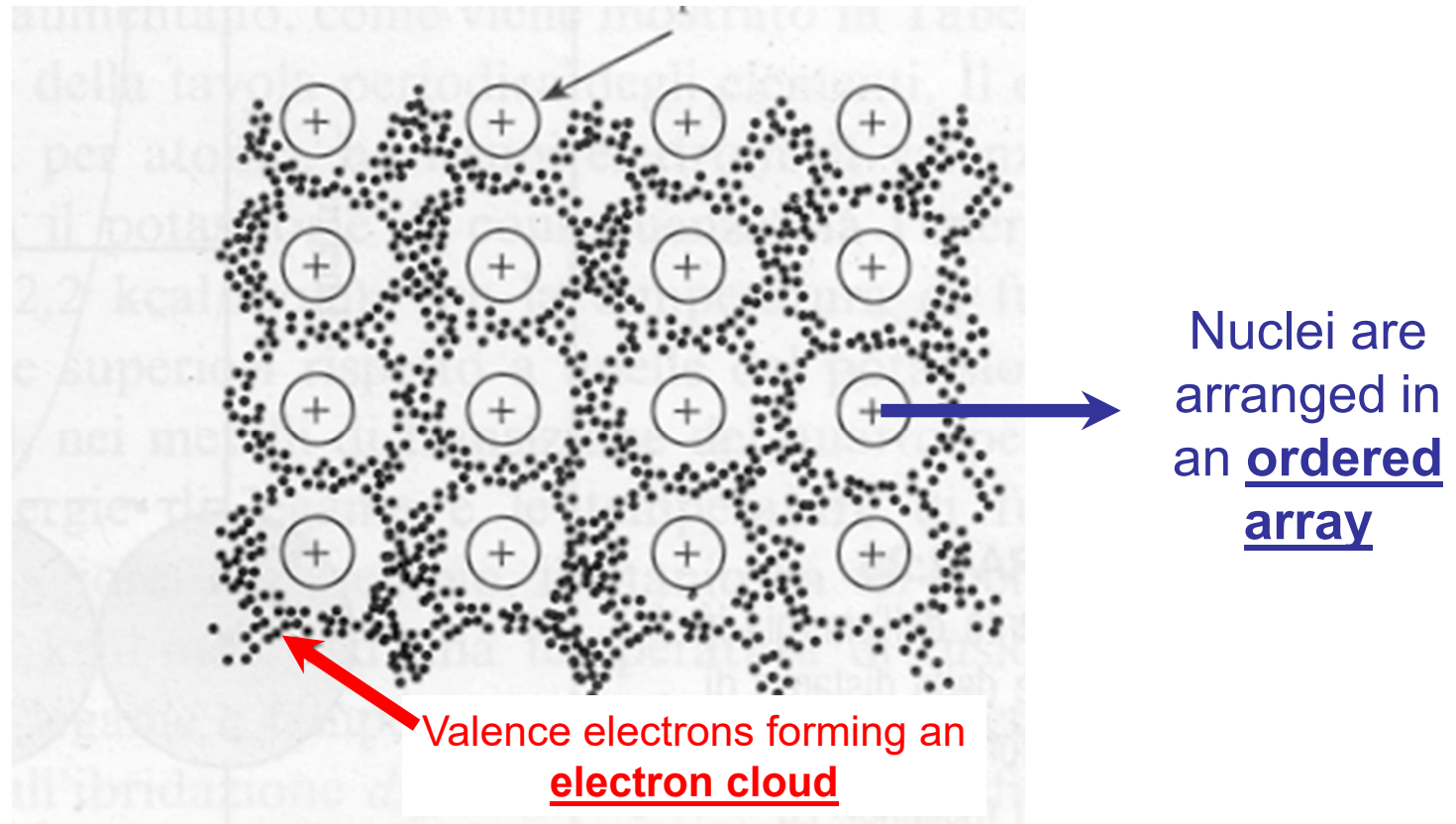
Examples of metal alloys are:

- **Cu-Zn** (brass)
- **Fe-C** (steel)
- **Sn-Pb** (an alloy used for soldering)

Alloys are classified according to the majority element that is present in it.

The main classes of alloys are:

- **iron-based alloys** for structures (e.g. reinforced concrete);
- **copper-based alloys** for piping, utensils, thermal conduction, electrical conduction, etc.;
- **aluminum-based alloys** for lightweight structures and metal-matrix composites.



Strong bond with non-localized electrons

Each nucleus is bonded to the others by the global electric field, but it is NOT bonded to another specific nucleus.

Electrons are NOT bonded to a specific nucleus.

Properties of metals

Optical properties

No transparency to visible light due to the electron energy of the metallic chemical bond (all visible radiations are absorbed); metals are shine after polishing

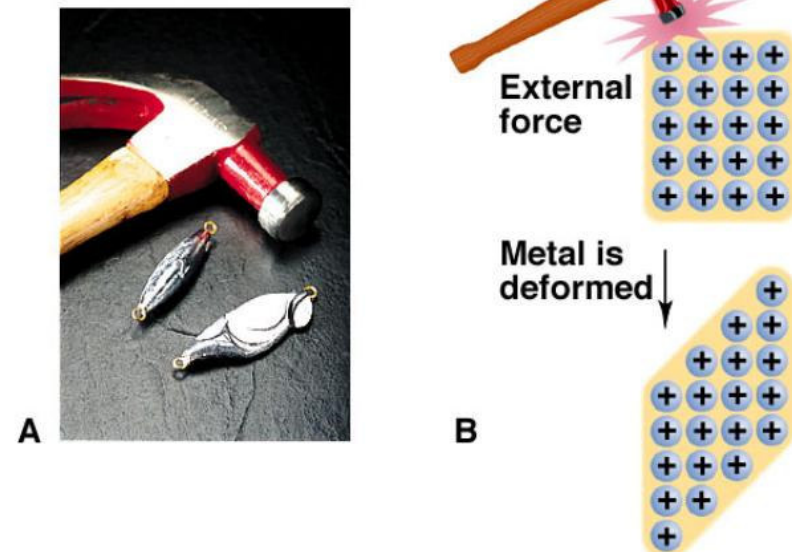
Electrical and heat transfer properties

Ability of conducting electricity and heat due to the presence of mobile electrons (electron cloud) inside metallic chemical bond

Mechanical properties

- High plasticity (rough definition: “high tendency to deform”), due to the non-directional character of metal bond
- High toughness (rough definition: “resistance to crack propagation and mechanical shocks”)

The Reason Metals Deform



The **bonding strength** in metal bonds is **quite variable**

Bonding strength (BS) deeply affects:

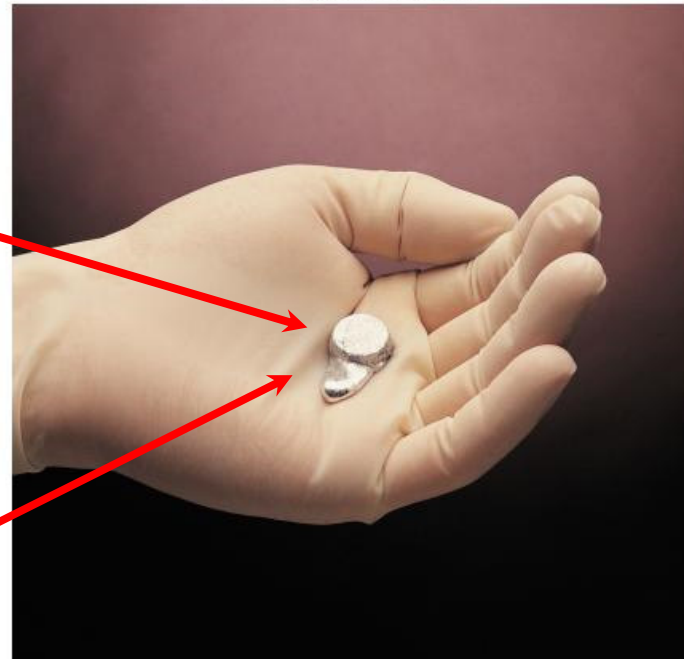
- hardness (H)
- **melting point (MP)**

Note: if BS $\uparrow$, then H and MP $\uparrow$

Examples:

- **Ga melts at room temperature**
- Hg is liquid at room temperature
- W is extremely hard and melts at a very high temperature (3430 °C)

The Unusually Low Melting Point of Gallium



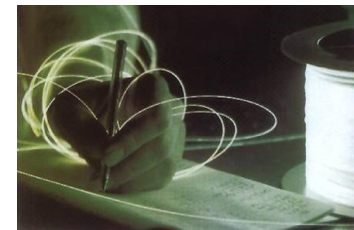
OVERVIEW ON CERAMICS

- **Compounds** (**oxydes**, carbides, nitrides, ..)(Al_2O_3 , SiC , Si_3N_4 , ...)(density ranging between 3 and 5 g/cm³)
- **hard and brittle, difficult to be machined**
- **Insulators** (heat and electricity)
- High melting point
- Prepared by **sintering** or other processes, never melting
- *examples: bricks, concrete, tiles, porcelains,*



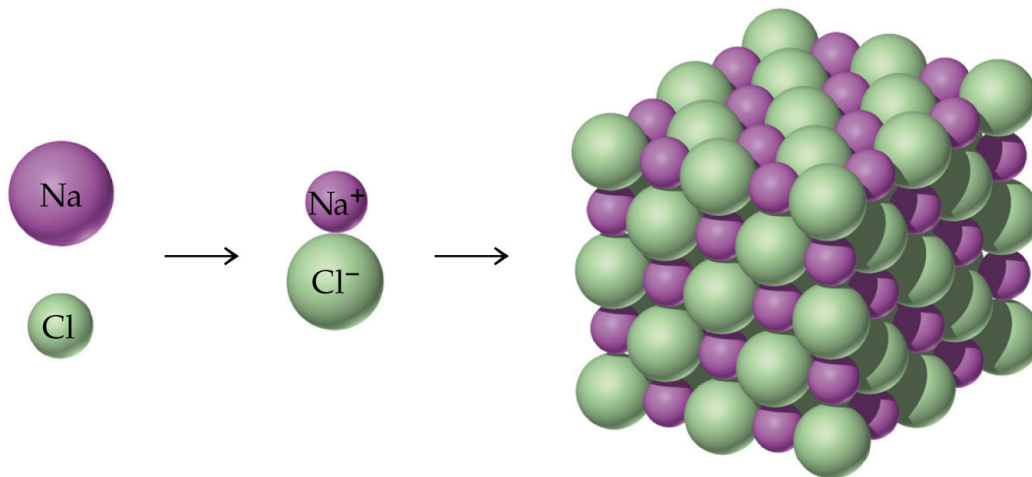
OVERVIEW ON GLASSES

- Prepared by **melting of oxides**
(SiO_2 , Na_2O , CaO , Al_2O_3 , K_2O ,...)
- Hard, brittle, *can be deformed only at suitable temperature, cannot be machined*
- **insulators** (heat and electricity)
- Density of about **2,5 g/cm³**
- *examples: window and car glasses, optical fibers*



Ionic bond

Ionic bond occurs between highly electropositive atom (METALS) and highly electronegative atoms (NON-METALS)

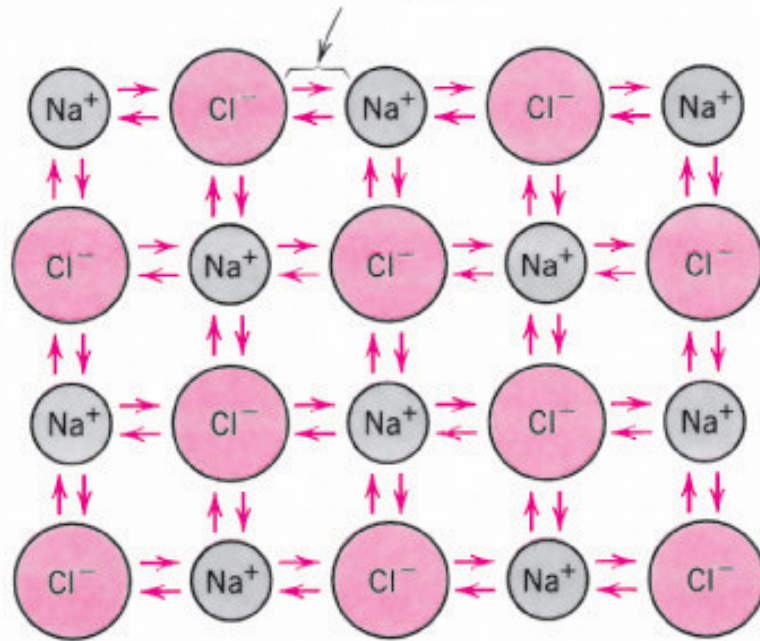


Reticular positions (= the positions of atoms inside the crystals) are occupied by charged atoms (IONS) with different size

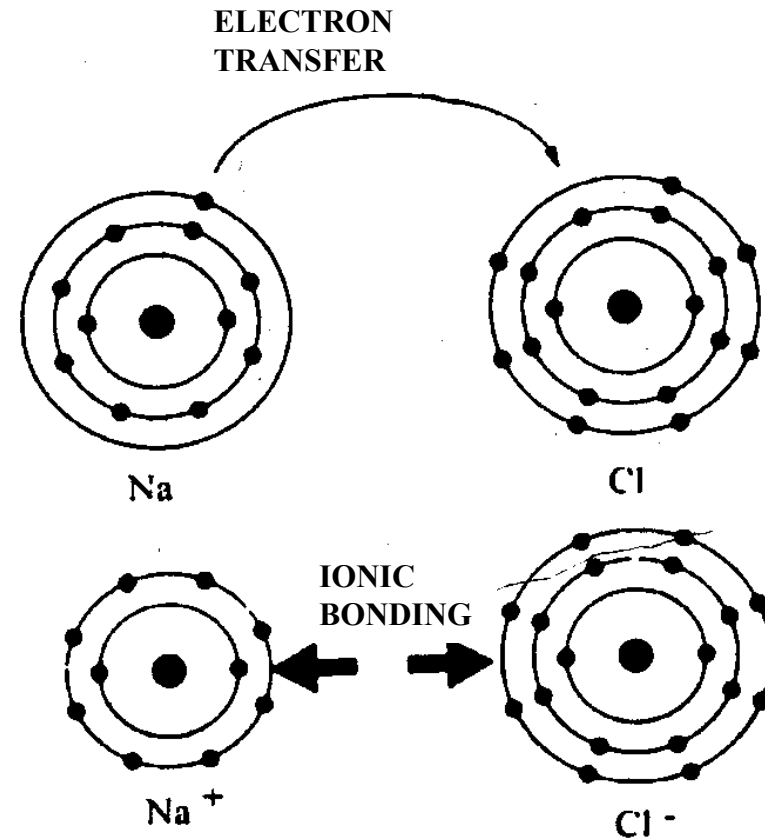


Forces of ionic bond are due to attractive interactions (Coulomb force) between ions with opposite charge

Coulomb bonding force



Schematic representation of ionic bonding in sodium chloride (NaCl)



- Crystal stability is assured by the electrostatic interactions between ions of opposite (+ or -) charge

- **Crystal charge = 0**

- Atoms size (atomic radius) influences the crystal structure

Properties of ionic solids

Optical properties

Some frequencies of visible light can interact with the electrons involved in the ionic bond and, accordingly, be absorbed; ionic solids can be partially transparent and coloured

Electrical properties

Ionic solids are good insulators; they can be conductors in aqueous solution.

Physical properties

- **high melting points**
due to the **high bonding strength**

- Low to mid density

Melting and Boiling Points of Some Ionic Compounds

Compound	mp(°C)	bp(°C)
CsBr	636	1300
NaI	661	1304
MgCl ₂	714	1412
KBr	734	1435
CaCl ₂	782	>1600
NaCl	801	1413
LiF	845	1676
KF	858	1505
MgO	2852	3600

Transition from ionic to covalent bond: mixed bond

IA																		IIIA	IVA	VA	VIA	VIIA	He		
H	2.1	IIA																B	C	N	O	F	-		
Li	1.0	Be																2.0	2.5	3.0	3.5	4.0	-		
Na	0.9	Mg																1.5	1.8	2.1	2.5	3.0	-		
K	0.8	Ca	21	Ti	23	Cr	25	Fe	27	Ni	29	Zn	31	32	As	34	Br	Kr							
Rb	0.8	Sr	Sc	V	Mn	Co	Cu	Ga	Ge	Se	I	Xe	-	-	-	-	-	-							
Cs	0.7	Ba	1.3	1.5	1.6	1.5	1.8	1.8	1.9	1.8	1.6	1.8	2.0	2.4	2.8	-	-	-							
Fr	0.7	Ra	1.2	1.4	1.6	1.8	1.9	2.2	2.2	2.2	1.9	1.7	1.7	1.8	1.9	2.1	2.5	-							
		57-71		72	73	74	75	76	77	78	79	80	81	82	83	84	At	Rn							
		La-Lu		Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	-	-							
		1.1-1.2		1.3	1.5	1.7	1.9	2.2	2.2	2.2	2.4	1.9	1.8	1.8	1.9	2.0	-	-							
		80-102																							
		Ac-No																							
		1.1-1.7																							

CaF₂: large

SiC: small

Table of Electronegativities

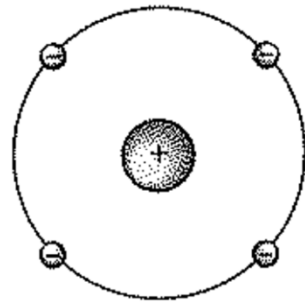
Table of Electronegativities

Adapted from Fig. 2.7, Callister 6e. (Fig. 2.7 is adapted from Linus Pauling, *The Nature of the Chemical Bond*, 3rd edition, Copyright 1939 and 1940, 3rd edition. Copyright 1960 by Cornell University.

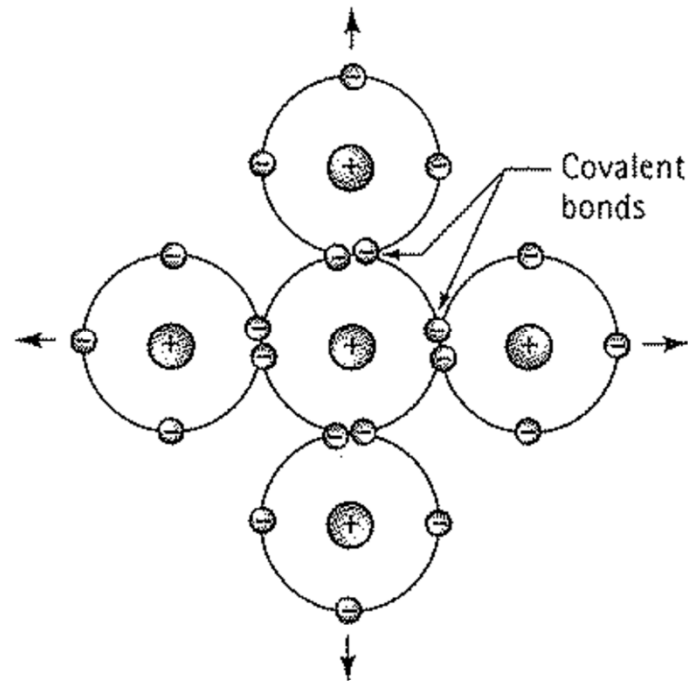
ELECTRONEGATIVITY:
“avidity of electrons”
 (practical definition)

Table 3.2 For Several
Ceramic Materials, Percent
Ionic Character of the
Interatomic Bonds

Material	Percent Ionic Character
CaF ₂	89
MgO	73
NaCl	67
Al ₂ O ₃	63
SiO ₂	51
Si ₃ N ₄	30
ZnS	18
SiC	12



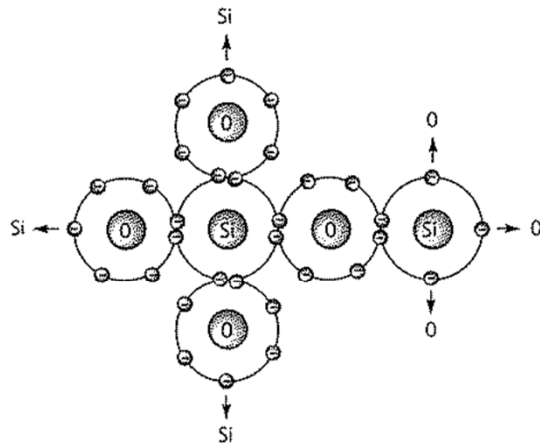
Silicon atom



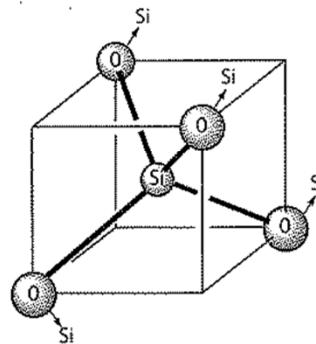
Note: the valence of one atom can be roughly viewed as the number of electrons available for the bonds, and accordingly as the number of bonds one atom can form with other atoms.

Figure 2.6

Covalent bonding requires that electrons be shared between atoms in such a way that each atom has its outer *sp* orbital filled. In silicon, with a valence of four, four covalent bonds must be formed.



Silica (SiO_2)



**COVALENT BOND
is
HIGHLY DIRECTIONAL**

Figure 2.8

The tetrahedral structure of silica (SiO_2), which contains covalent bonds between silicon and oxygen atoms.

Properties of covalent solids

Optical properties

Transparent and colourless, as the bonding energy is higher than the energy of visible light that so does not interact with the material but passes through it.

Electrical properties

Low electrical conductivity as electrons are strongly bonded in specific sites

Physical properties

- High melting point due to the high bonding strength
- Low density

Mechanical properties

- High hardness due to the high bonding energy
- Brittle fracture due to the high bonding energy and bond directionality

OVERVIEW ON POLYMERS

- Organic macro-molecules,
natural or synthetic
- Prepared by organic synthesis
- Low density, less than 1 g/cm^3
- Easy to machine
- Insulators (sound, heat and electricity)
- Low thermal and mechanical properties
- *examples: tyres, adhesives, paints, bitumen,..*



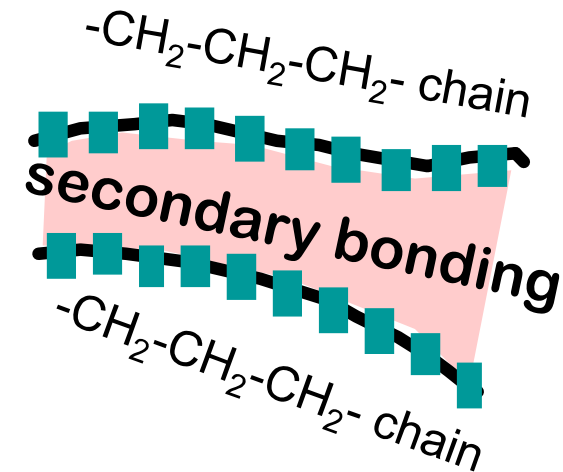
Bonds in POLYMERS: covalent + Van der Waals

Polymers (e.g. nylon, polyethylene, polyvinylchloride, rubber, etc.), commonly referred to as “plastics”, basically consist of $-\text{CH}_2-\text{CH}_2-\text{CH}_2-$ molecular chains that have covalent bonding within each chain (e.g., C-C bond or C-H bond) and Van der Waals' forces between the chains.

Due to the bonding, polymers are typically electrical and thermal insulators. However, conducting polymers can be obtained by doping, and conducting polymer-matrix composites can be obtained by the use of conducting fillers.

Most of polymers:

- cannot sustain high temperature,
- are insulating to the passage of electricity,
- show low mechanical resistance,
- are flexible



because of the presence of weak secondary bonds

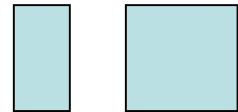
Comparison of some properties dependent on bond

Van der Waals

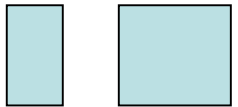
Metallic

Ionic

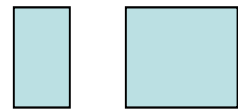
Covalent



STRENGTH OF THE CHEMICAL BOND



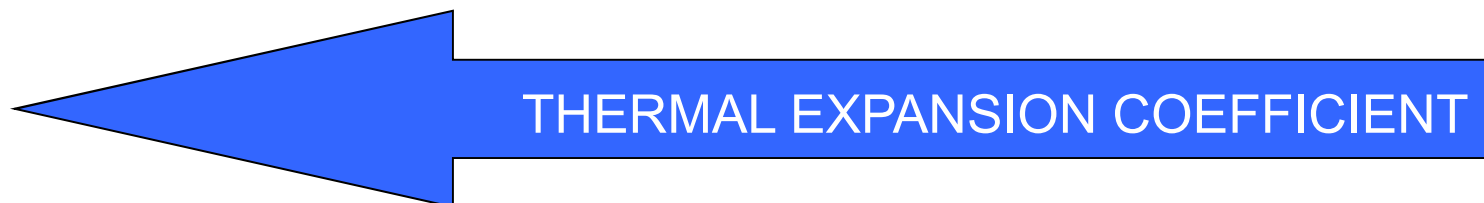
MELTING TEMPERATURE



STIFFNESS



HARDNESS



THERMAL EXPANSION COEFFICIENT



OVERVIEW ON COMPOSITES

- TWO PHASES:

MATRIX and SECOND PHASE

- Classified according to the matrix:

- Metal matrix composites

- Polymer matrix composites

- Ceramic matrix composites

- Glass and glass-ceramic matrix composites

- examples: *wood, reinforced concrete, bones, ...*



OVERVIEW ON COMPOSITES

- Can be classified according to the second phase morphology:
- Fiber-reinforced composites (long fibers, short fibers)
- Particle-reinforced composites
- The second phase can be: metal, polymer, ceramic, glass



STRUCTURE OF MATERIALS

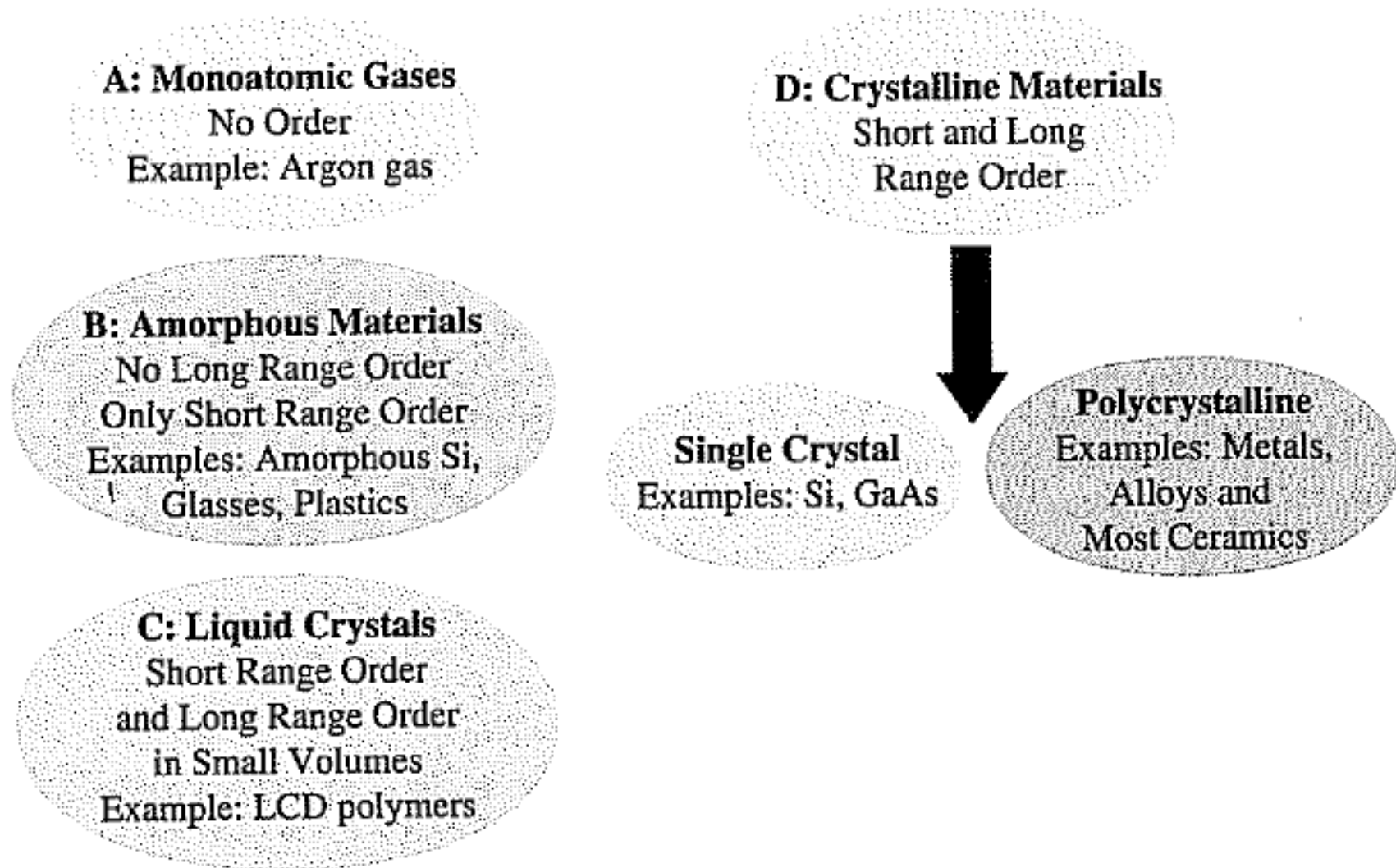
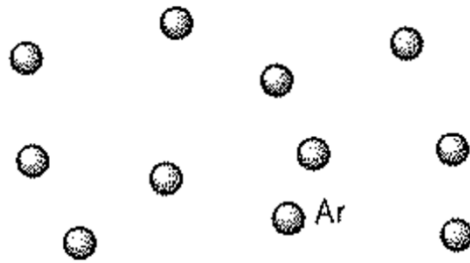


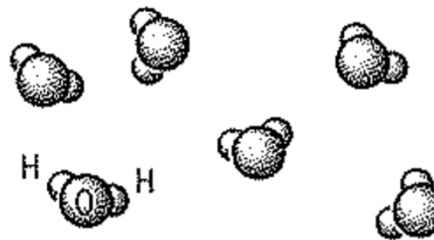
Figure 3-4 Classification of materials based on the type of atomic order.

Argon (gas)

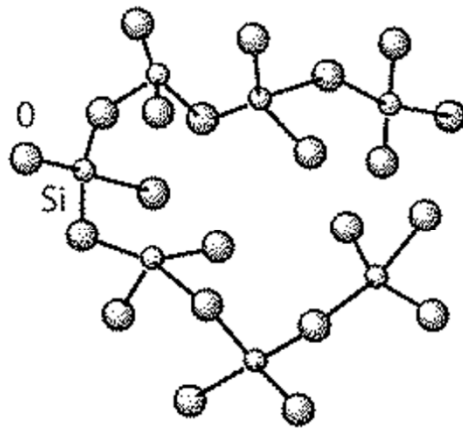


(a)

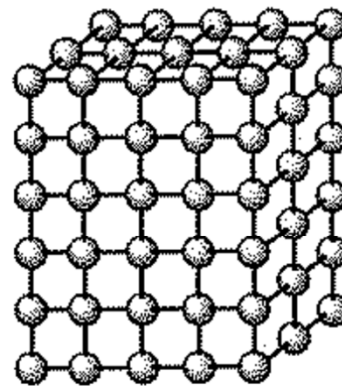
Water (in gas state)



(b)



(c)



(d)

Figure 3.1

The levels of atomic arrangement in materials:

(a) Inert gases have no regular ordering of atoms.

(b, c) Some materials, including steam and glass, have ordering only over a short distance.

(d) Metals and many other solids have a regular ordering of atoms that extends through the material.

Silicate glass

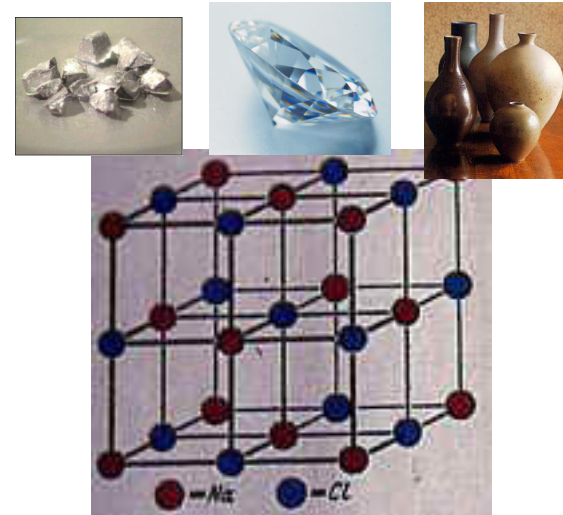
(each Si atom is surrounded by four O atoms)

Metals and ceramics

From a structural viewpoint, solid materials can be divided into:

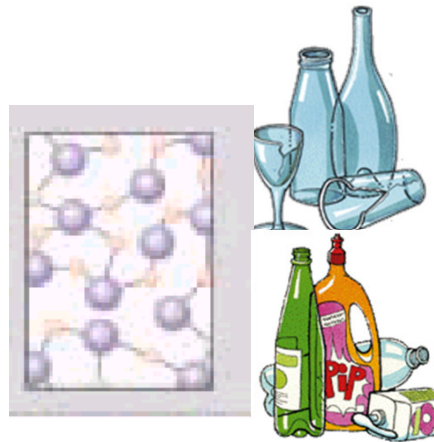
CRYSTALLINE

Metals, ceramics



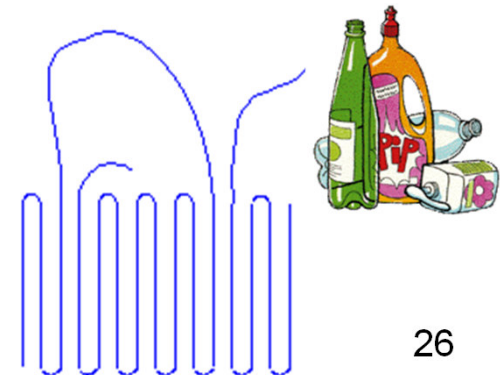
AMORPHOUS

Glasses,
polymers



SEMI-CRYSTALLINE

Polymers, glass-ceramics



CORRELATION STRUCTURE- PROPERTIES

- ...the aim of Materials Science



- CRYSTALLINE STRUCTURES

- Melting Temperature

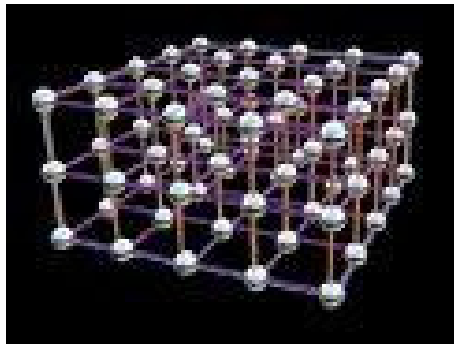
- AMORPHOUS STRUCTURES

- They do not have a defined melting temperature, but undergo progressive softening



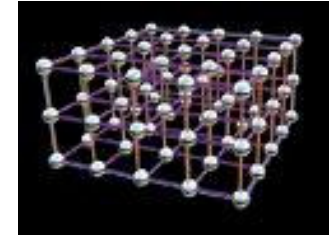
CRYSTALLINE STRUCTURE: MELTING TEMPERATURE

- Melting temperature $\rightarrow$ Solid to liquid (T_m)
- directly proportional to the material bond strength

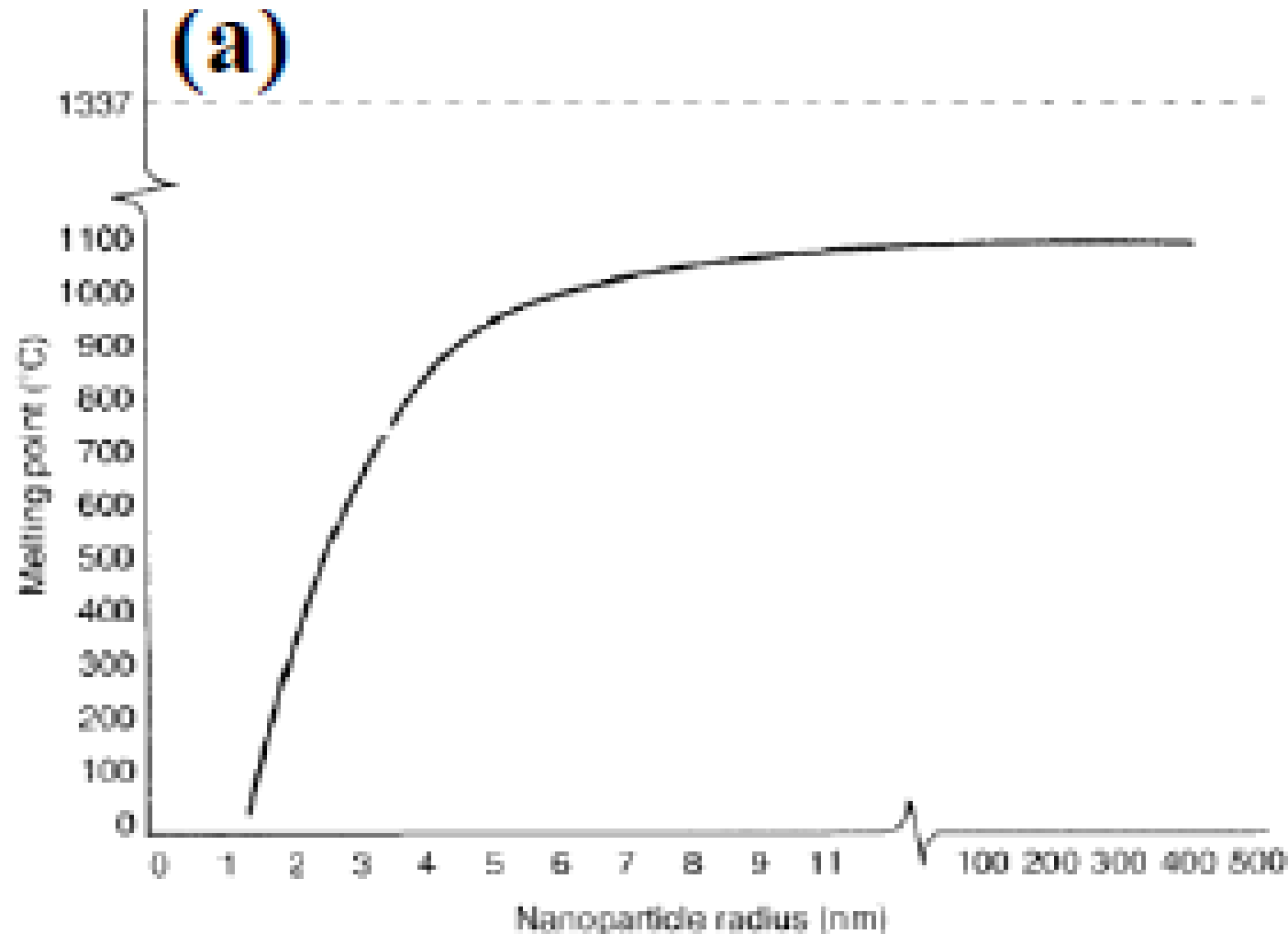


MELTING TEMPERATURE

	T _m (°C)		T _m (°C)		T _m (°C)
W	3410	Mg	651	Al ₂ O ₃	2045
Ti	1675	Pb	270	SiO ₂	1730
Fe	1536	Sn	232	MgO	2800
Ni	1453	Al	660	CaCO ₃	1339
Cu	1083				

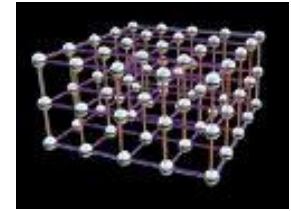


(a) Melting point temperature of gold as a function of nanoparticle radius

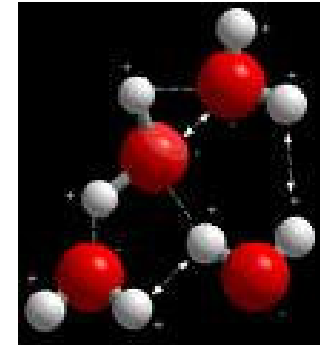


Introduction to Nanoscience and Nanotechnology' Authors Gabor L Hornyak, Harry F, Tibbals, Joydeep Dutta and John J. Moore, CRC Press, NY 978-1-4200-4805-6, 2009.

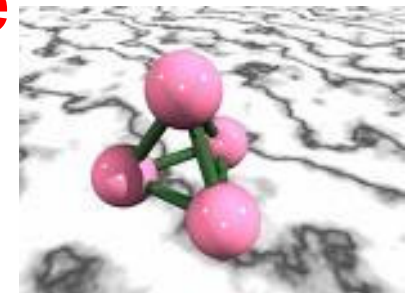
CRYSTALLINE STRUCTURE



- **Chemical bond** and crystalline structure: atoms, ions or molecules in crystalline structures are in ordered positions in **crystalline cells**



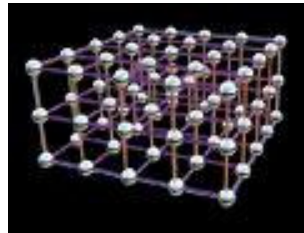
- atoms, ions or molecules are at the equilibrium distance between attraction and repulsion forces: **bond length or bond distance**



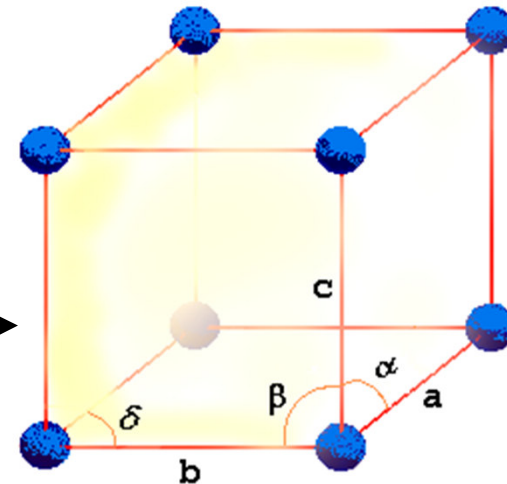
CRYSTALLINE STRUCTURE

- “CRYSTALLINE materials: ordered positioning of **atoms** (ex: metals), or **ions** (ex: ceramics) or **molecules** (ex: polymers)”

- CRYSTALLINE
CELL



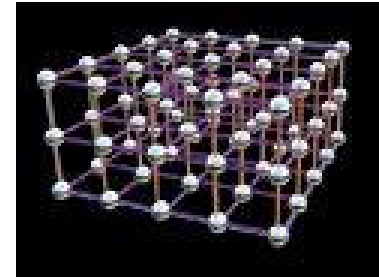
- a, b, c
0,1-0,26 nm



CRYSTALLINE CELLS

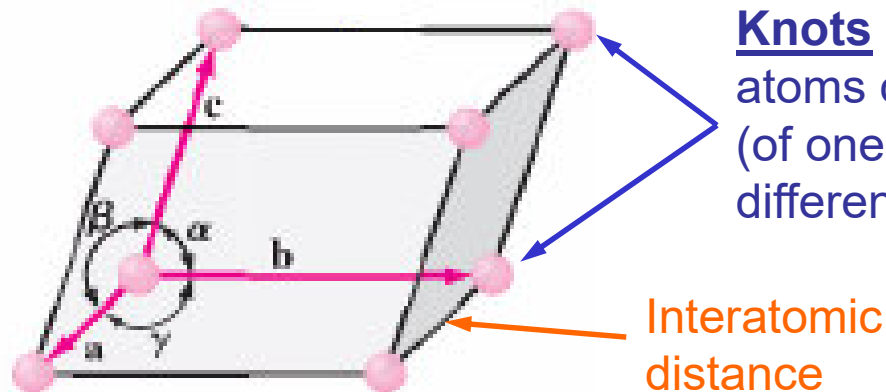
a, b, c cell parameters

α, β, γ cell angles



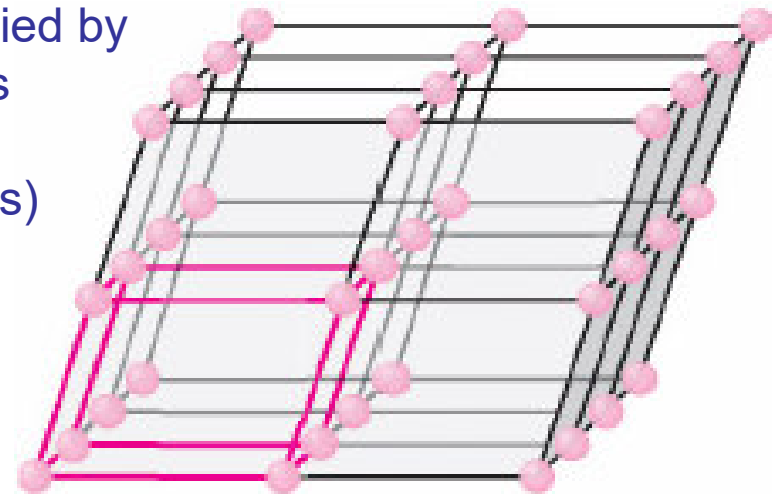
- identify the structure of crystalline materials by cell parameters
- a, b, c about 0,1-0,26 nm at room T and without ext. applied forces

From the unit cell...



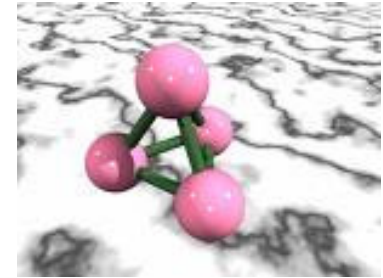
a, b, c are the distance parameters
 α, β, γ are the angular parameters

...to the entire crystal



Inside crystals the atoms or ions move around their lattice knot (= equilibrium position in the lattice)

CRYSTALLINE CELLS

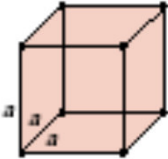
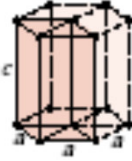
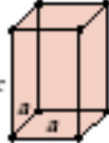
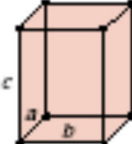
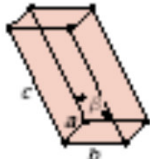


- a, b, c change with T and applied forces
- At 0 K they are at the equilibrium distance
- When T increases, vibration around the equilibrium distance cause **thermal expansion, then melting.**



SEVERAL CRYSTALLINE CELLS

$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$	CUBIC
$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	TETRAGONAL
$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	ORTHORHOMBIC
$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$	ROMBOEDRIC
$a = b \neq c$	$\alpha = \beta = 90^\circ, \gamma = 120^\circ$	HESAGONAL
$a \neq b \neq c$	$\alpha = \beta = 90^\circ \neq \gamma$	MONOCLINE
$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$	TRICLINE

<i>Crystal System</i>	<i>Axial Relationships</i>	<i>Interaxial Angles</i>	<i>Unit Cell Geometry</i>
Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$	
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ, \gamma = 120^\circ$	
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	
Rhombohedral	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$	
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	
Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90^\circ \neq \beta$	
Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$	

**There are 7 basic
lattice cells**

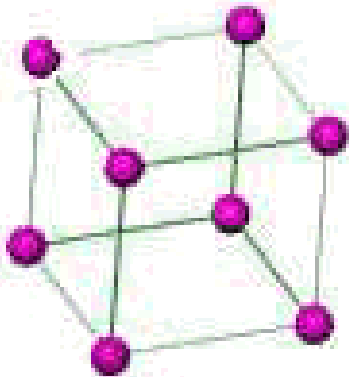
Note:

a, b, c are typically within 0.1-0.29 nm
at $T = 25^\circ\text{C}$ and without external
forces applied upon the crystal.

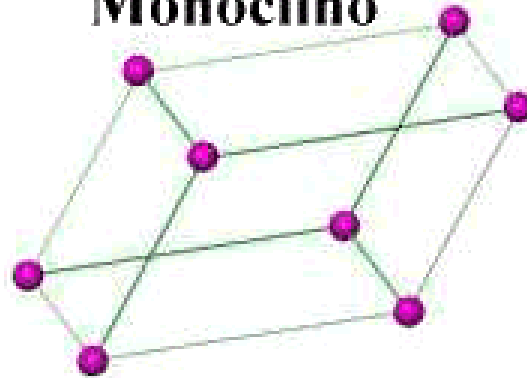
If $T \uparrow \rightarrow \uparrow$ vibrations of atoms/ions
around the knots

Crystalline cells

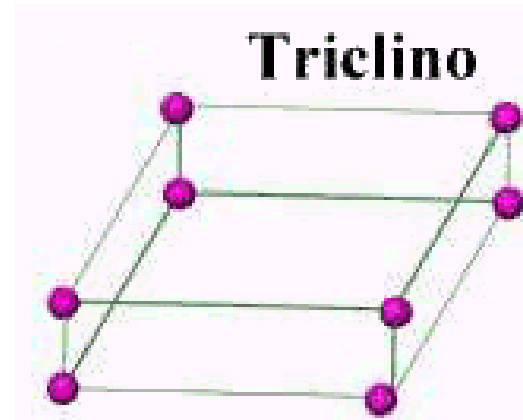
Cubico



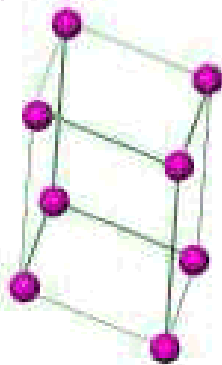
Monoclino



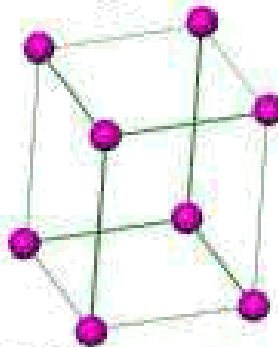
Triclino



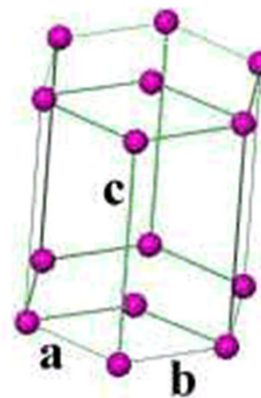
Romboedrico



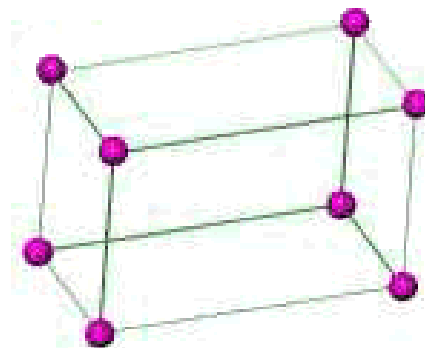
Tetragonale

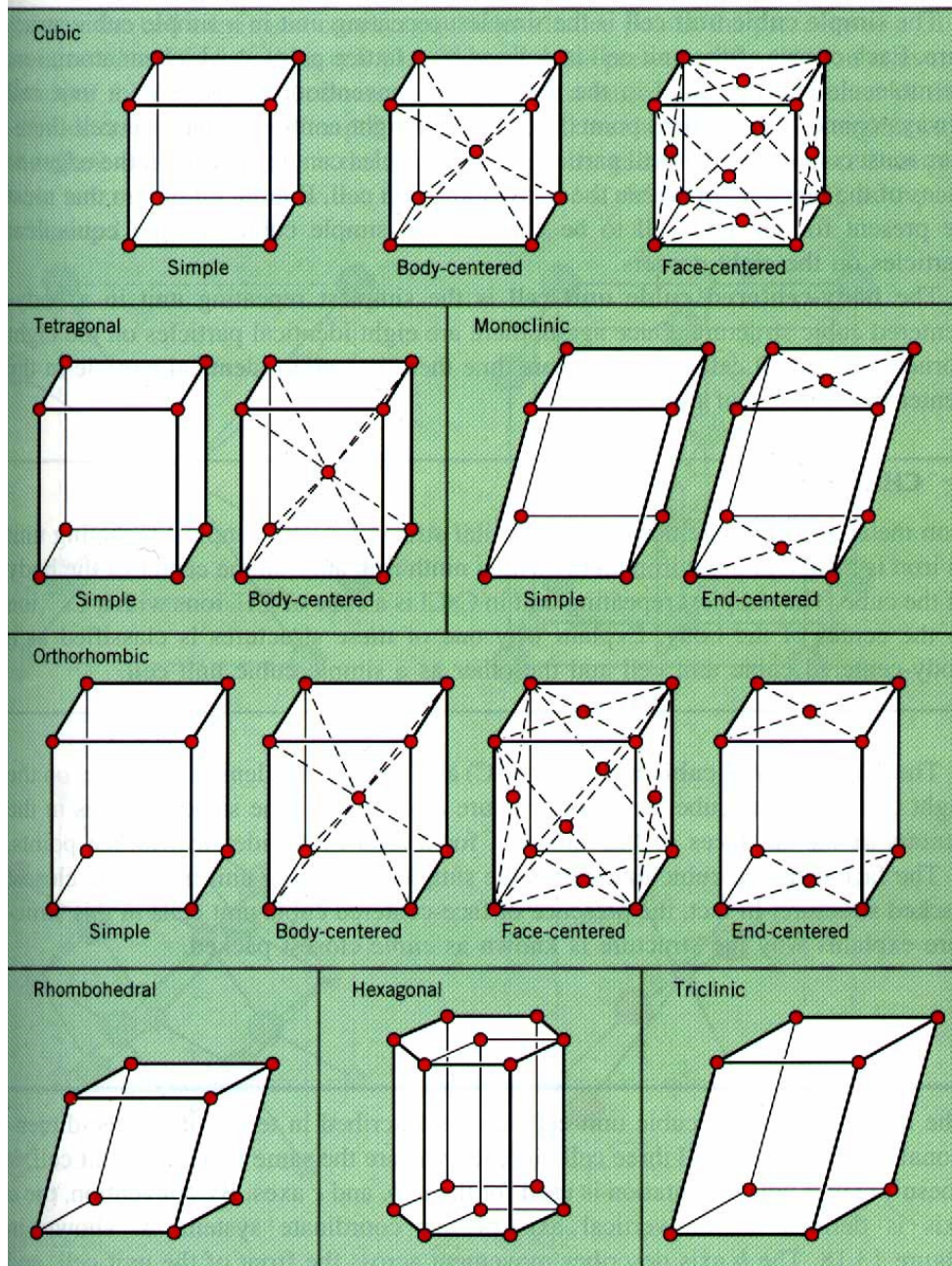


Esagonale



Ortorombico





There are other 7 composed lattice cells derived from the basic ones.

These new 7 lattice cells may have knots on some of their faces (**face-centred cells, FCC**) or inside the cell (**body-centred cell, BCC**).

All these 14 lattice cells are called **Bravais's lattice cells**.

Face-centered cubic (FCC) cell

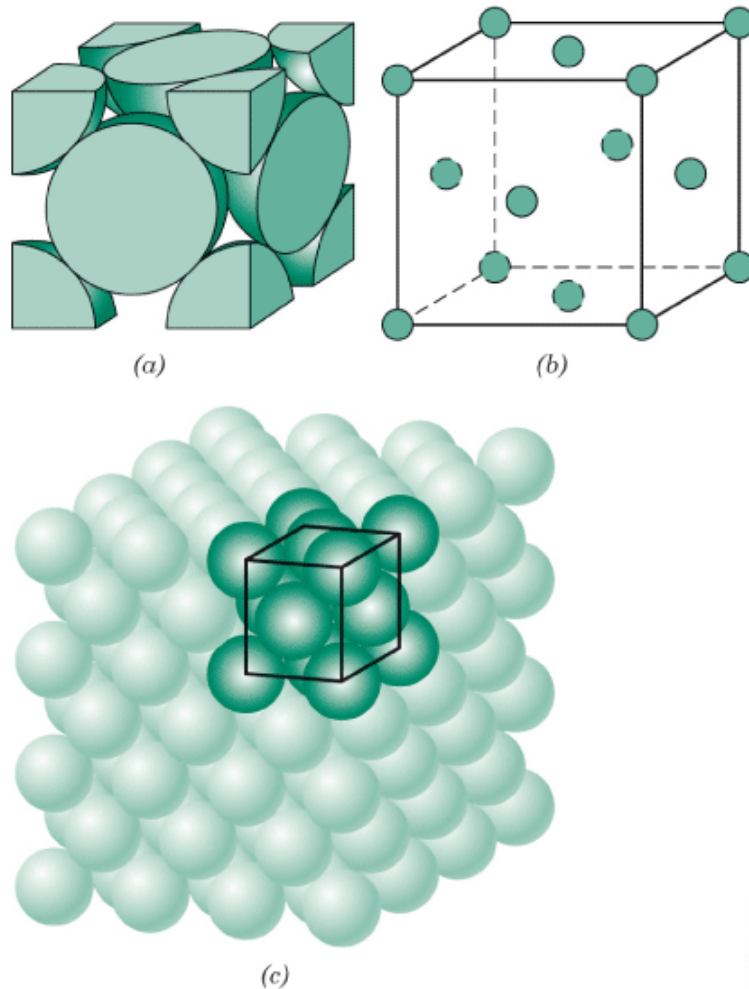


Figura 3.1 Struttura cristallina cubica a facce centrate (cfc): (a) rappresentazione della cella unitaria con il modello a sfere rigide, (b) rappresentazione con il modello a sfere ridotte e (c) aggregato di parecchi atomi. [Figura (c) da W. G. Moffatt, G. W. Pearsall, and J. Wulff, *The Structure and Properties of Materials*, Vol. I, *Structure*, p. 51. Copyright © 1964 by John Wiley & Sons, New York. Riproduzione autorizzata da John Wiley & Sons, Inc.].

Body-centered cubic (BCC) cell

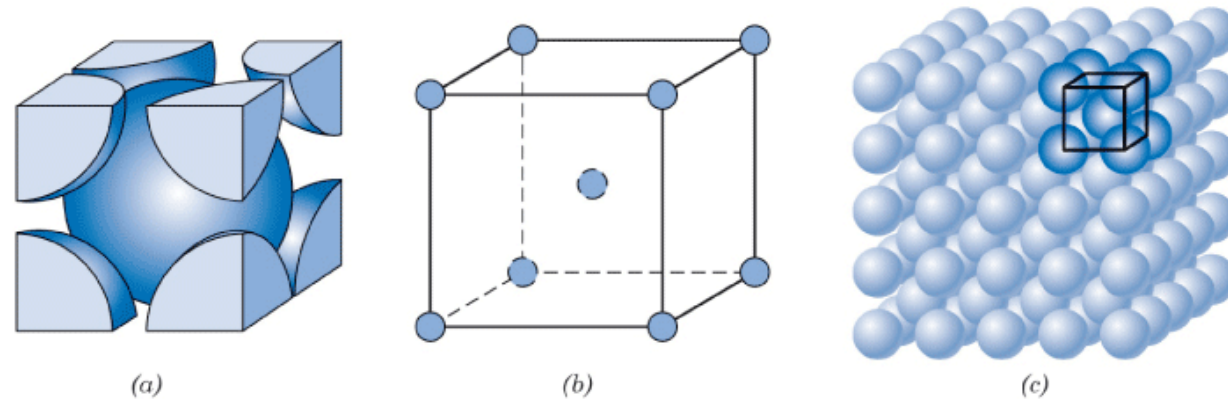
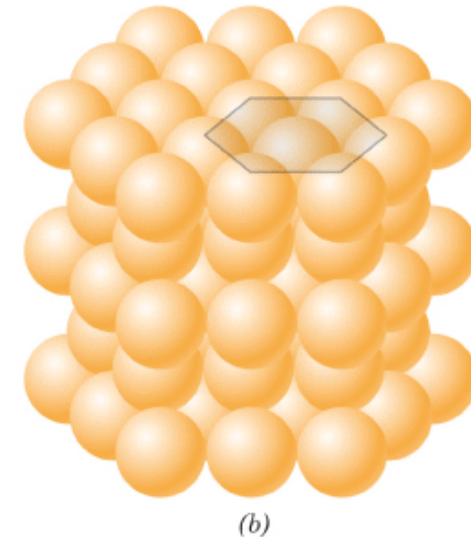
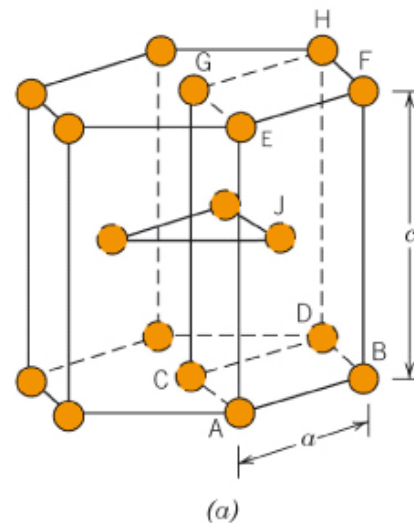


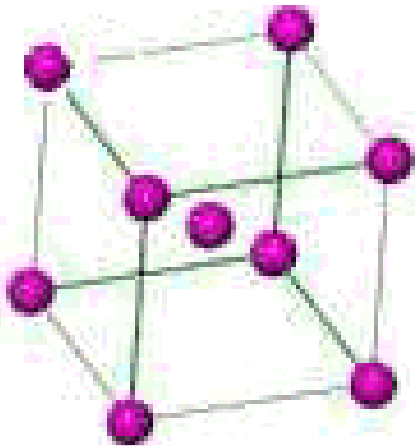
Figura 3.2 Struttura cristallina cubica a corpo centrato, (a) rappresentazione della cella unitaria a sfere rigide, (b) cella unitaria rappresentata con sfere ridotte, (c) un aggregato di molti atomi. [Figura (c) da W. G. Moffatt, G.W. Pearsall, and J.Wulff, *The Structure and Properties of Materials*, Vol. I, *Structure*, p. 51. Copyright © 1964 by John Wiley & Sons, New York. Riprodotta con il permesso di John Wiley & Sons, Inc.].

Hexagonal compact (HC) cell

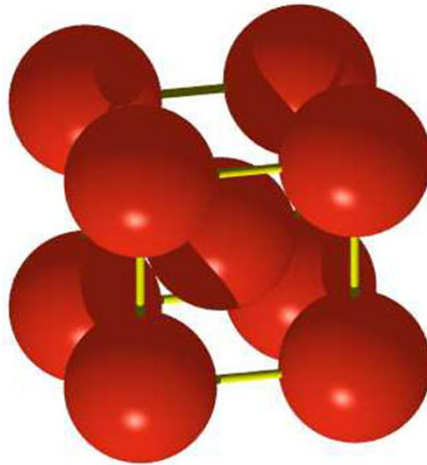
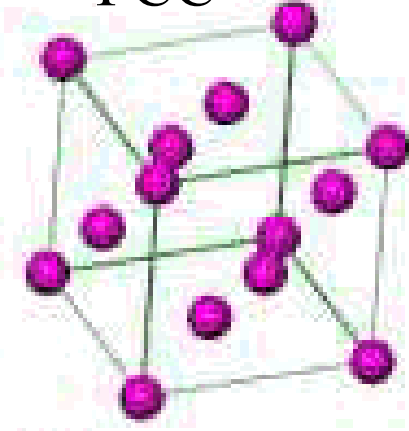
Figura 3.3 Struttura cristallina esagonale compatta, (a) cella unitaria rappresentata con sfere ridotte (a e c rappresentano, rispettivamente, le lunghezze del lato della base e dell'altezza del prisma), e (b) un aggregato di molti atomi. [Figura (b) da W. G. Moffatt, G.W. Pearsall, and J. Wulff, *The Structure and Properties of Materials*, Vol. I, *Structure*, p. 51. Copyright © 1964 by John Wiley & Sons, New York. Riproduzione autorizzata da John Wiley & Sons, Inc.].



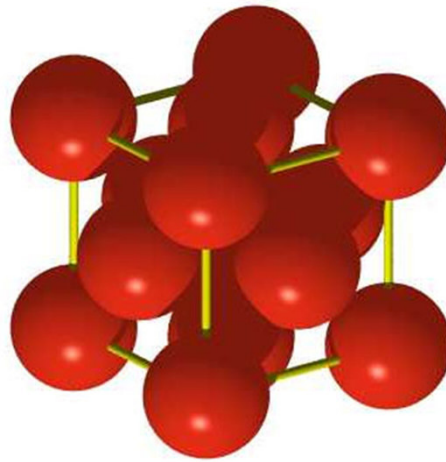
BCC



FCC

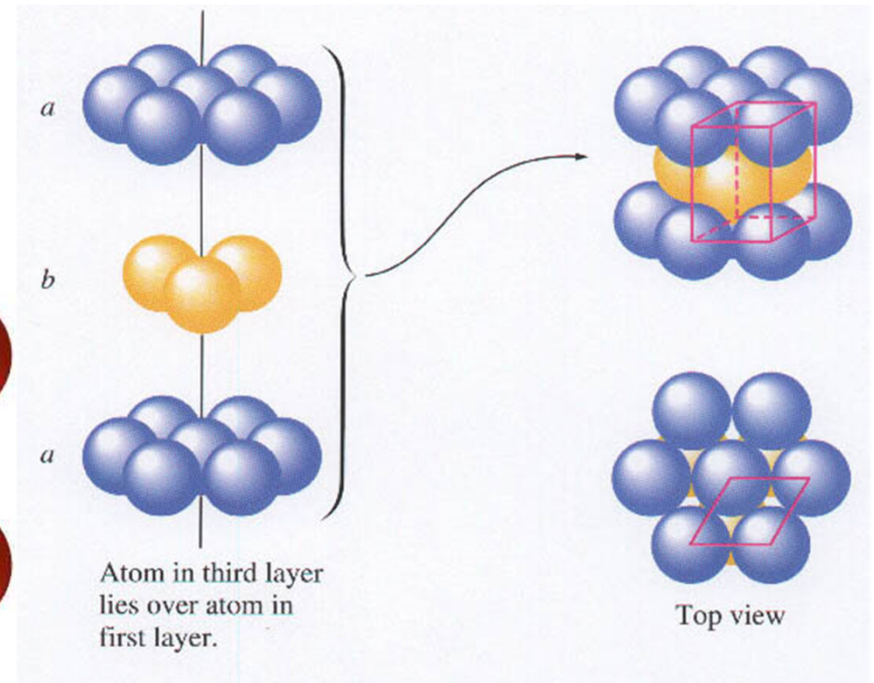


Es: Fe, W, Cr

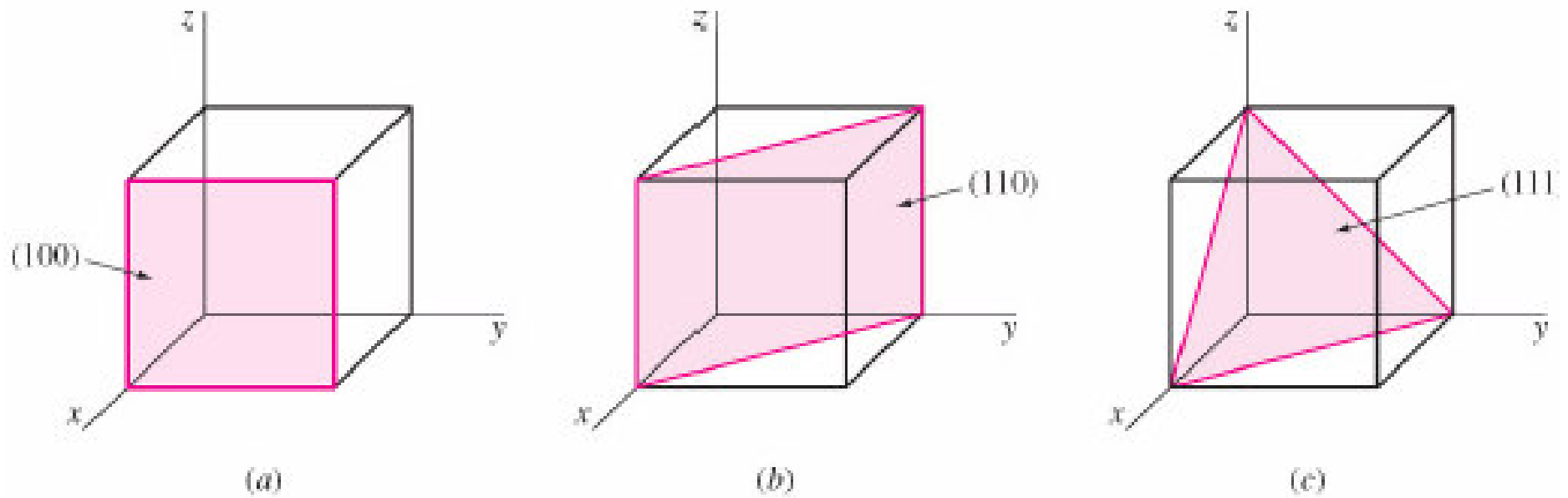


Es: Cu, Al, Ag, Au

HC

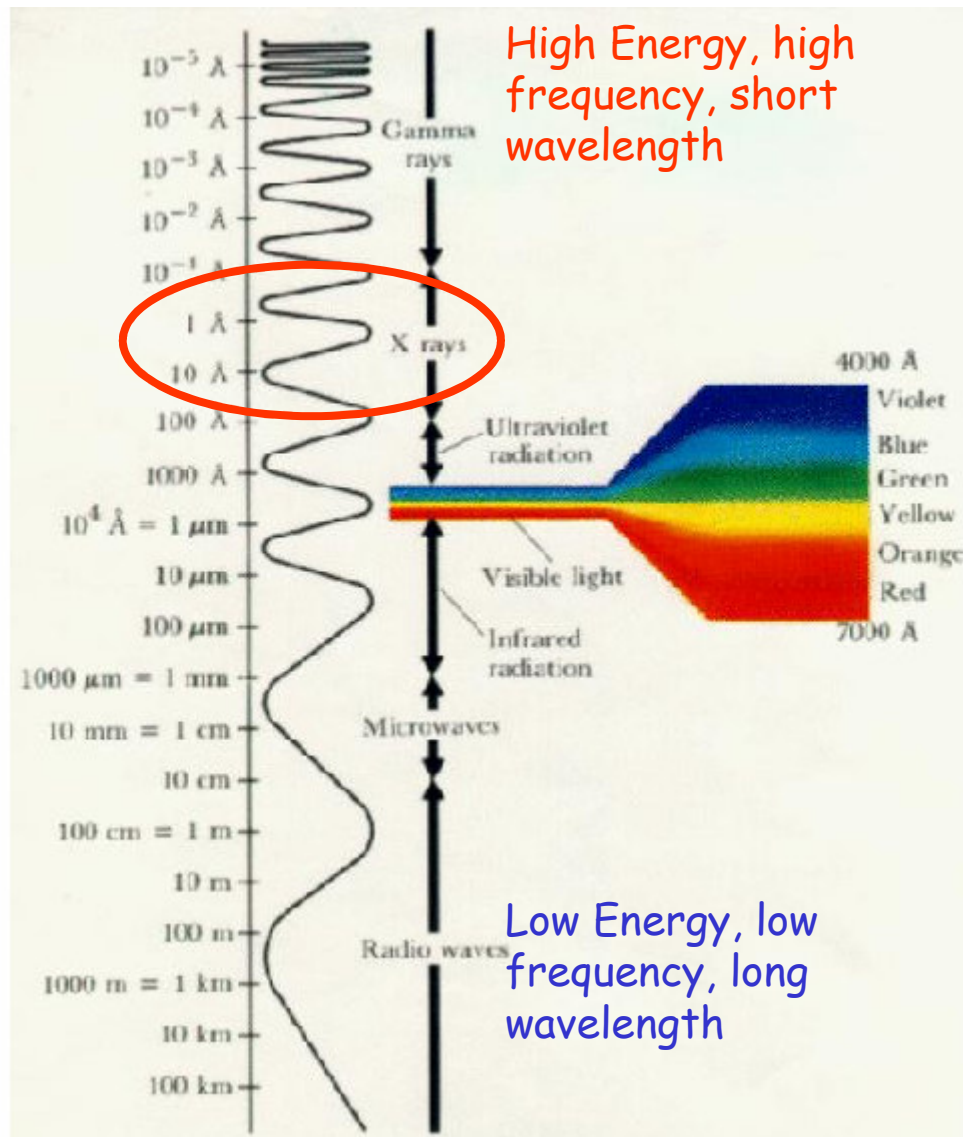


Es: Mg, Ti, Zn



$(h\ k\ l)$ are called Miller indexes

Energy, frequency



Electromagnetic radiation spectrum

$$E = h\nu$$

$$\lambda = c / \nu$$

E = energy

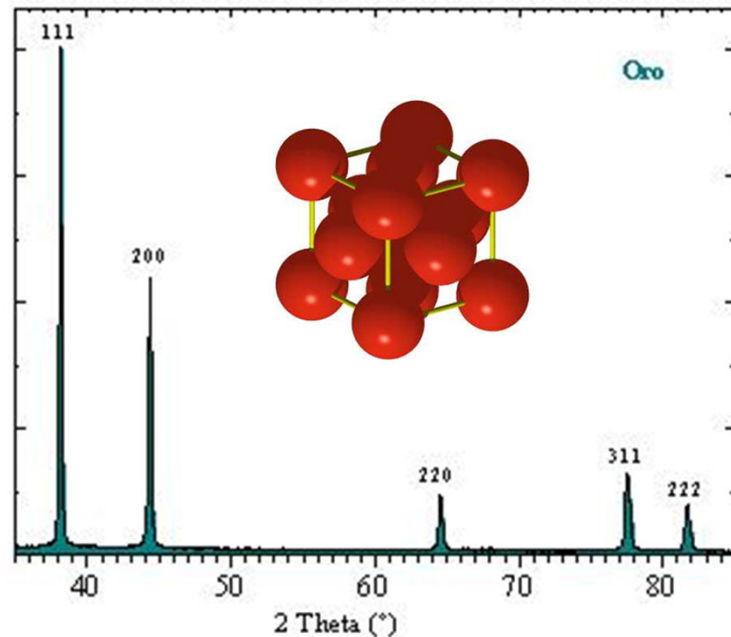
ν = frequency

λ = wavelength

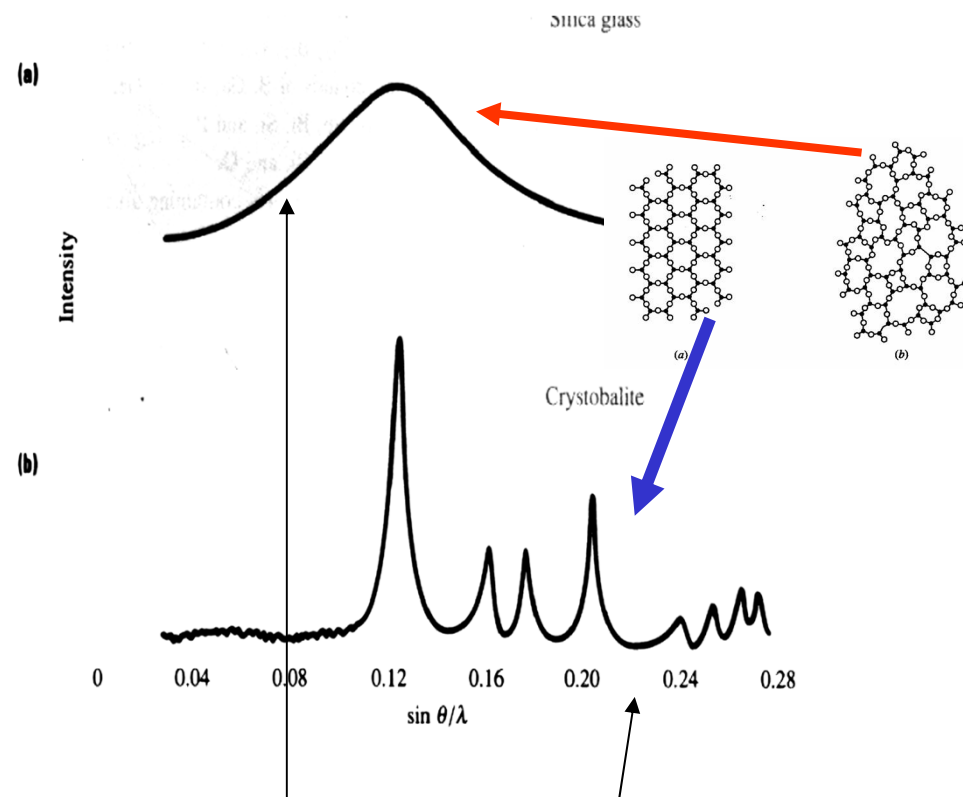
c = light speed

wavelength

X-ray diffraction (XRD)



XRD of a crystalline material (Au)

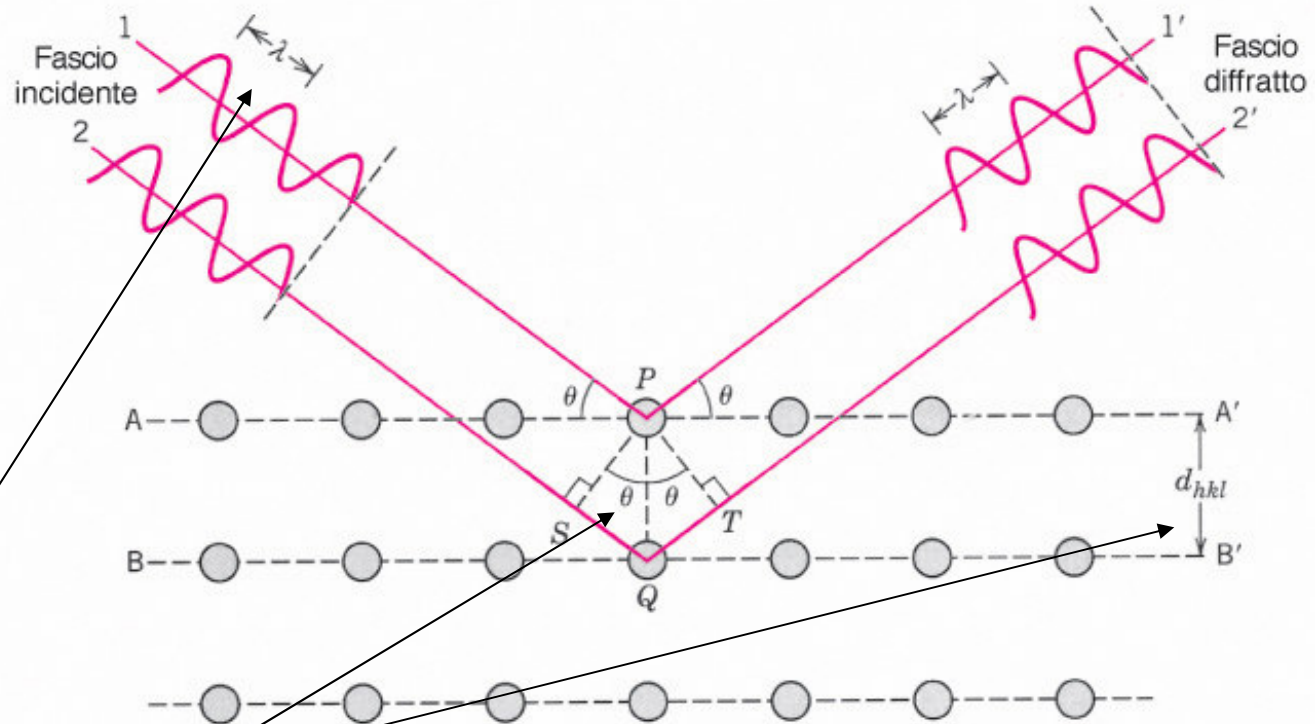


XRD of **amorphous silica** and **crystalline silica** (SiO_2)

X ray diffraction

FIGURA 3.18

Diffrazione a raggi x da parte dei piani di atomi (A-A' e B-B').



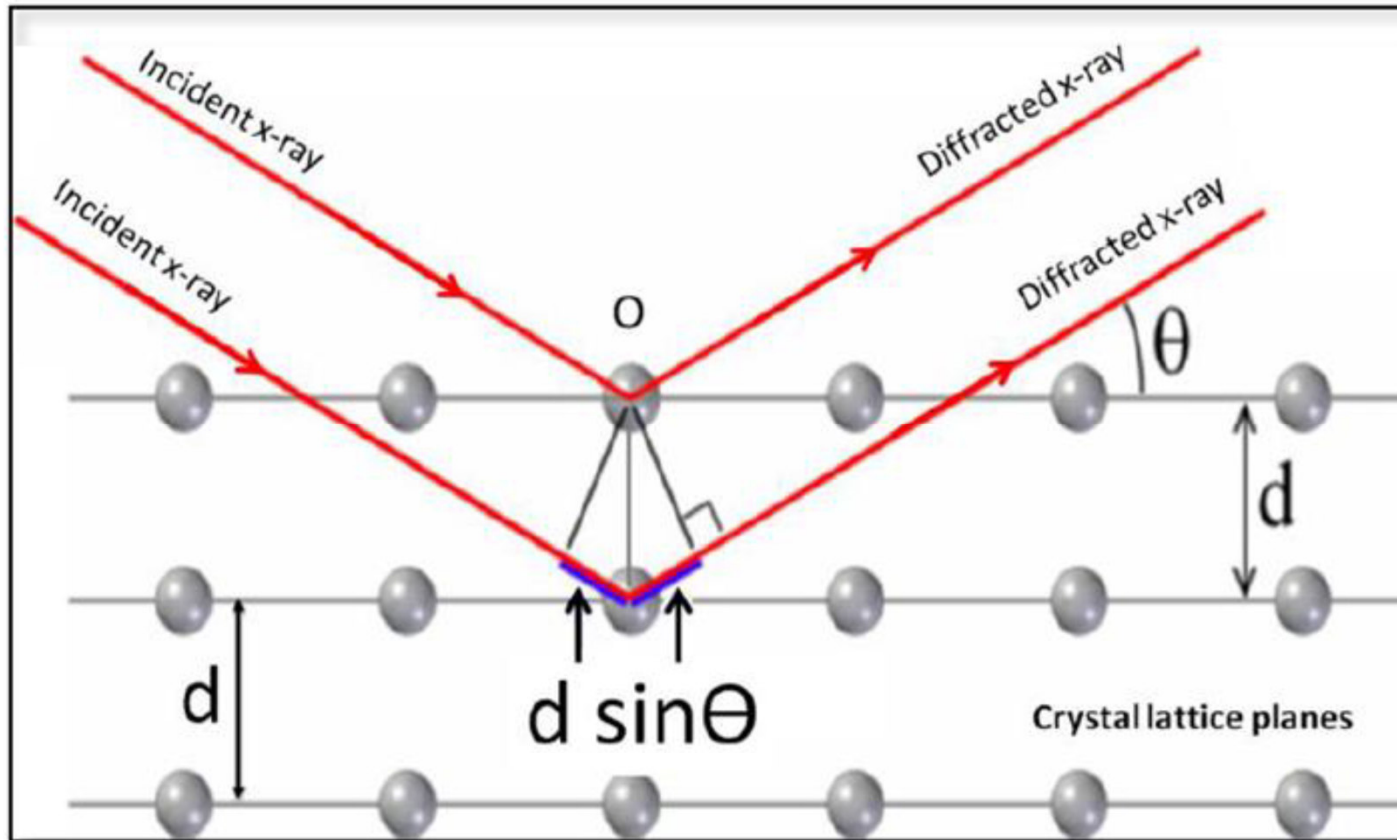
$$n \lambda = 2 d \sin \theta$$

(Bragg law)

n = number, 1, 2, 3...

$$n\lambda = 2d\sin\theta$$

where n is an integer, λ is the wavelength of the beam, d is the spacing between the diffracting planes, and θ is the incident angle of the beam.



X-ray diffraction of silicon carbide (SiC)

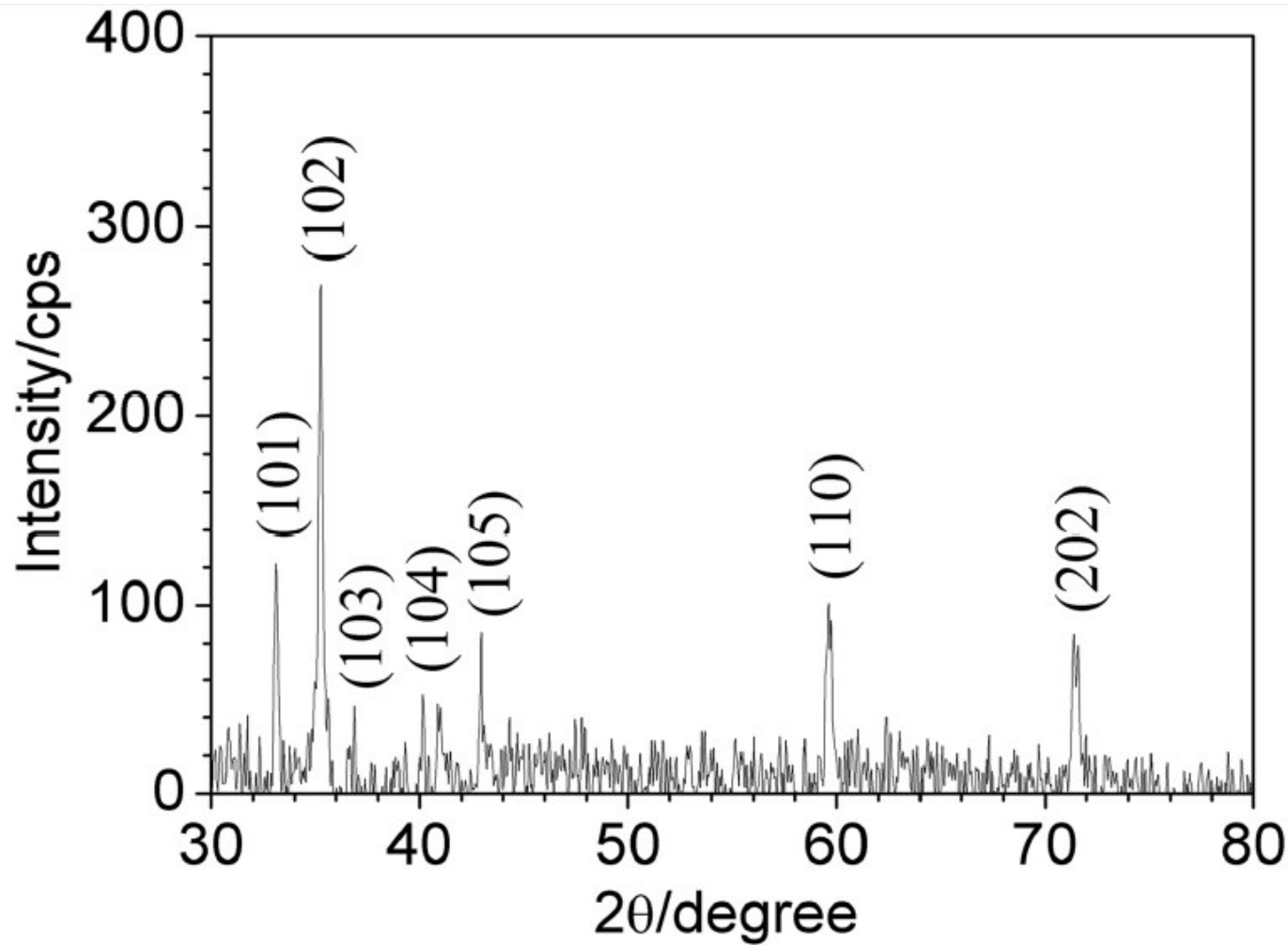


Fig. 3 XRD pattern of the in situ synthesized SiC NWs.

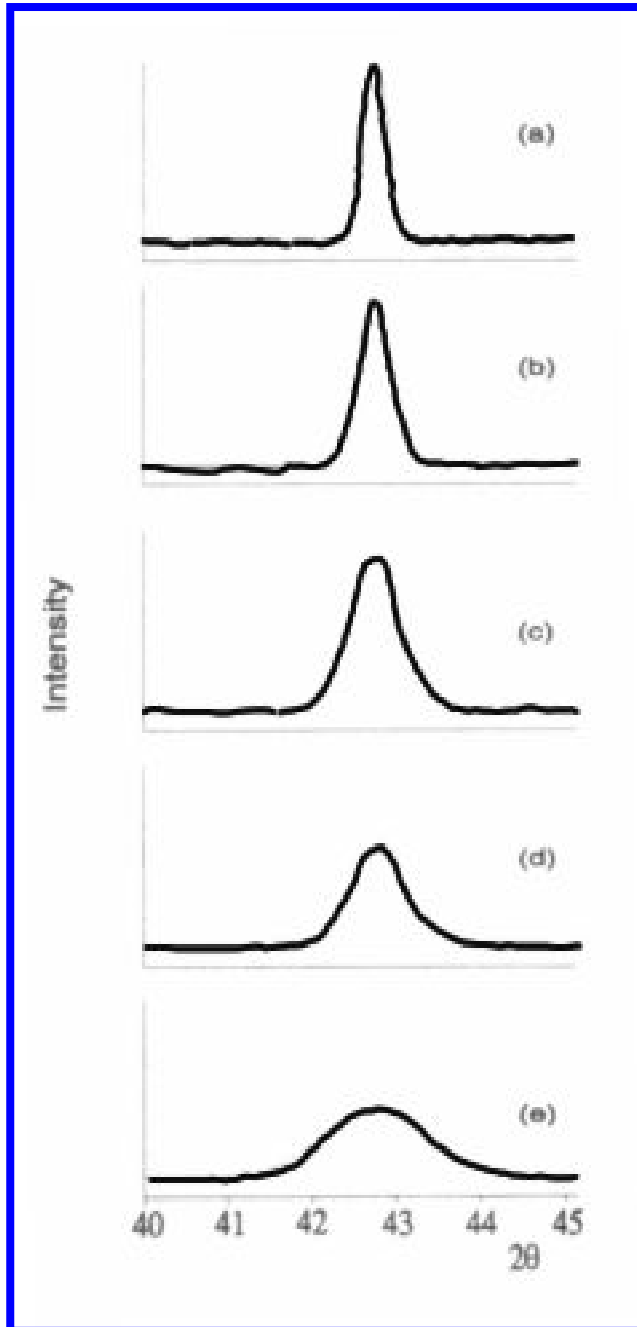
The interplanar distance d depends on:

- lattice parameters
- Miller indexes ($h\ k\ l$)

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

for a cubic cell (in this case $a = b = c$)

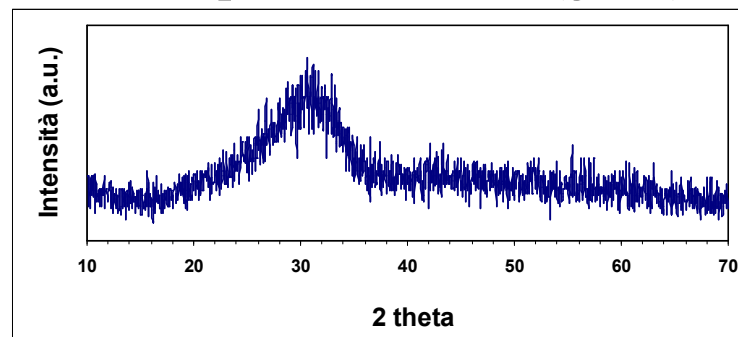
There are other (more complex) mathematical correlations for the other types of crystalline cells.



Assessment of
crystal size by
estimating the
broadness of the
diffraction peak:

- a) 50 μm
- b) 1 μm
- c) 100 nm
- d) 10 nm
- e) 5 nm

Amorphous material (glass)

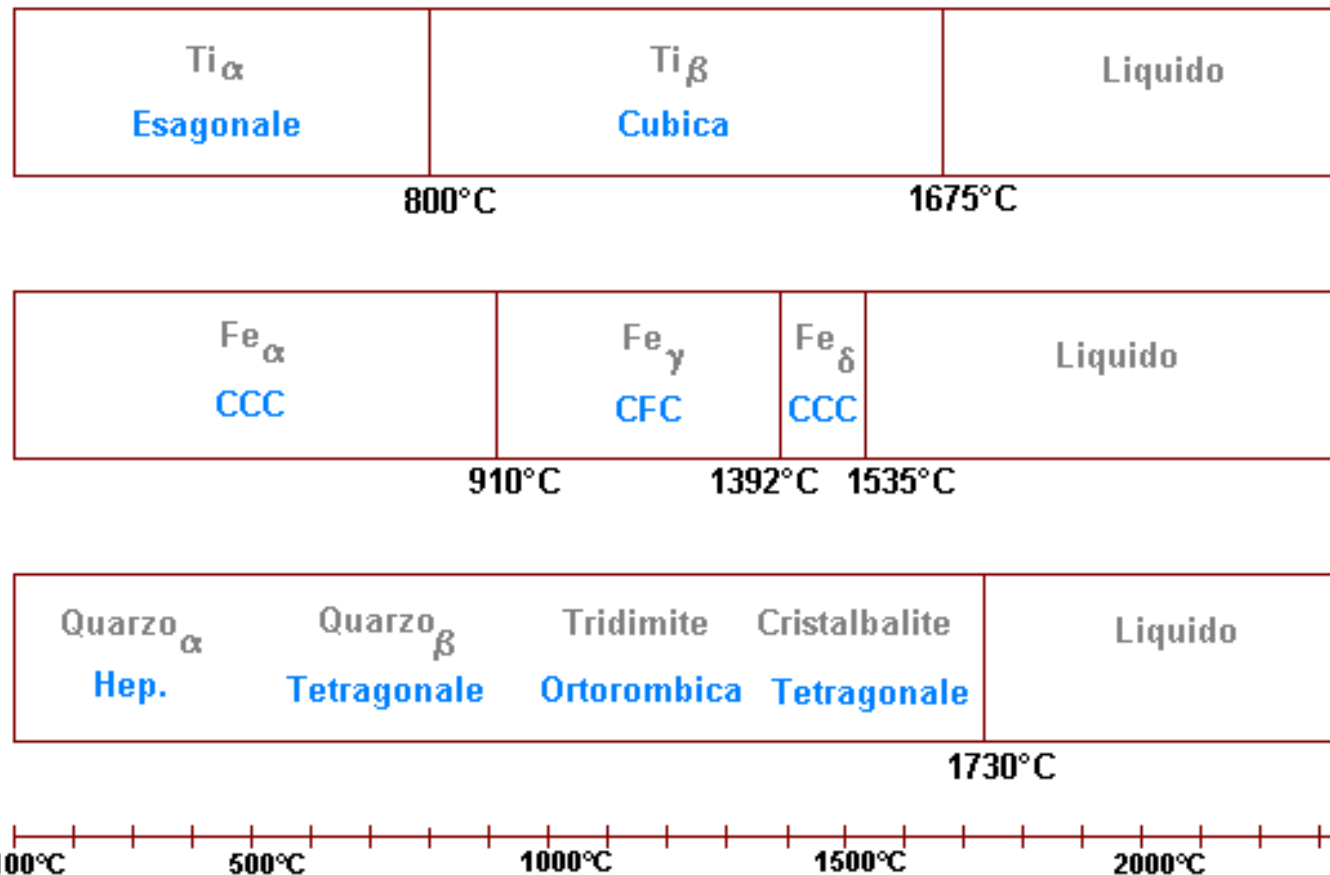


CRYSTALLINE STRUCTURE:

ALLOTROPY and POLYMORPHISM

- A crystalline solid melts when heated
- Allotropy or Polymorphism: change of crystalline structure during heating (solid state reaction)
- **examples:** Ti, Fe, SiO₂
- Allotropy/Polymorphism → variation of materials properties: (V, α , k, E, d,....)

Examples



CRYSTALLINE STRUCTURE: DEFECTS

- Crystalline structure is not completely ordered
→ LATTICE DEFECTS:
- ZERO-DIMENSIONAL DEFECTS
- MONODIMENSIONAL DEFECTS
- BIDIMENSIONAL DEFECTS
- THREE-DIMENSIONAL DEFECTS

Lattice defects have influence on:

- **physical properties** (e.g. density)
- **mechanical properties** (e.g. strength)
- **technological properties** (e.g. cold working ability, electronic conductivity for semiconductors, resistance to corrosion).

**SOME PROPERTIES OF MATERIALS
(at the macro-scale)
CAN BE PROPERLY TAILORED BY
CONTROLLING CRYSTALLINE
DEFECTS (at the micro-scale)**

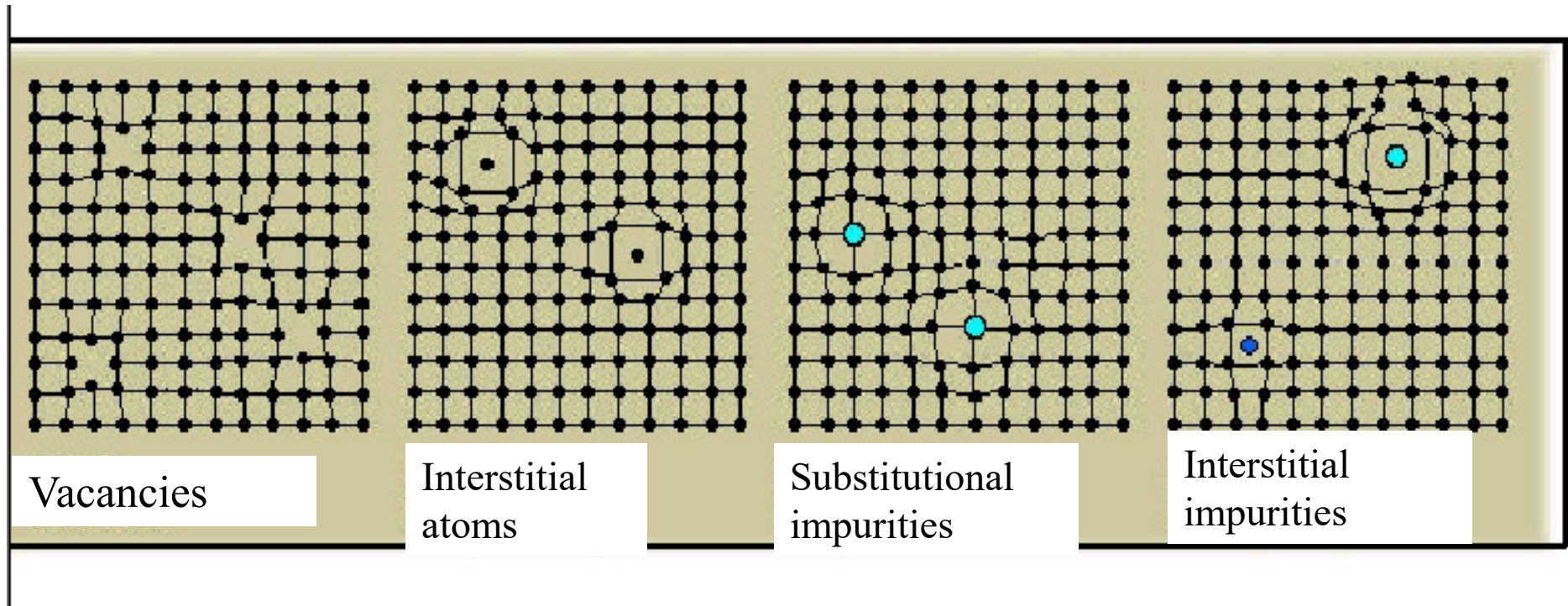
Zero-dimensional defects or point defects

Lattice defects of **atomic dimensions** (1 atom)

Actually, a part of crystal of about 100 atomic radii around the defect is perturbed!

4 types of point defects:

- vacancies
- interstitial atoms
- substitutional atoms
- defects in ionic solids («couple»)



Rough (but practical) definition: we have point defects when the atoms are not in their “correct” places (i.e. the knots) in the crystalline lattice!

ALL THESE LATTICE DEFECTS INDUCE HIGH PERTURBATION IN THE CRYSTAL:

ATOMS AROUND THE POINT DEFECT(S) ARE MOVED AWAY FROM THEIR EQUILIBRIUM POSITION, AND SO THERE IS A STRESS INSIDE THE CRYSTAL

Vacancies

A lattice site (knot) is unfilled and it remains **EMPTY**

Vacancies allow atomic motion from a lattice site to another one (diffusion)

Sources of vacancies

- chemical impurities (e.g. ions with different charges (+) or (-))
- **presence of one element with 2 different oxidation numbers (valences) (e.g. iron, Fe: we can have Fe^{2+} or Fe^{3+})**
- solidification processes (especially in metals)
- atomic vibrations induced by T

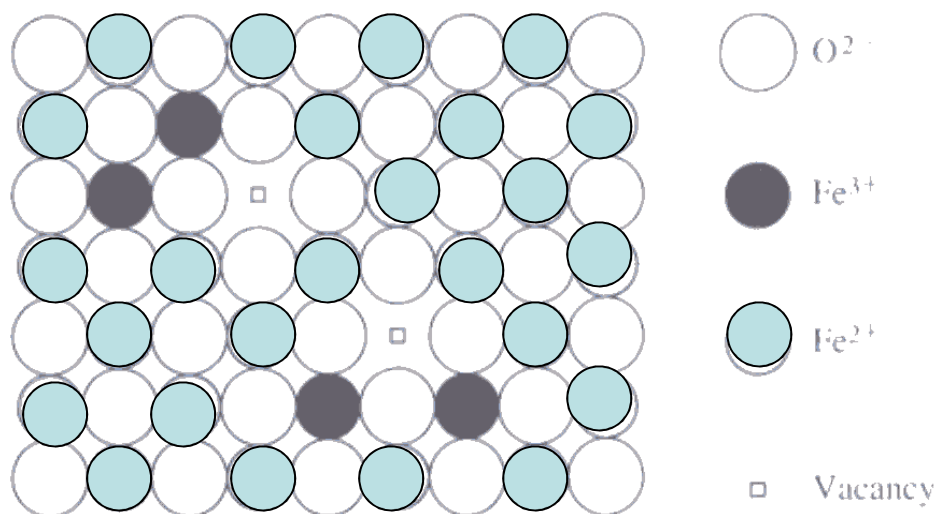


FIGURE 4-7 Iron oxide, Fe_{1-x}O with $x \approx 0.05$, is an example of a non stoichiometric compound. Similar to the case of Figure 4-6, both Fe^{2+} and Fe^{3+} ions occupy the cation sites with **one Fe^{2+} vacancy occurring for every two Fe^{3+} ions present.**

Vacancies concentration depends on temperature according to **Arrhenius's law**:

$$\frac{N_v}{N_{tot}} = C \cdot e^{-E_v / KT}$$

N_v = equilibrium number of vacancies
 N_{tot} = total number of atomic sites
 C = constant
 E_v = energy required for the formation of a vacancy
 K = Boltzmann's constant

Metals at room T at equilibrium: $N_v/N_{tot} = 10^{-17}$

→ (1 vacancy every 100 million billion atoms)

Metals at high T (just below $T_{melting}$) at equilibrium: $N_v/N_{tot} = 10^{-4}$

→ (1 vacancy every 10,000 atoms)

N_v/N_{tot} would be = 0 (NO VACANCIES) if $T = 0$

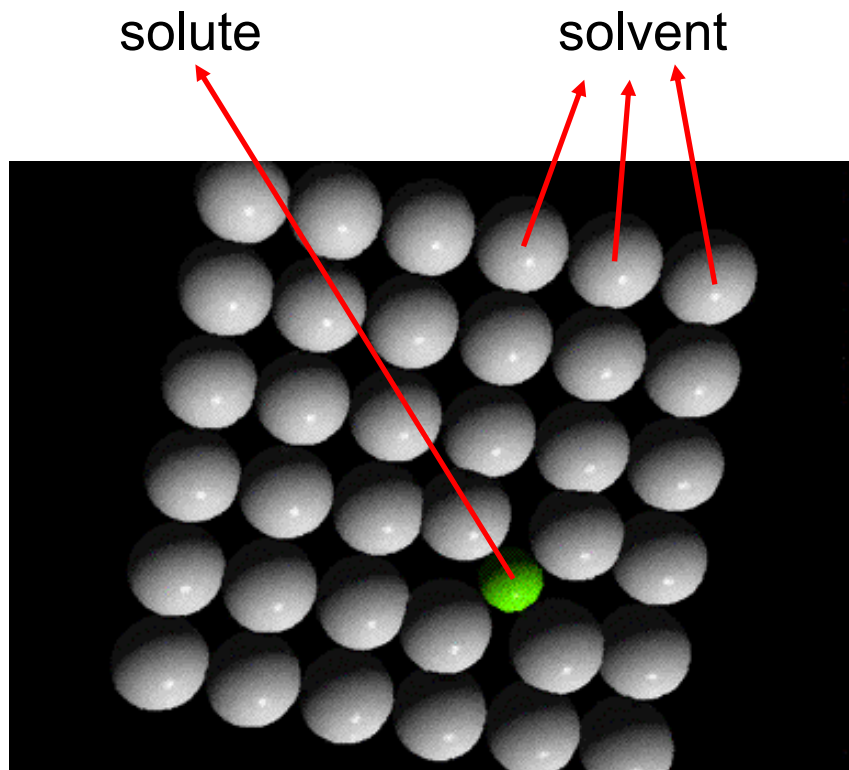
BUT: It is impossible to get to the absolute zero
AND THEREFORE it is impossible to obtain a perfect crystal without
vacancies!

Interstitial atoms

An atom is crowded into an interstitial site (i.e., it does NOT occupy a LATTICE KNOT).

Atom from the crystal $\Rightarrow$ self-interstitial (unusual)

Foreign atom $\Rightarrow$ interstitial impurity atom.

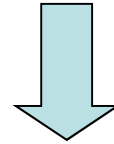


The material can be viewed as a **SOLID SOLUTION** in which the interstitial atoms are the solute.

Metallic lattices: common interstitial impurity atoms are H, C, N, O.

Oxides (e.g. ceramics): small-size metallic ions

In most cases, interstitial elements show atomic radius **BIGGER** than the radius of the lattice holes.



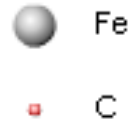
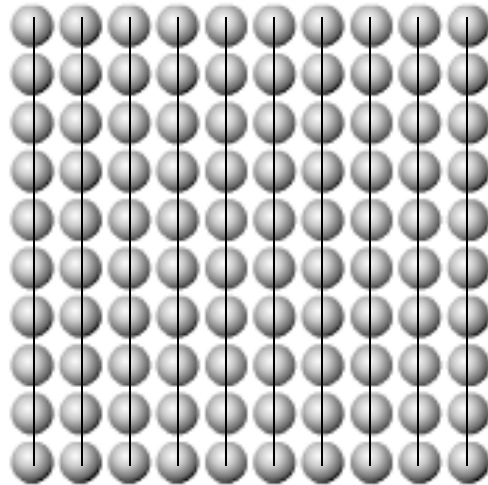
The introduction of an interstitial atom causes perturbation and stress in the crystal → the atoms around the interstitial atom are moved away from their equilibrium positions (or lattice knots).

The crystal can accept only a limited amount of stress and distortion.



Interstitial solid solutions cannot be obtained for every composition, but there is a **LIMIT OF SOLUBILITY** of the interstitial atom in each lattice → this is related to the maximum stress “accepted” by the crystal

Pure Fe lattice

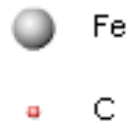
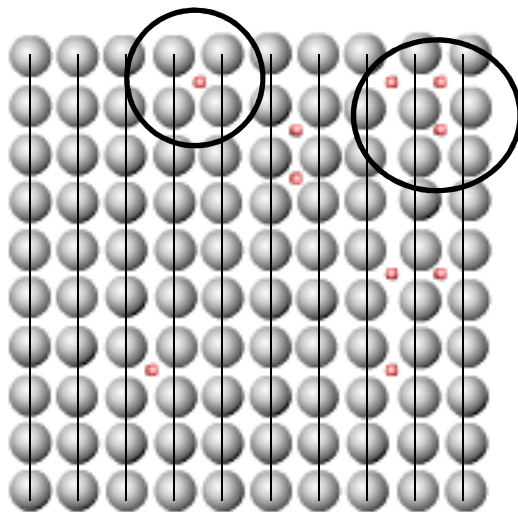


Example: STEEL

(it can be viewed as a solid solution of C in Fe)

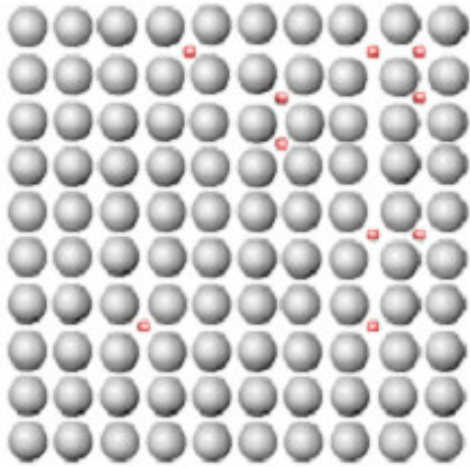
Fe atoms occupy the lattice **knots**
(equilibrium positions)

STEEL

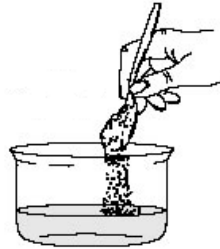


The positions of the Fe atoms are
SLIGHTLY DISPLACED
as C is added

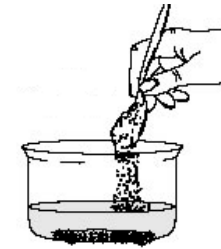
Carbon atoms are introduced into Fe lattice
in interstitial positions to increase the
mechanical properties of iron (beneficial
internal compressive stress)



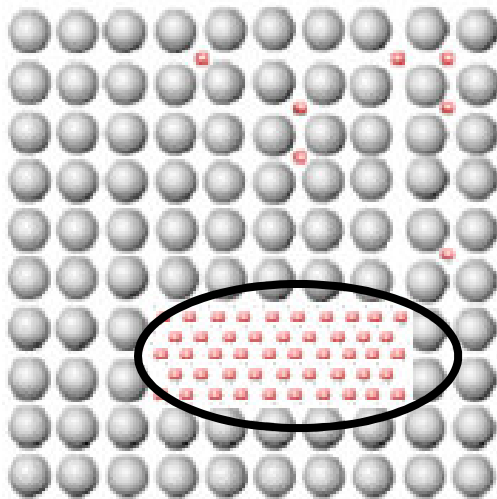
The interstitial atom is added in an amount below the limit of solubility → an interstitial **SOLID SOLUTION** is formed



if you put sugar into water up to the solubility limit you will dissolve it



If you put too much sugar into water, solid sugar will appear as a second phase

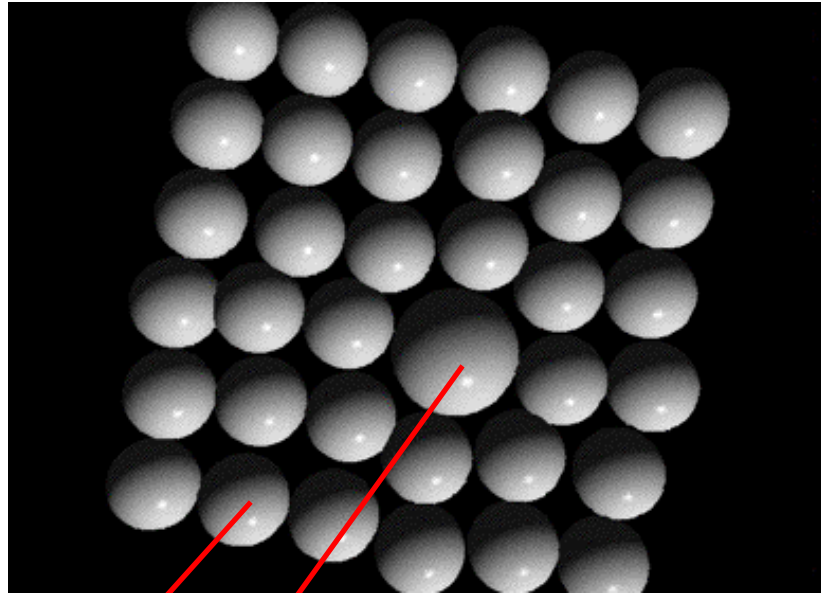


The interstitial atom is added in an amount above the limit of solubility → an **interstitial solid solution** and a second phase (with different crystalline structure and composition) are formed.

Nature follows this way because of thermodynamics reasons → a second phase presenting lower energy (i.e. **MORE STABLE**) is obtained

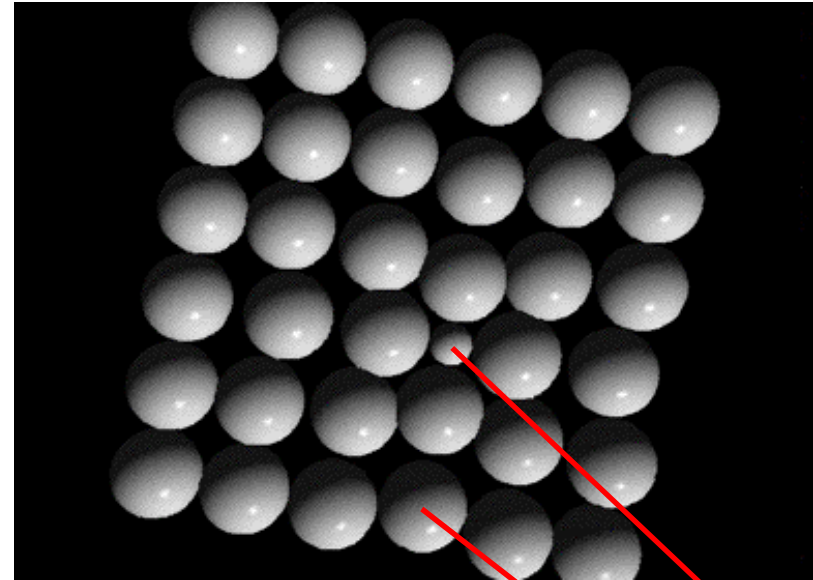
Substitutional atoms

Substitutional atom is smaller or bigger than host atoms and occupies a lattice knot.



solvent

solute

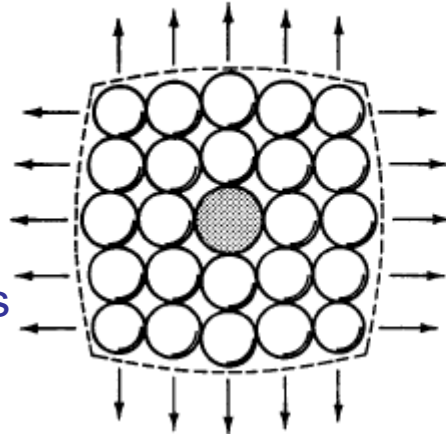


solute

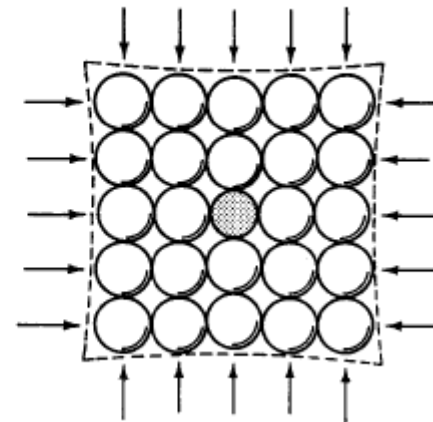
solvent

Local stress in substitutional solid solution

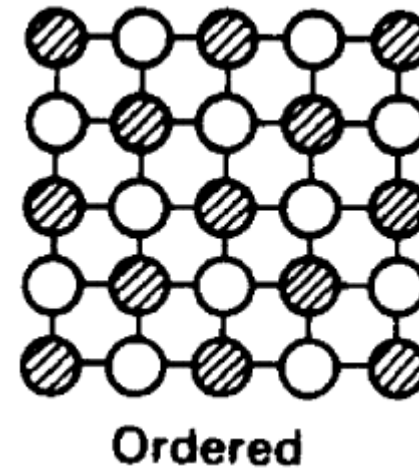
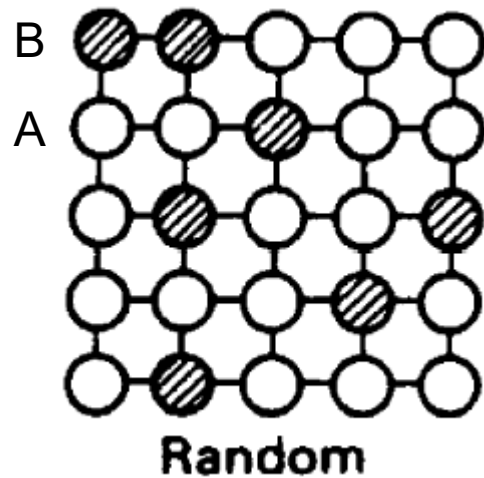
Compressive stress



Tensile stress



In general, two different classes of substitutional solid solutions can be created:

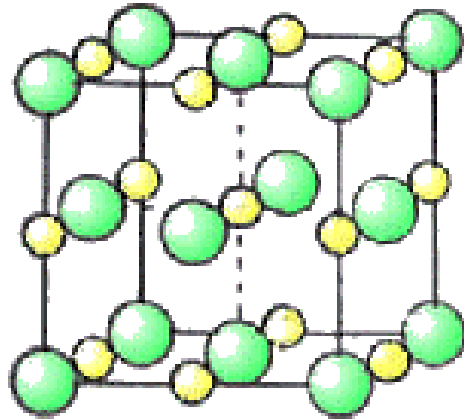


Atoms of A and B can arrange in a **random** way or can form an **ordered solid solution**

An ordered structure will be destroyed at high temperature because of **thermal agitation**

Defects in ionic crystals

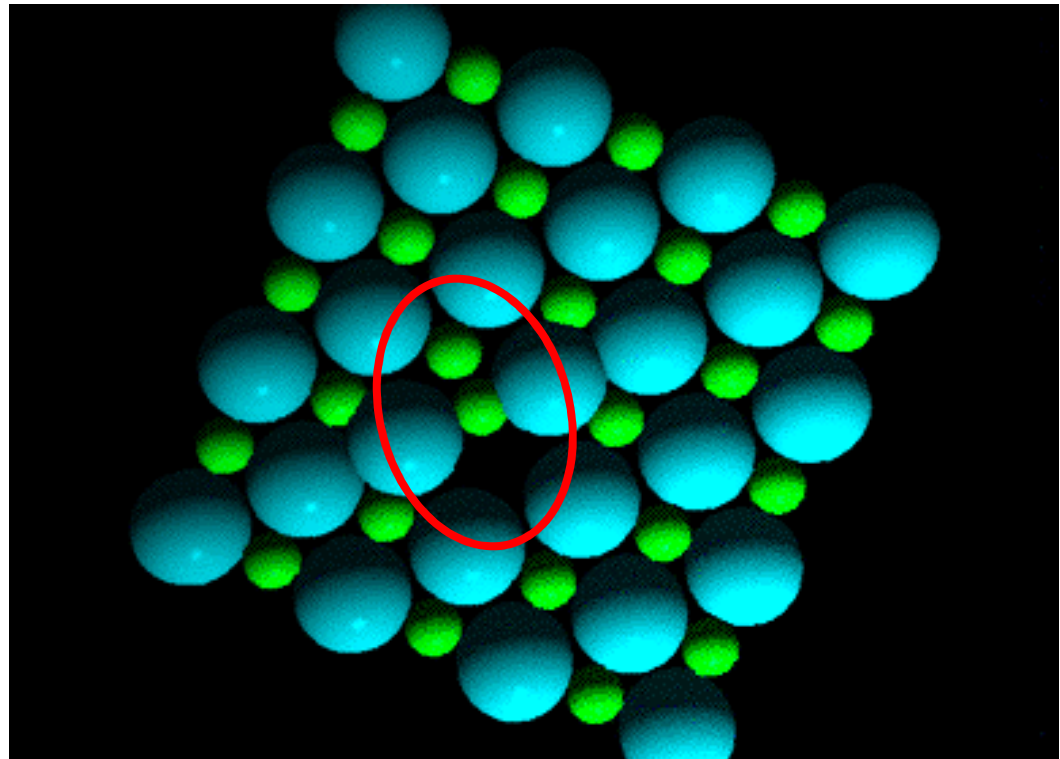
Defects are caused by ionic couple with **opposite charge**.



Examples of ionic solids are many ceramics (e.g. alumina, Al_2O_3 , used as refractory material in furnaces; zirconia, ZrO_2 ,...) and salts (e.g. **NaCl**)

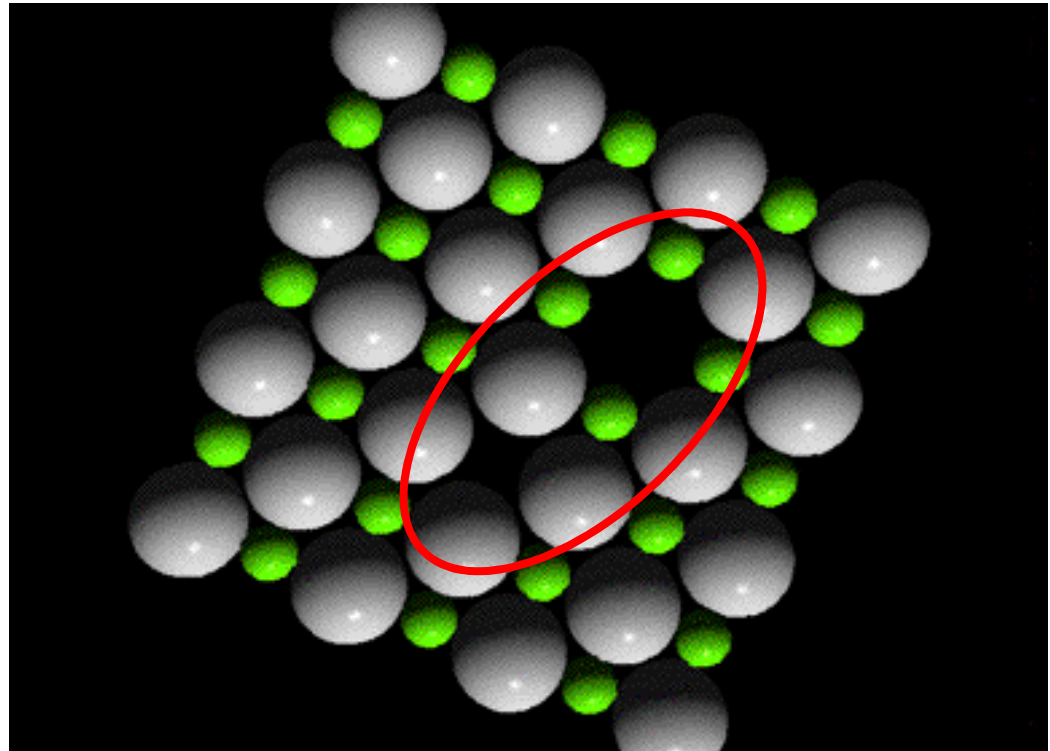
Frenkel defect:

One vacancy (cation or anion)
+
one interstitial ion



Schottky defect:

anion vacancy
+
cation vacancy



**Defects in ionic crystal
increase
electrical conductivity**

ARE POINT DEFECTS «BAD THINGS»?

- **ALLOYS** and in general **solid solutions** are possible because of interstitials and substitutional defects
- presence of substitutional impurities are necessary for charge movement in semiconductor science

....

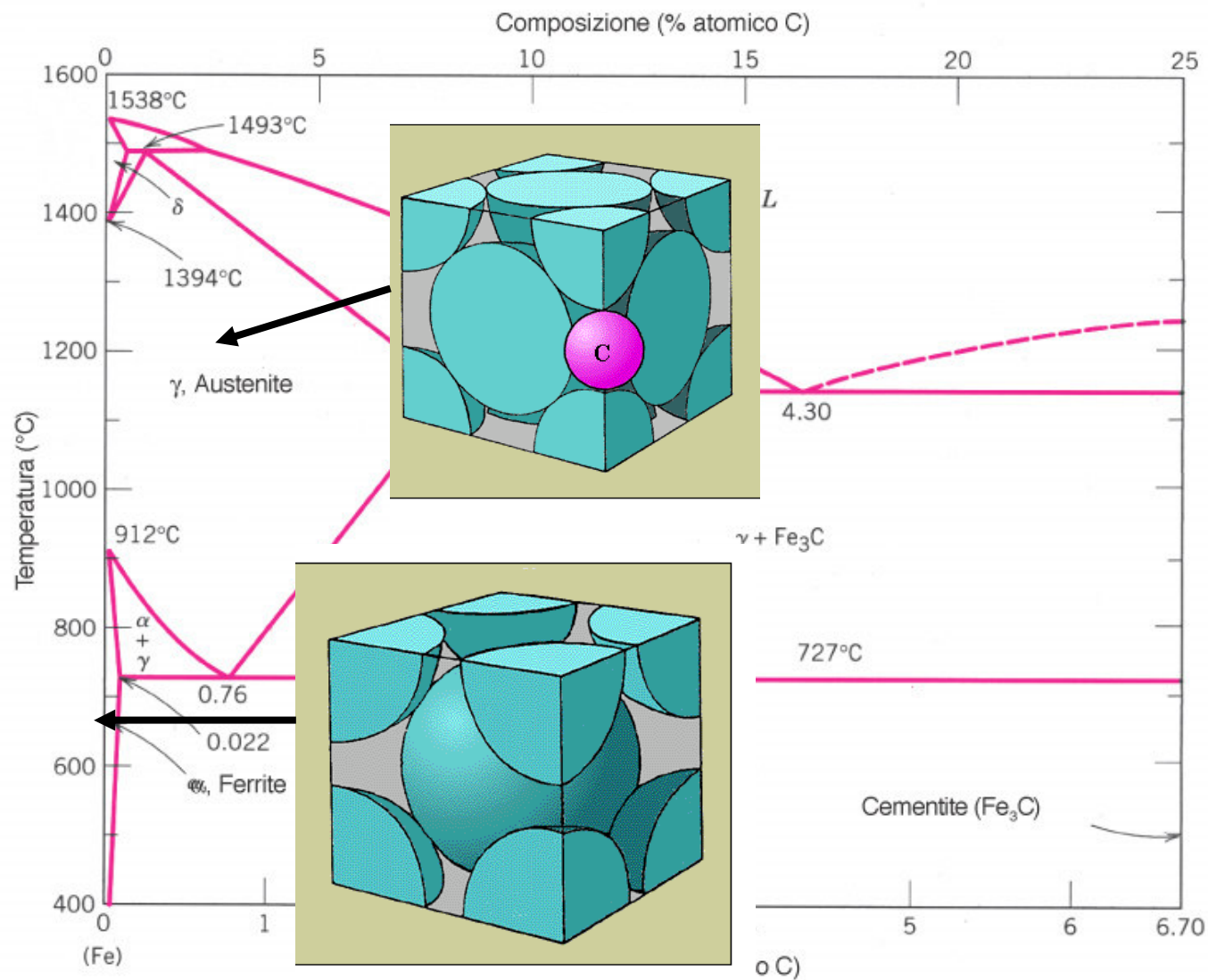


FIGURA 9.22 Diagramma di fase ferro-carburo di ferro. (Da *Binary Alloy Phase Diagrams*, 2nd edition, Vol.1, T.B. Massalski, Editor-in-Chief, 1990. Ristampa autorizzata da ASM International, Materials Park, OH.)

Solid solutions: substitutional alloys

Atoms with similar radius (difference <15%), crystalline cell, electronegativity and valence → substitutional solid solution

These are the four **HUME-ROTHERY** rules

- **Cu-Ni Alloys**

Rule 1: $r_{\text{Cu}} = 0.128 \text{ nm}$ and $r_{\text{Ni}} = 0.125 \text{ nm}$.

$$\Delta R\% = 2.3\%$$

favorable ✓

Rule 2: Ni and Cu have the FCC crystal structure.

favorable ✓

Rule 3: $E_{\text{Cu}} = 1.90$ and $E_{\text{Ni}} = 1.80$. Thus, $\Delta E\% = -5.2\%$

favorable ✓

Rule 4: Valency of Ni and Cu are both +2.

favorable ✓

Expect Ni and Cu forms S.S. over wide composition range.

Is solid-solution favorable, or not?

- **Cu-Ag Alloys**

Rule 1: $r_{\text{Cu}} = 0.128 \text{ nm}$ and $r_{\text{Ag}} = 0.144 \text{ nm}$.

$$\Delta R\% = 9.4\%$$

favorable ✓

Rule 2: Ag and Cu have the FCC crystal structure.

favorable ✓

Rule 3: $E_{\text{Cu}} = 1.90$ and $E_{\text{Ni}} = 1.80$. Thus, $\Delta E\% = -5.2\%$

favorable ✓

Rule 4: Valency of Cu is +2 and Ag is +1.

NOT favorable

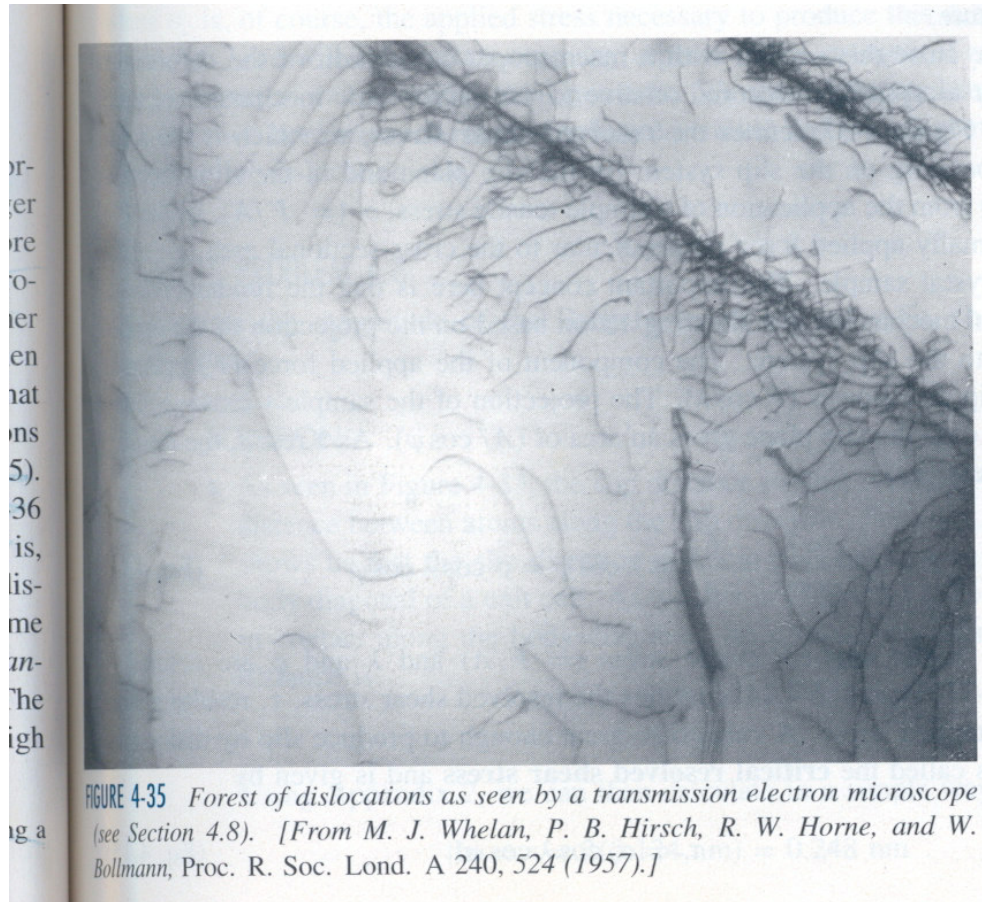
Expect Ag and Cu have limited solubility.

Solid solutions: interstitial alloys

- Atoms as interstitial defects different from those of the main lattice
- Only small atoms can enter a metallic cell as interstitial defects (max 10%)
- Ex.: carbon is an interstitial defect in the iron lattice in steels, up to about 2%
- $R_{\text{Fe}} = 1.24 \text{ \AA}$, $R_{\text{C}} = 0.71 \text{ \AA}$

Linear defects: dislocations

DISLOCATIONS can be seen as irregular lines on the cross-section of a material at very high magnifications.



TEM analysis of a metal (magnification > 500,000 X)

Dislocations caused distortion of the crystalline lattice around a row of atoms.

Dislocations are produced:

- during solidification
- during plastic deformation

Dislocations are usually present in **commercial materials**

Real dislocations have a quite irregular and complex shape

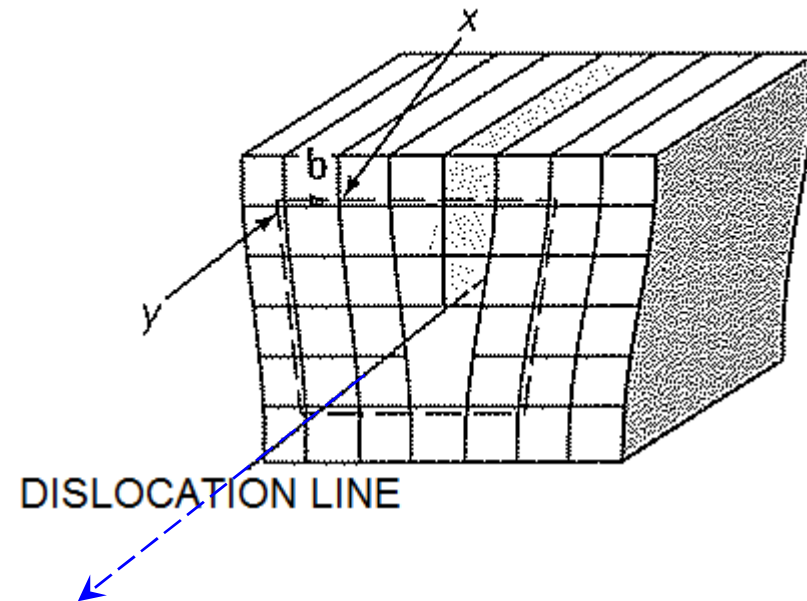
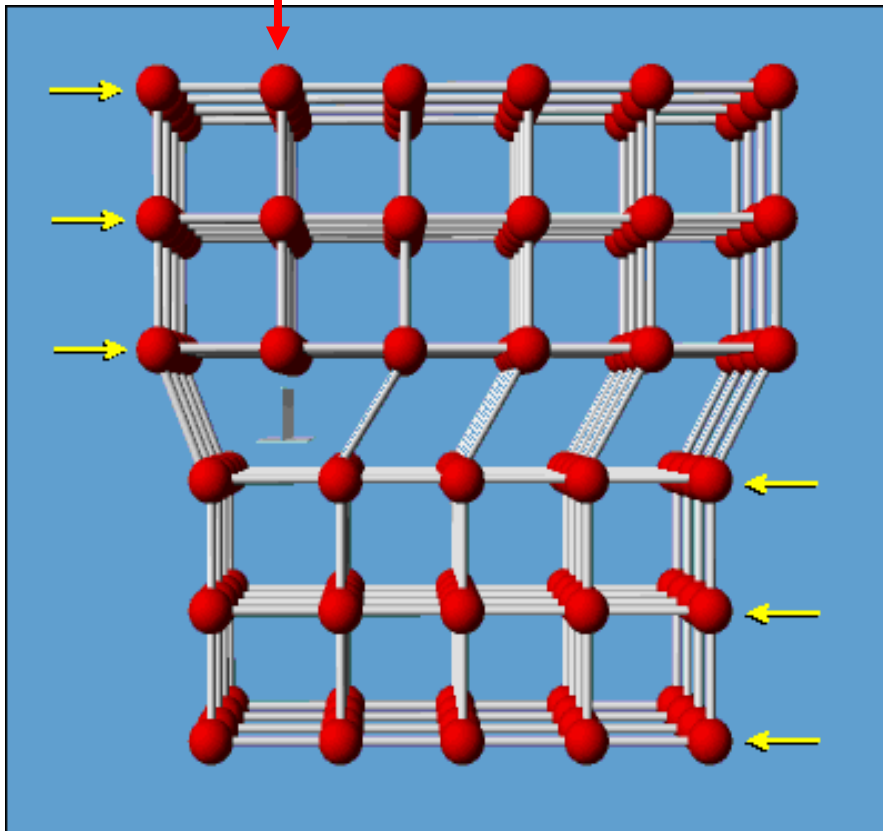
We can describe them by separating a dislocation into 2 different components:

- edge or linear
- screw

Real dislocations have both edge and screw components, so they are **mixed dislocations**

Edge dislocations

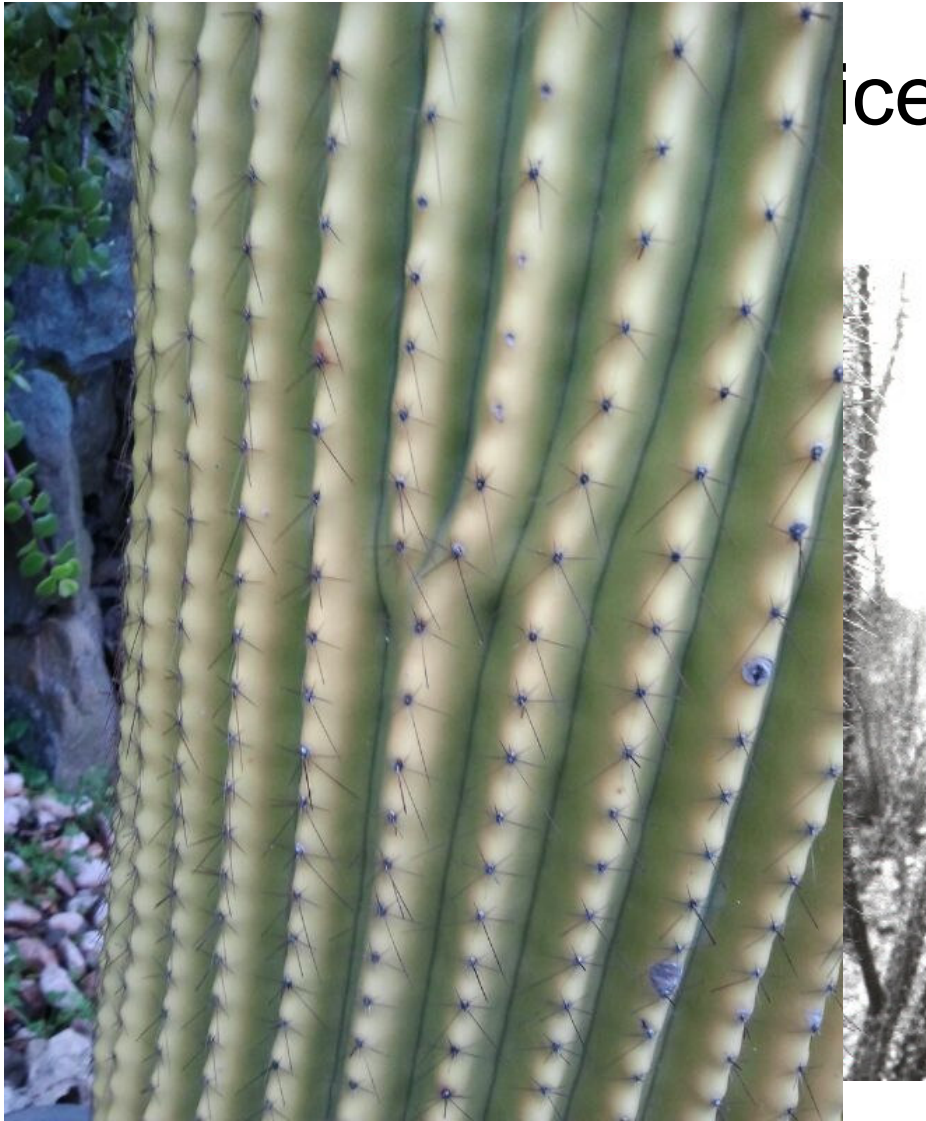
An extra half-plane is present in a region of the crystal



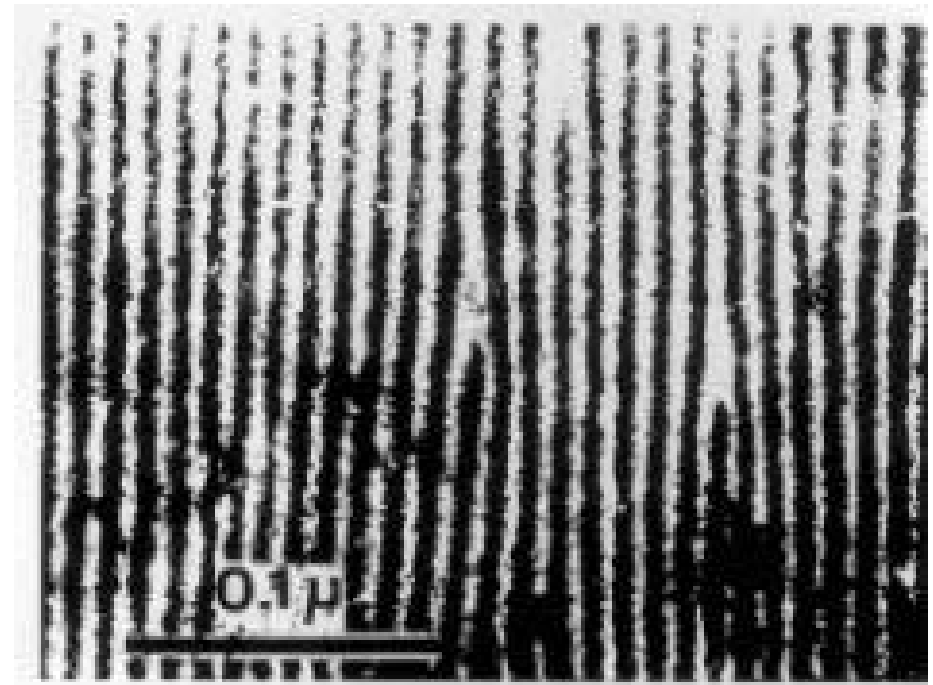
The DISLOCATION LINE of an edge dislocation (like that in the image) is a vector perpendicular to the paper plane and pointing out of it.

Dislocations in nature....!

ice plane more....



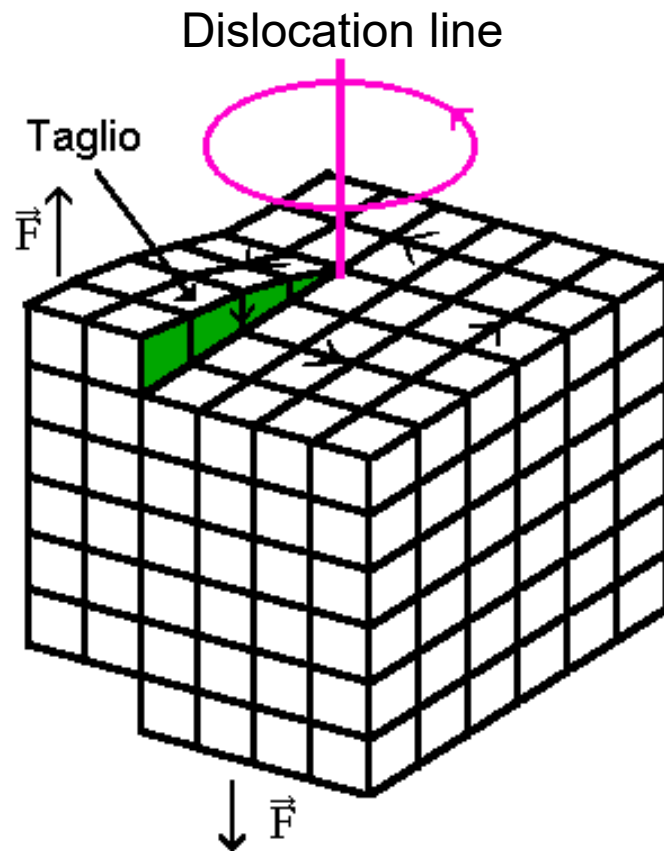
cactus!



Copper sulphide (CuS)

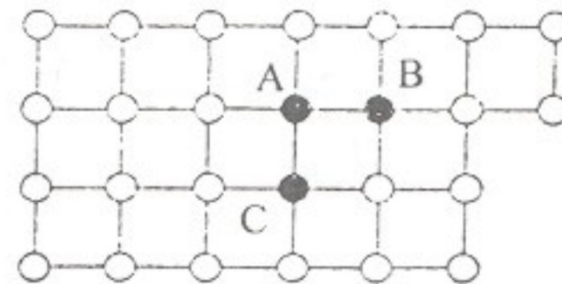
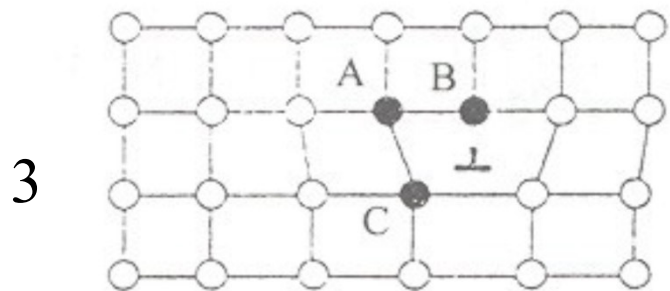
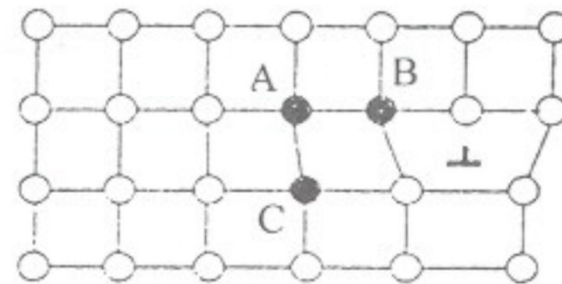
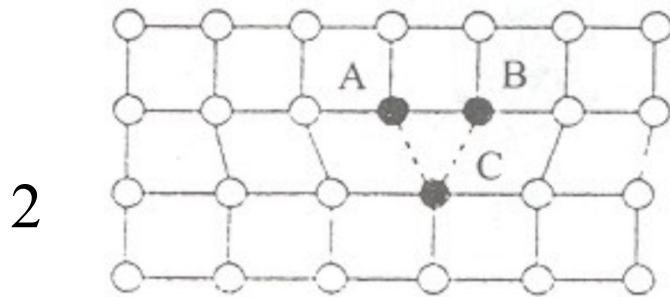
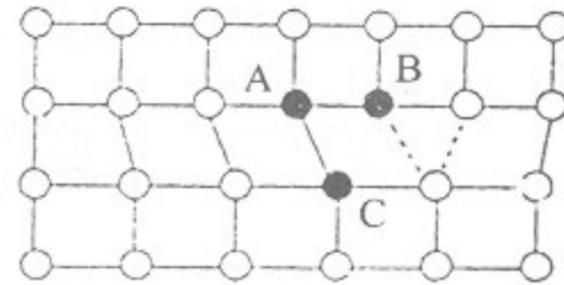
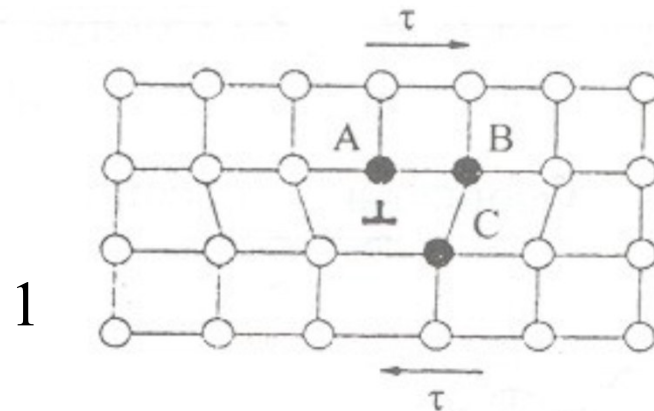
Screw dislocations

A screw dislocation does not include an extra plane: the planes perpendicular to the dislocation line show small distortion and they trace a spiral path around the dislocation line.

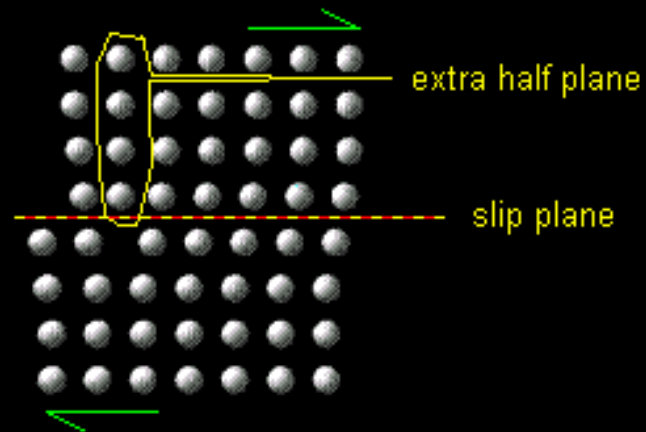


Superficie di un cristallo organico: in evidenza la spirale dovuta all'accrescimento del cristallo per la presenza di una dislocazione a vite.

Dislocation motion

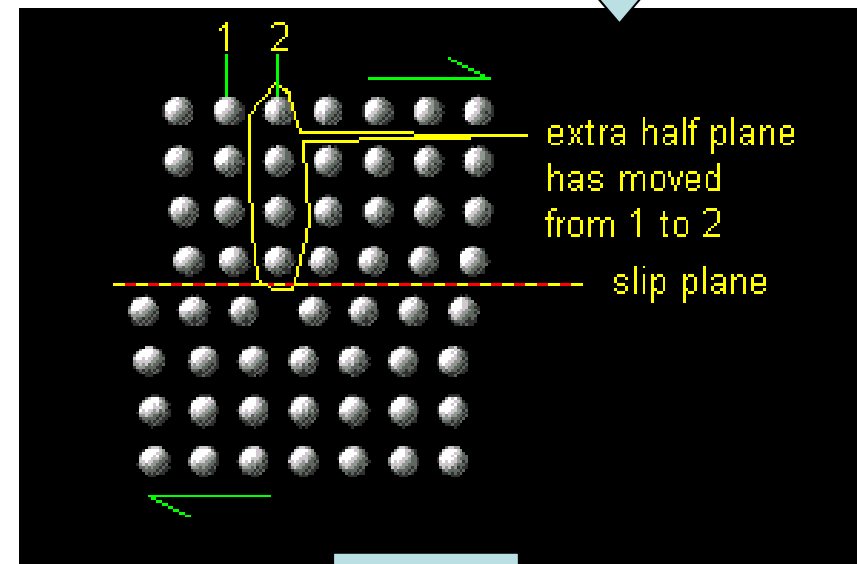
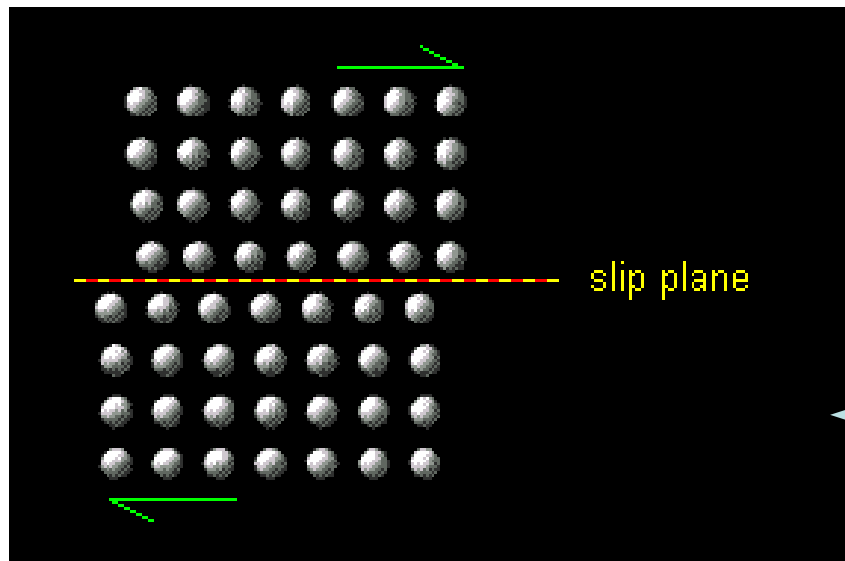


Glide of many dislocations result in slip, which is the most common manifestation of deformation in crystalline solids. Near the dislocation line, some atoms spacings are far from the ideal values and the required **shear stress** to overcome the lattice resistance to the movement is much smaller than the theoretical shear stress of the perfect **lattice**.



Dislocation motion

PLASTIC DEFORMATION

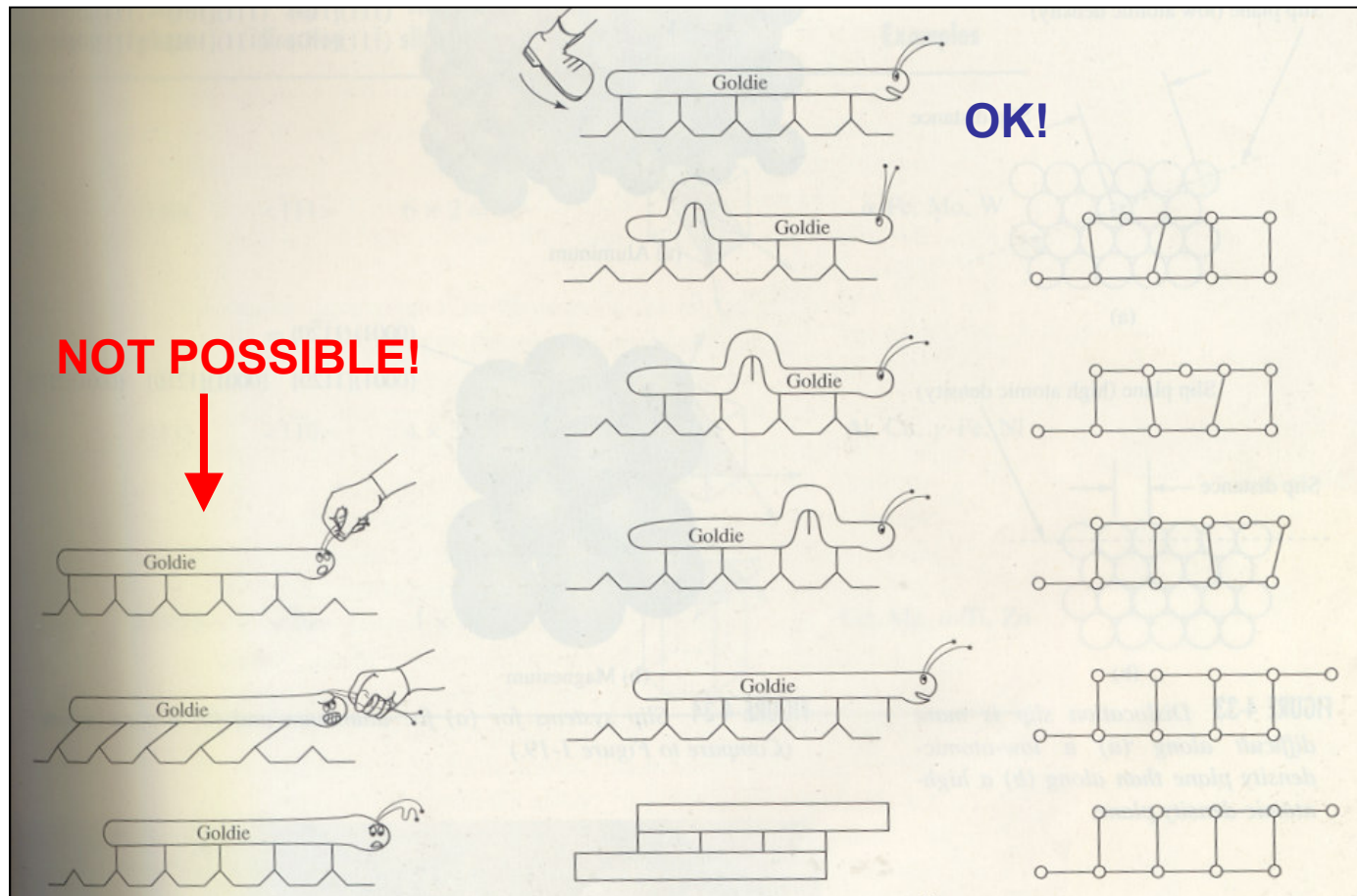


Dislocations motion: simple analogies

Caterpillars

Caterpillar has a lot of legs; it cannot move them all together, but it moves a couple of legs at the same time.

Leg movement looks like a wave from the caterpillar's hind legs to its forefeet.



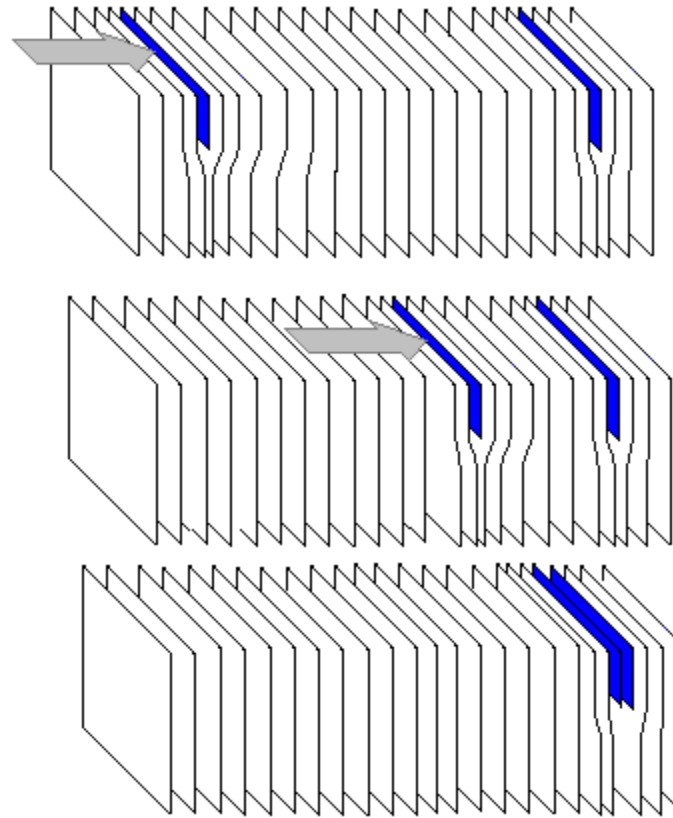
Motion of carpet fold

It is difficult to let slide a big carpet pulling it from the end, because of the friction with the floor; but if there is a fold (= the dislocation) the carpet can be moved by fold translation.



Fig. 5.12 Mott's analogy of the carpet.

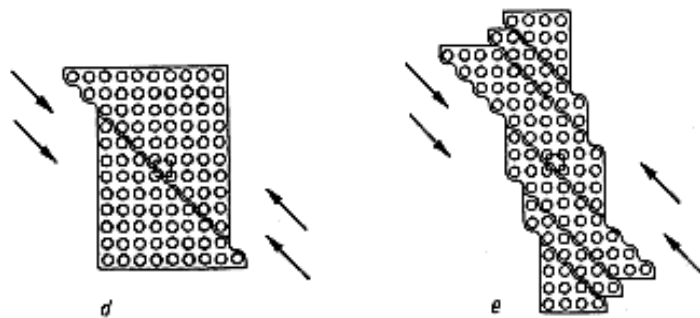
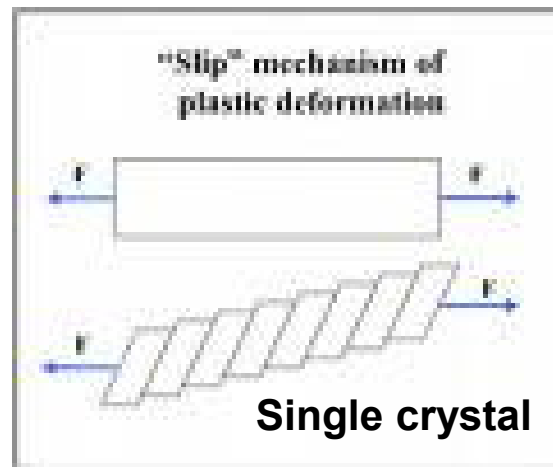
Dislocations can move and accumulate



Dislocation movement makes easier the
PLASTIC DEFORMATION
(see the behaviour of metals...)

Dislocations motion and plastic deformation

Dislocation motion begins as yield strength has been reached → PLASTIC DEFORMATION of a material is due to this phenomenon



The beginning of plastic deformation upon a slip plane
Deformation Proceeding upon further slip plane.

Single crystal

Dislocation are particularly significant in **PURE METALS AND ALLOYS** since they provide a mechanism for plastic deformation, which is the cumulative effect of **slip of many dislocations**.

Plastic deformation refers to **IRREVERSIBLE DEFORMATION or CHANGE IN SHAPE** that occurs when the force (stress) that cause it is removed. This is because **the applied stress causes dislocation motion that, in turn, causes permanent deformation.**

The plastic deformation is to be distinguished from elastic deformation, which is a **TEMPORARY CHANGE IN SHAPE** that **occurs while a force or stress remains applied** to a material. In elastic deformation, the shape change is a result of stretching of interatomic bonds, but **NO MOTION OF DISLOCATIONS OCCURS.**

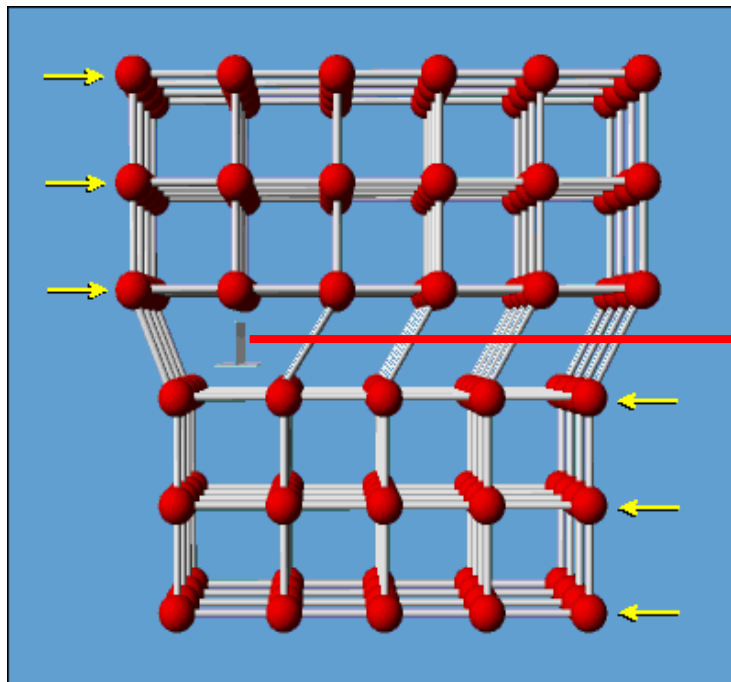
Perfect crystal deformation:

A lot of strong bonds are broken to allow relative sliding of atomic planes

Resistance of a perfect crystal is about 1000-10000 times higher than that observed one in real crystals (with defects)

Real crystal deformation:

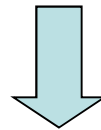
Less amount of strong bonds are broken when planes are sliding because of the presence of **DISLOCATIONS** (weak points where sliding of planes is easy because some chemical bonds are missing)



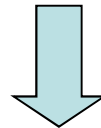
A lot of chemical bonds are missing along the dislocation line

The movement of dislocations
generates new dislocations:
a dislocation “forest” is produced since the sample shows
PLASTIC DEFORMATION

There is repulsion between the stress fields around each
dislocation; therefore the average distance between
dislocations decreases and dislocations start blocking the
motion of each other



DISLOCATIONS ACCUMULATION



**HIGHER RESISTANCE OF THE MATERIAL
TO FURTHER DEFORMATION**

Hardening

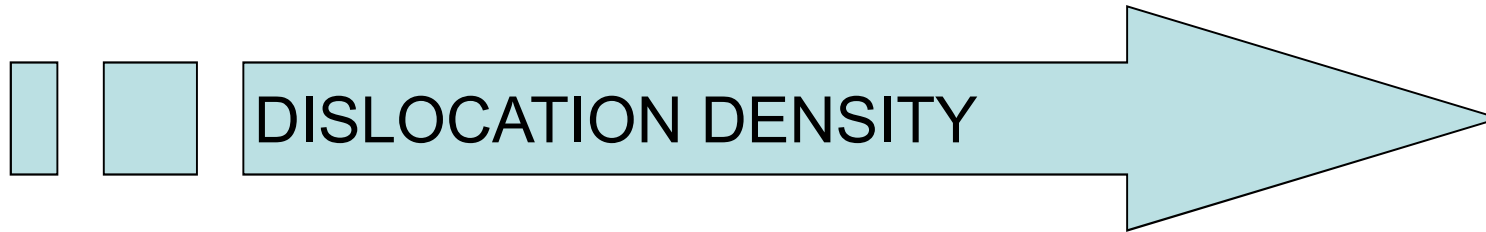
HARDENING is obtained during deformation of a metallic material by increasing the number of **dislocations**.

Before deformation, the dislocation density is about 10^6 cm of dislocation line per cubic centimeter of material. When a stress greater than the yield strength is applied, dislocation begin to slip (**PLASTIC DEFORMATION** occurs).

Eventually, a dislocation moving on its slip plane can encounter obstacles that pin the ends of the dislocation line; as we continue to apply stress, the dislocation can create additional dislocations.

The dislocation density can increase to about 10^{12} cm of dislocation line per cubic centimeter of material: when we have too many dislocations, **they interfere with their own motions**.

The result is an **increased strength** but reduced ductility (i.e., the material is less prone to be wrought).



Perfect crystal
(0 dislocations/cm³)



Very high
yield
strength

Real
polycrystalline
material
(10⁶ cm/cm³)



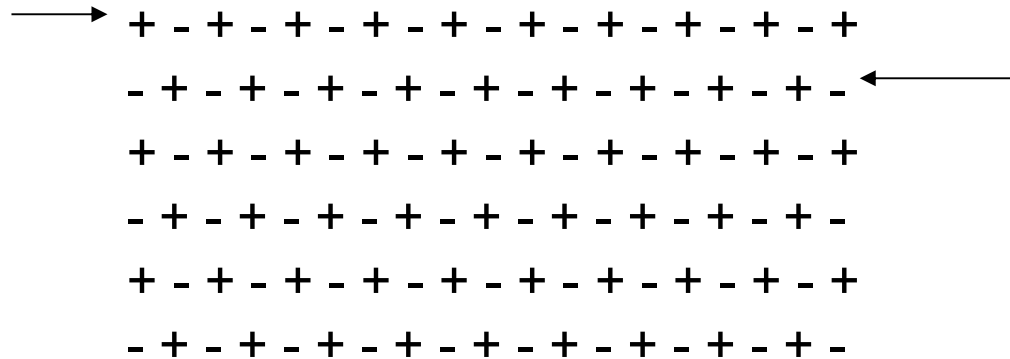
Low yield
strength

Real polycrystalline
material **after plastic
deformation** (work-
hardened)
(10¹² cm/cm³)



High yield
strength

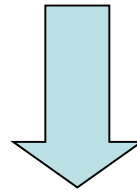
Ionic solids



Example: NaCl

There are
NO PREFERRED sliding directions

Two planes of the ionic crystal are slid into non-preferred direction → overlap of ions with equal charge → they reject to each other → **CRYSTAL FRACTURE**.



**Ionic crystals are
BRITTLE MATERIALS**

In metals there is no repulsion due to the presence of the delocalized electronic cloud!

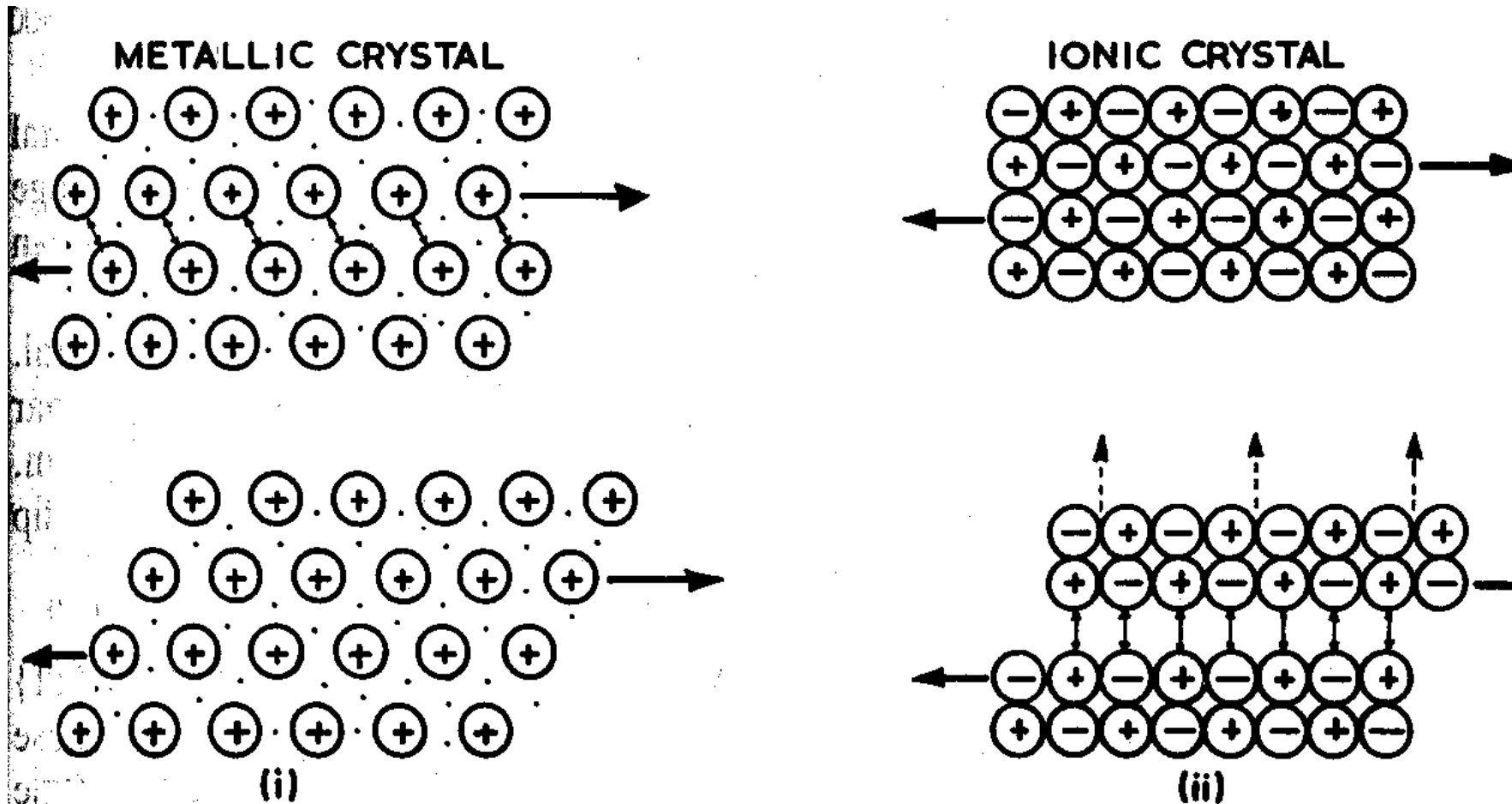


Fig. 5.2 Deformation in crystals. (i) 'Slip' occurs in a metallic crystal. (ii) 'Cleavage' occurs in ionic crystals.

Remember that:

Covalent solids (e.g. diamond, SiC...):

→ directional bonding is UNFAVORABLE to dislocation motion

COVALENT SOLIDS ARE BRITTLE MATERIALS!

Metallic solids:

→ dislocation motion OCCURS SMOOTHLY (structures with dense atomic packing, small lattice parameters, non-directional bonding) → ductile materials.

METALS HAVE SIGNIFICANT PLASTIC DEFORMATION ABILITY

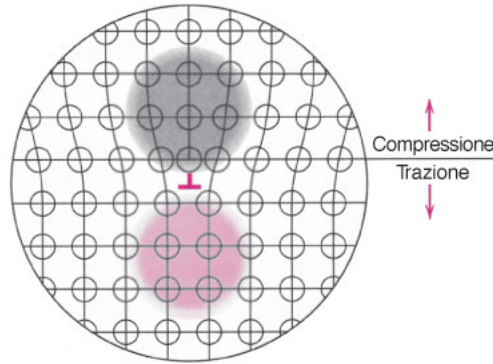


FIGURA 7.4 Zone di compressione (scure) e di trazione (colorate) localizzate intorno ad una dislocazione a spigolo. (Da W.G. Moffatt, G.W. Pearsall e J. Wulff, *The Structure and Properties of Materials*, Vol. I, *Structure*, p. 85. Copyright © 1964 by John Wiley & Sons, New York. Per concessione di John Wiley & Sons, Inc.)

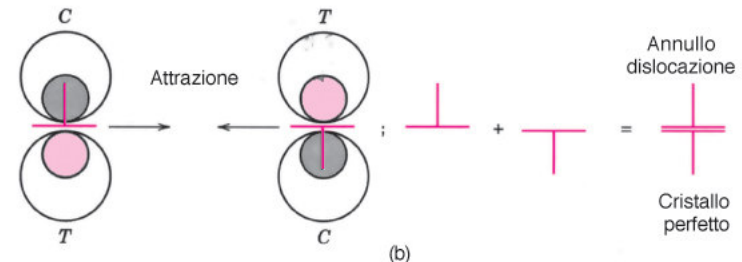
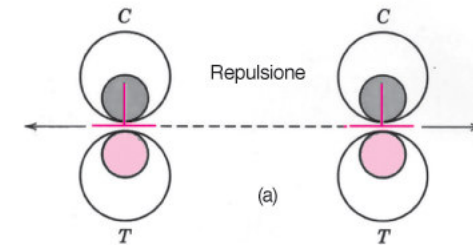
Callister
Scienza e Ingegneria dei materiali, una introduzione
EdiSES

Interaction between dislocations and point defects

Mechanical hardening, cold working

FIGURA 7.5

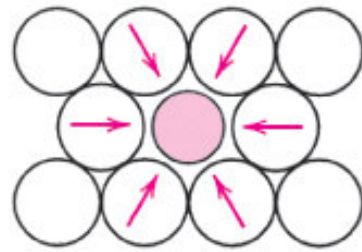
(a) Due dislocazioni a spigolo dello stesso segno e giacenti sullo stesso piano di scorrimento manifestano uno sforzo repulsivo reciproco; C e T indicano regioni di compressione e di trazione, rispettivamente. (b) Dislocazioni a spigolo di segno opposto e giacenti sullo stesso piano di scorrimento manifestano una mutua forza attrattiva. Dopo l'incontro, si annullano reciprocamente lasciando una regione cristallina perfetta. (Da H.W. Hayden, W.G. Moffatt e J. Wulff, *The Structure and Properties of Materials*, Vol. III, *Mechanical Behaviour*, p. 75. Copyright © 1965 by John Wiley & Sons, New York. Per concessione di John Wiley & Sons, Inc.)



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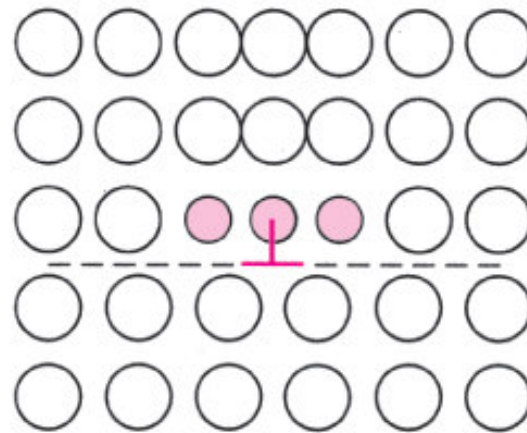
INTERACTION between DISLOCATIONS AND POINT DEFECTS

- DISLOCATIONS INTERACT with themselves (cold working, hardening).
- Dislocations interact with point defects: dislocation motion is slower in alloys than in pure metals
- Better mechanical properties of alloys vs pure metals
- Without dislocations, the plastic deformation of metals cannot be explained
- Dislocations can be seen with Scanning or Transmission electron Microscopy (SEM, TEM).



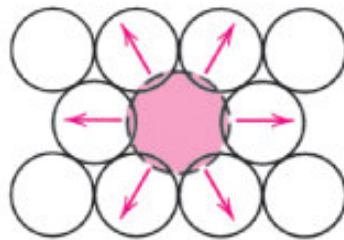
(a)

Callister

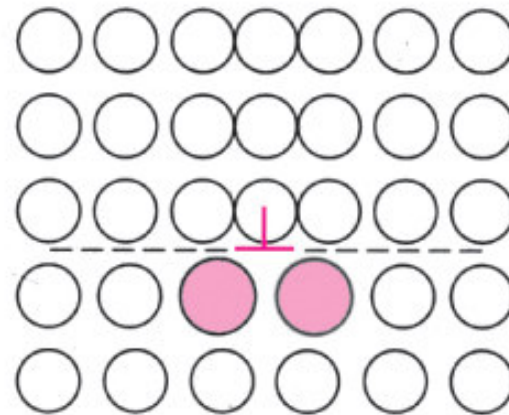


(b)

FIGURA 7.17 (a) Rappresentazione delle tensioni reticolari di trazione sugli atomi ospitanti per sostituzione di un atomo di impurezza più piccolo. (b) Possibile accomodamento di atomi di impurezza più piccoli in una dislocazione a spigolo in modo che vi sia una parziale eliminazione delle tensioni reticolari impurezza-dislocazione.



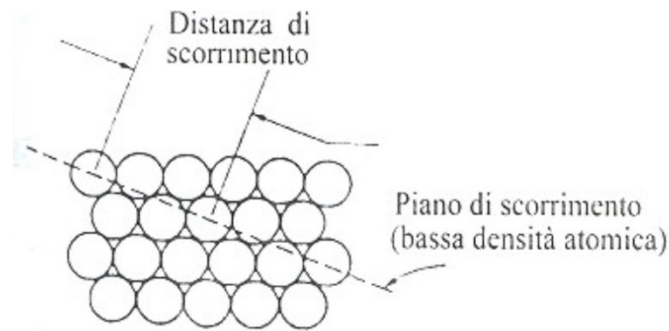
(a)



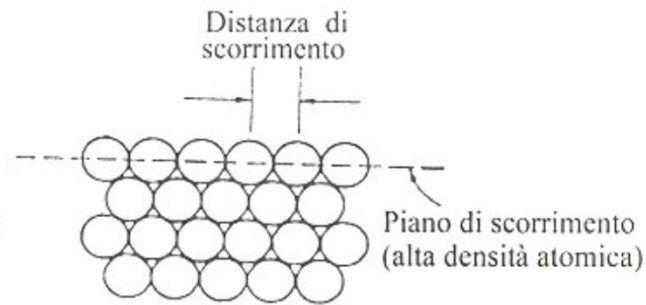
(b)

FIGURA 7.18 (a) Rappresentazione delle tensioni reticolari di compressione sugli atomi ospitanti per sostituzione di un atomo di impurezza più grande. (b) Possibile accomodamento di atomi di impurezza più grandi in una dislocazione a spigolo in modo che vi sia una parziale eliminazione delle tensioni reticolari impurezza-dislocazione.

Preferential sliding planes for dislocations



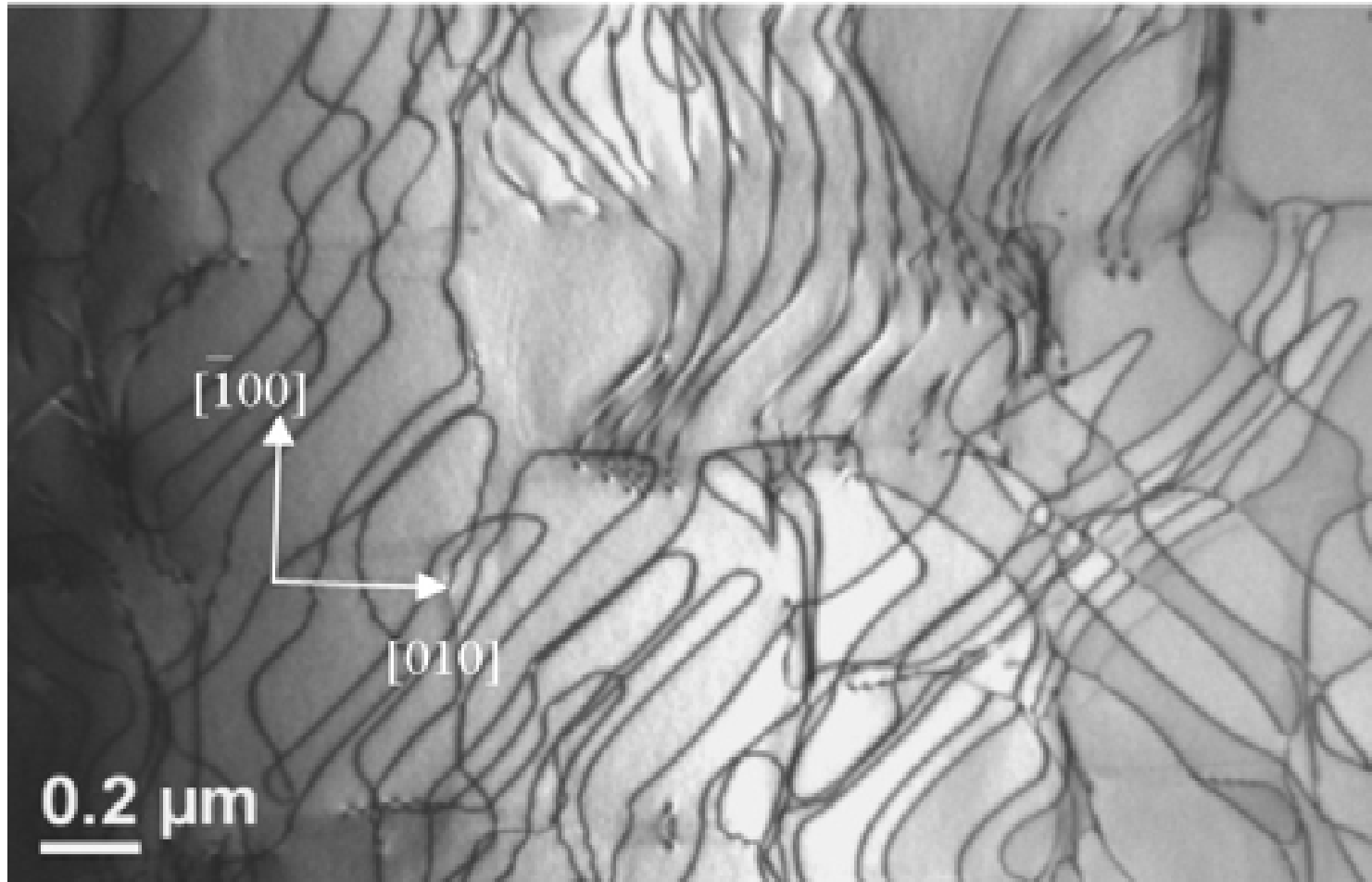
Low density plane



High density plane



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Dislocations in CMSX-4 single crystal superalloy in the primary creep range at temperature 950 °C and stress of 185 MPa. TEM bright-field image (B. Dubiel AGH-UST)

Aleksandra Czyrska-Filemonowicz, International Centre of Electron Microscopy for Materials Science & Faculty of Metals Engineering and Industrial Computer Science, AGH University of Science and Technology, Krakow, Poland

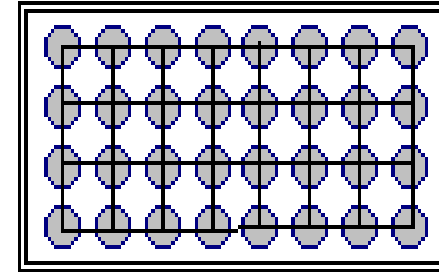
Materiali per l'Ingegneria
M.Ferraris

SOLIDIFICATION OF A LIQUID AND ITS STRUCTURE

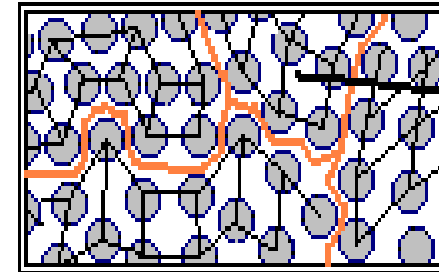
LIQUID
Si
1425 °C

or **SiO₂**
a **1730 °C**

SLOW SPEED
SOLIDIFICATION
MONOCRYSTALLINE
STRUCTURE

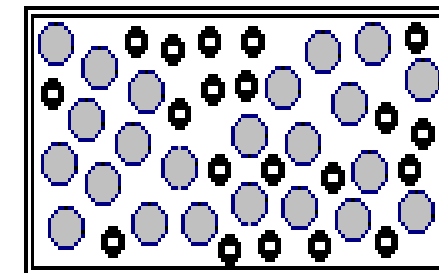


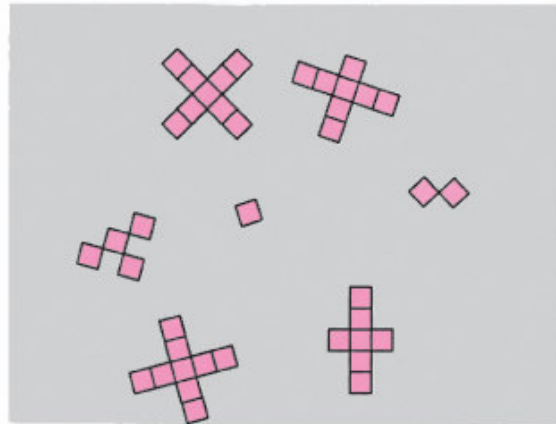
MEDIUM SPEED
SOLIDIFICATION
POLYCRYSTALLINE
STRUCTURE



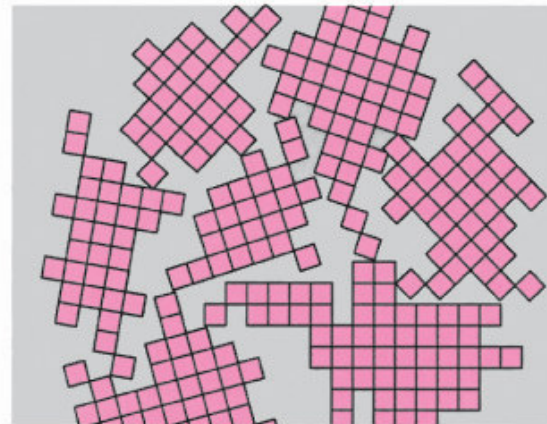
GRAIN
and GB

HIGH SPEED
SOLIDIFICATION
AMORPHOUS
STRUCTURE

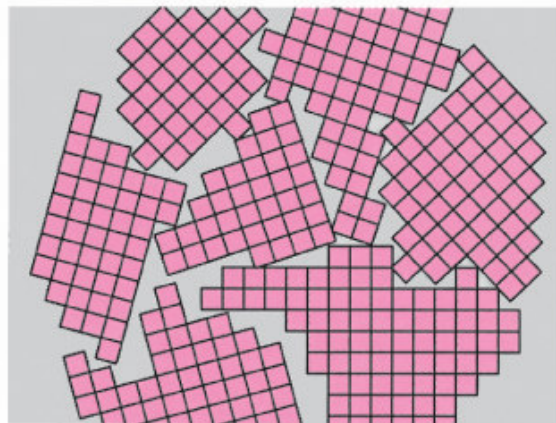




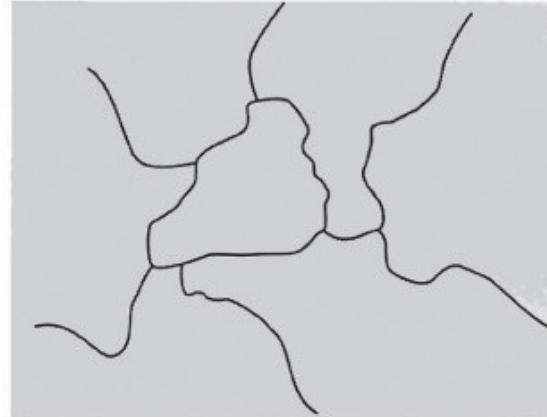
(a)



(b)



(c)



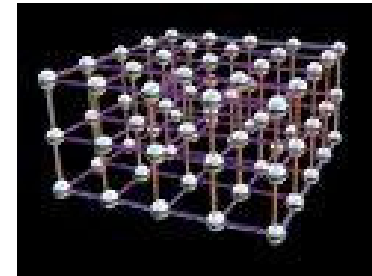
(d)

Nucleation and growth,
formation of GB
in
polycrystalline
materials

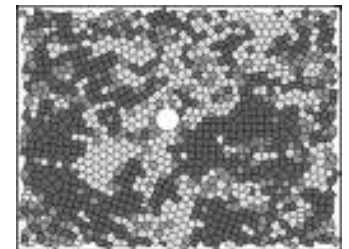
FIGURA 3.16 Diagrammi schematici dei vari stadi nella solidificazione di un materiale policristallino; i quadratini rappresentano le celle unitarie. (a) Piccoli nuclei di cristalliti. (b) Crescita dei cristalliti; vengono anche mostrati alcuni grani che sono tra loro adiacenti. (c) Al completamento della solidificazione, si sono formati grani di forma irregolare. (d) La struttura dei grani come appare al microscopio; le linee scure sono i bordi di grano. (Da W. Rosenhain, *An Introduction to the Study of Physical Metallurgy*, 2nd edition, Constable & Company Ltd., London, 1915.)

CRYSTALLINE STRUCTURE

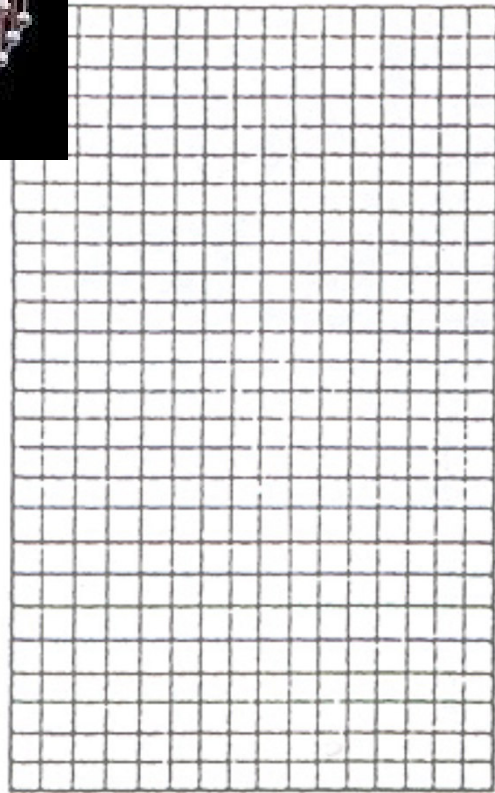
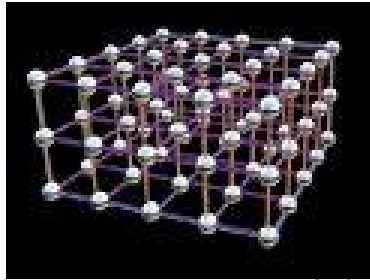
- MONOCRYSTALS: crystalline cells all oriented in the same way (ex. Si for microelectronics)



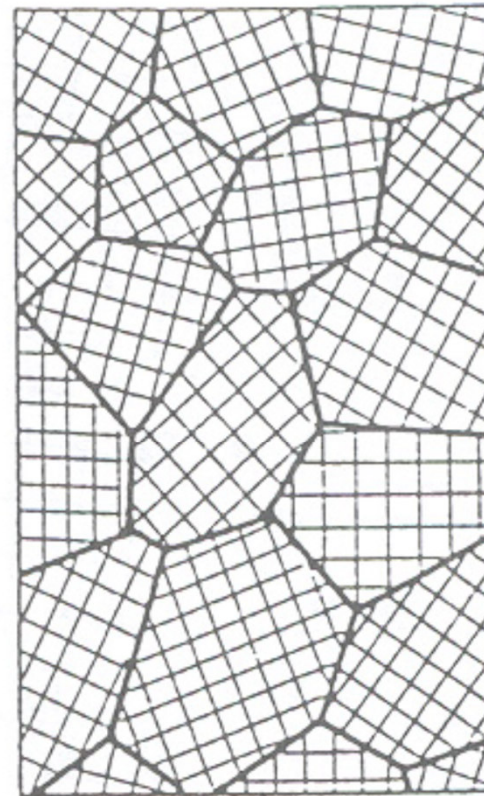
- POLYCRYSTALS: crystalline cells oriented in the same way ONLY inside grains: solidification too quick to allow same orientation in entire lattice (ex .metals, alloys, ceramics)



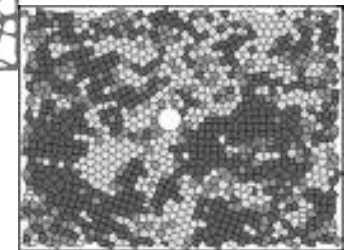
Monocrystal (a) vs. polycrystal (b)



(a)

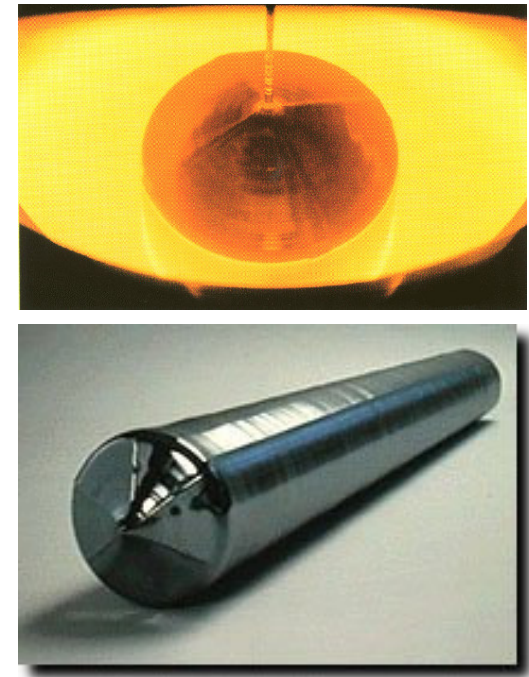


(b)



Monocrystals

- Difficult to obtain compared to polycrystalline materials
- Silicon for electronics
 - Extra slow molten silicon solidification (1-10 μm per second)
 - About 1.5 m length silicon cylinders 300 mm diameter, sliced as wafers
 - About 10 defects/cm²



Nickel-based Superalloys

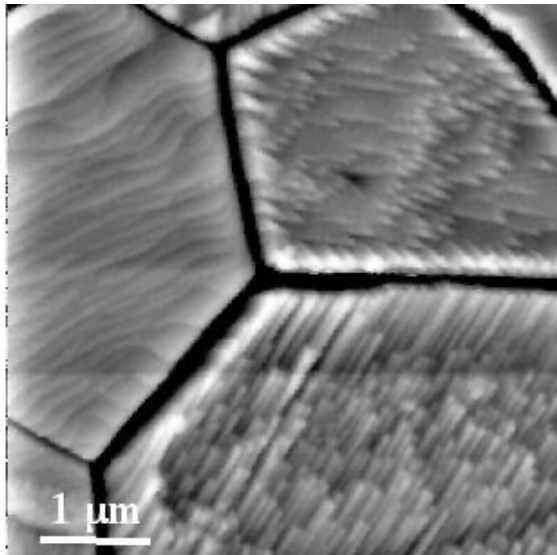




A single-crystal turbine blade.
Courtesy of Siemens

Polycrystalline materials

- Natural solidification from a liquid metal gives a polycrystalline metal
- Several grains, same cell orientation inside the grain, grains separated by grain boundaries



Grains in
 Al_2O_3



Grains
in steel

<http://mimp.mems.cmu.edu/~ordofmag/alumina.jpg>
<http://www.mse.nthu.edu.tw/jimages/Beuty/Steel1.jpg>

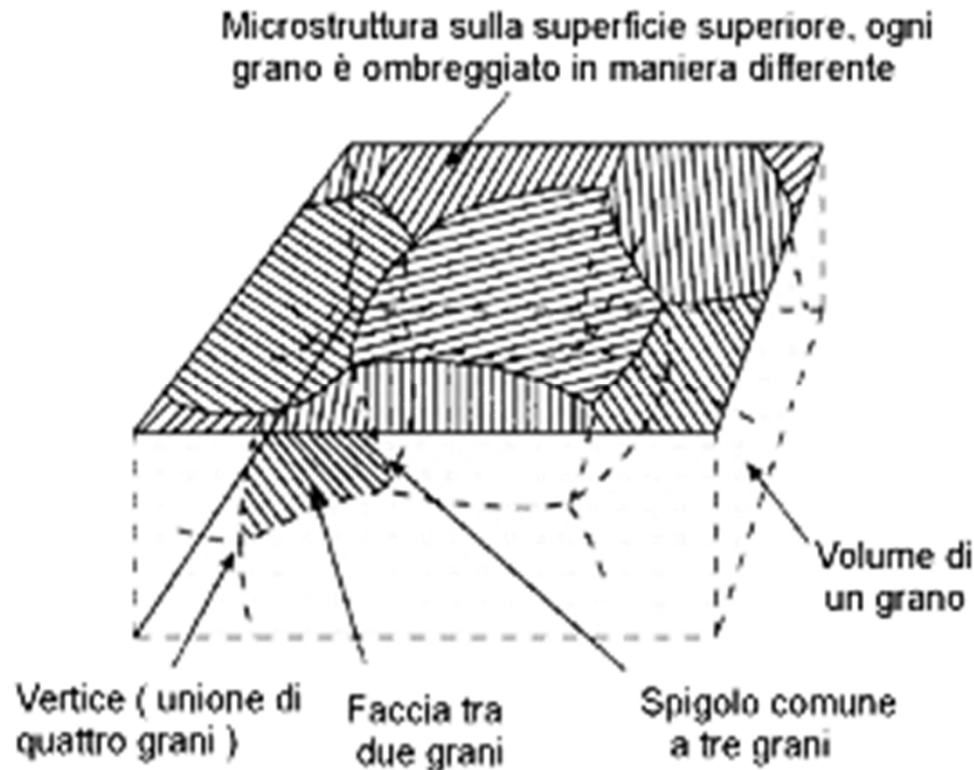
Surface defects: grain boundaries

Most materials are polycrystalline and are made of many single crystals

Each small grain has a different crystallographic orientation respect to the others

Grains arise from material solidification

GRAIN BOUNDARIES separate grains: in a 3-D view they are **SURFACES**

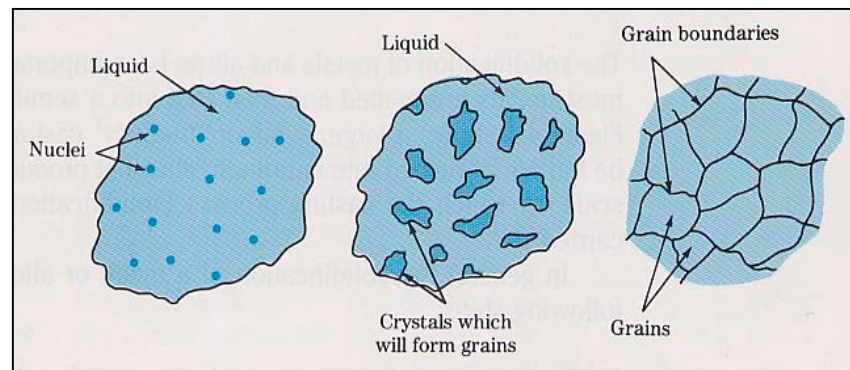


Grain boundaries:

area of about 2-5 atomic diameters, where crystalline order does not exist and the grains impinge on each other when the solidification is completed.

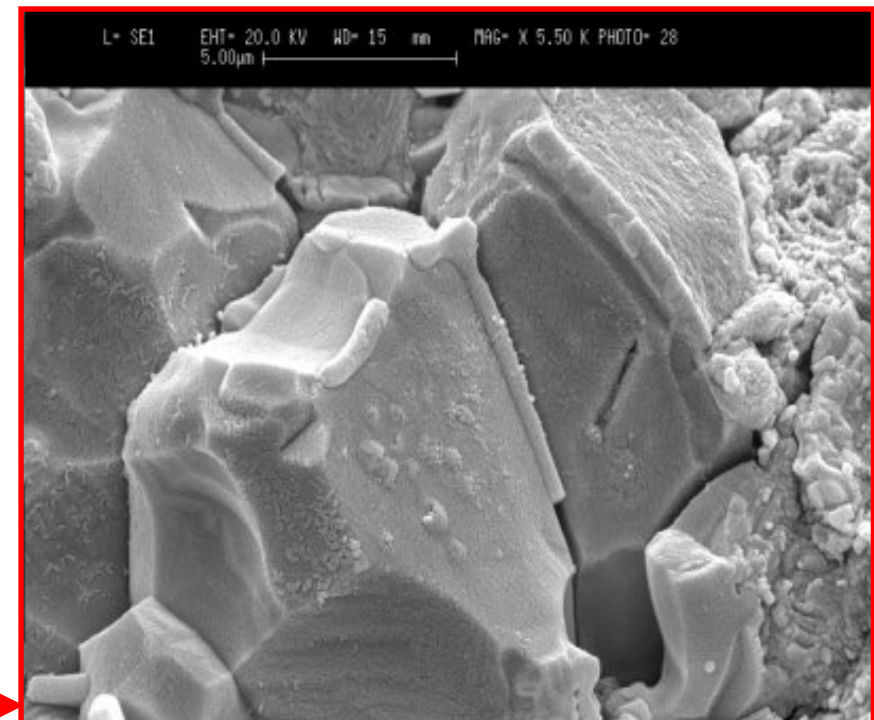
Grain boundaries are indicated as lines and they are the intersection between grain surfaces with the observed plane

- Grain size and orientation affect some material properties; for instance, **finer grain size, higher the mechanical strength of the material**
- Grain boundaries have high energy state as they can be viewed as regions in which dislocations accumulate (with their deformation energy); therefore, grain boundaries are **more chemically reactive** in comparison to grains
- Grain boundaries nucleate during **solidification**; geometrical shape of each crystal depends on restrictions due to adjacent grain growth.



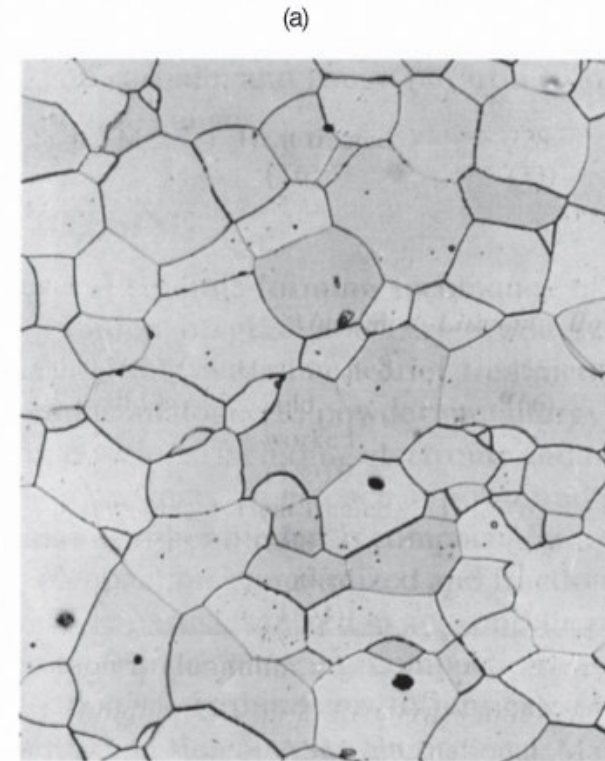
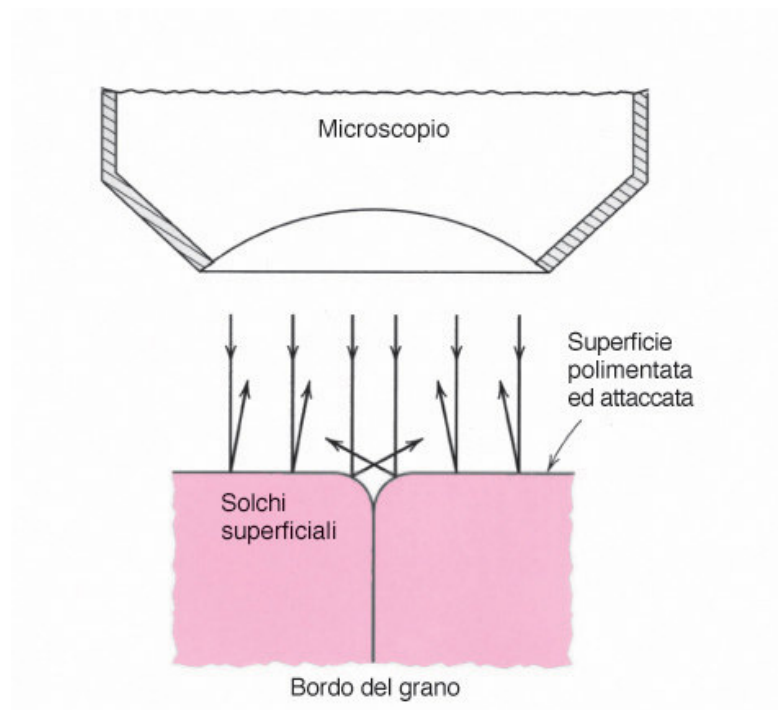
- In most cases, grain size varies from 1 μm to 1 cm (depending on the material, working route etc.).
- Grains can be found not only in metal but also in **ceramics**

Alumina (Al_2O_3)



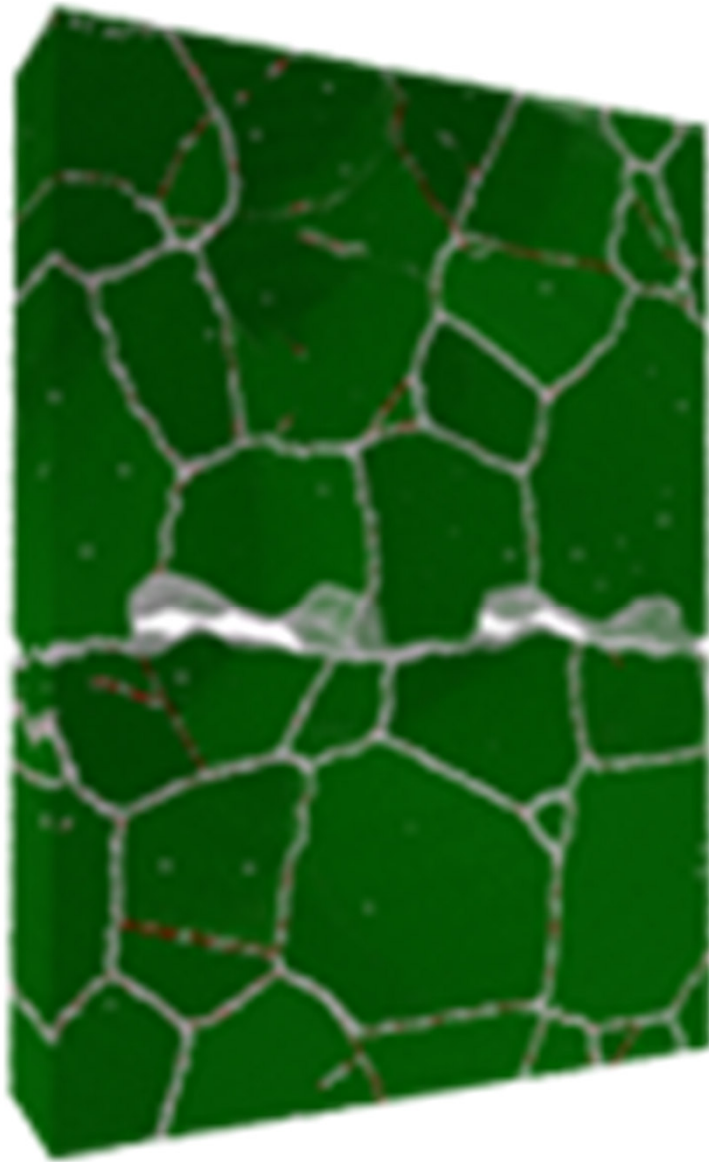
GRAIN BOUNDARIES

- Observable by optical microscopy after chemical etching (e.g. by acids)



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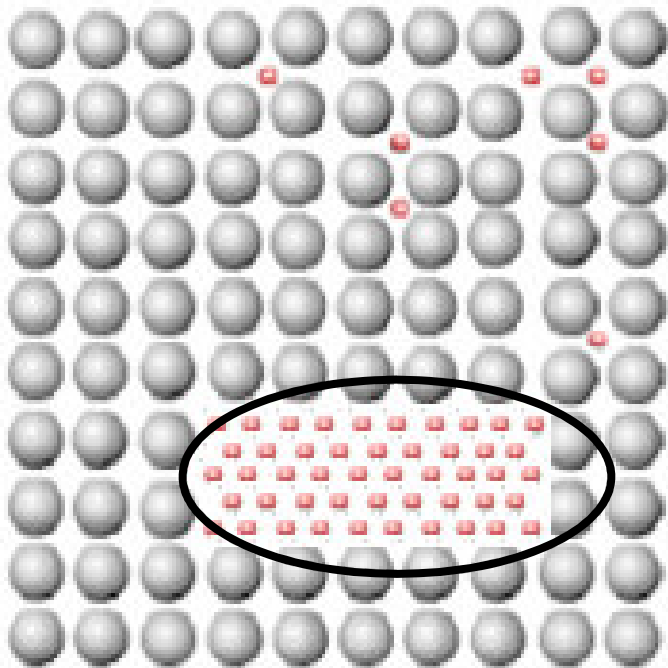
Grain boundaries and crack propagation



3-D defects

Part of the crystal is substituted by a different compound

We can have such a situation when, in making interstitial solid solution, **interstitial atoms are added an amount above the limit of solubility**



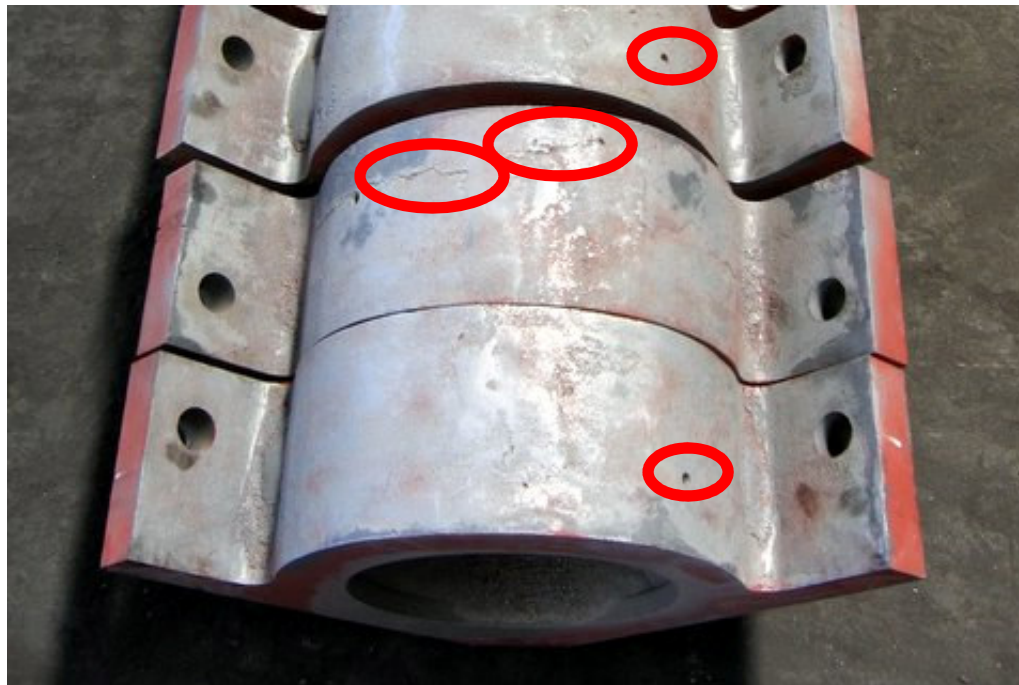
In this case, we will obtain an **interstitial solid solution** and a second phase (i.e. a **PRECIPITATE** with different crystalline structure and composition)

Nature follows this way because of thermodynamics criteria → the second phase has lower energy (i.e. MORE STABLE) is obtained

Macroscopical defects (NOT lattice defects)

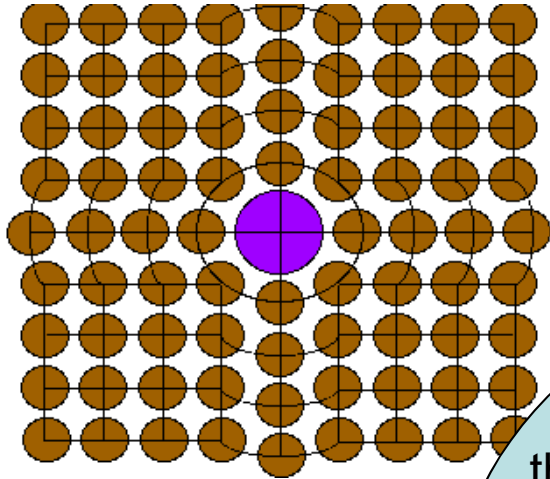
Pores and cavities can strongly decrease mechanical, electrical and optical properties of materials

Cracks can strongly decrease mechanical properties leading to CATASTROPHIC FAILURE (**fracture**) of the material → this should be taken in account particularly for ceramic-based materials (*in Architecture especially rocks and stones, bricks, glass...*)

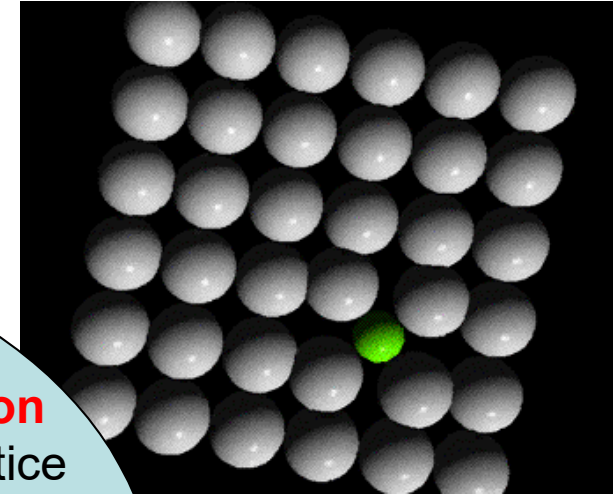


How to improve hardness and strength of materials?

Substitutional solid solution



Interstitial solid solution

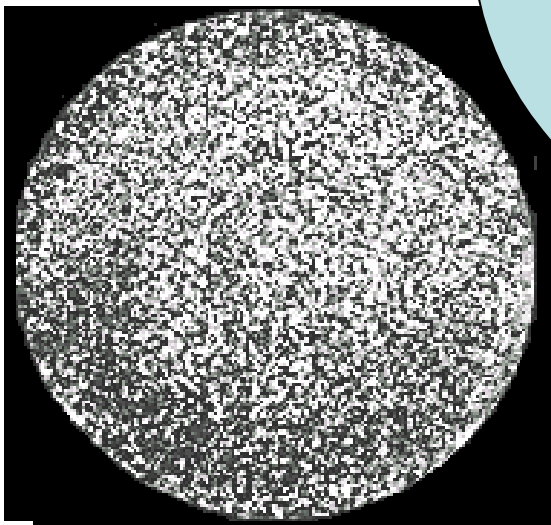


We try to induce **lattice distortion** that creates stress fields in the lattice opposing to plastic deformation and dislocation motion

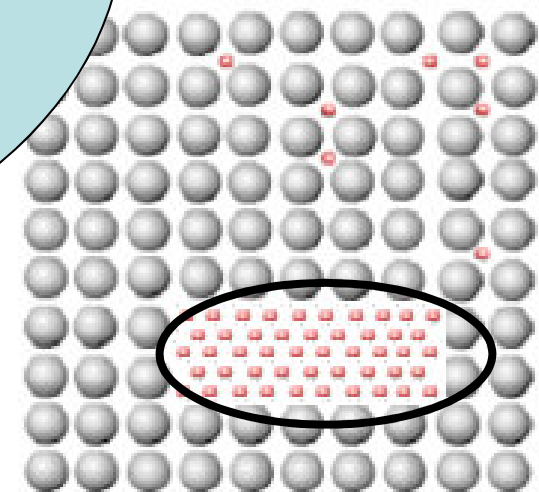


**INCREASE IN
YIELD STRENGTH**

Grain size reduction



Second phase

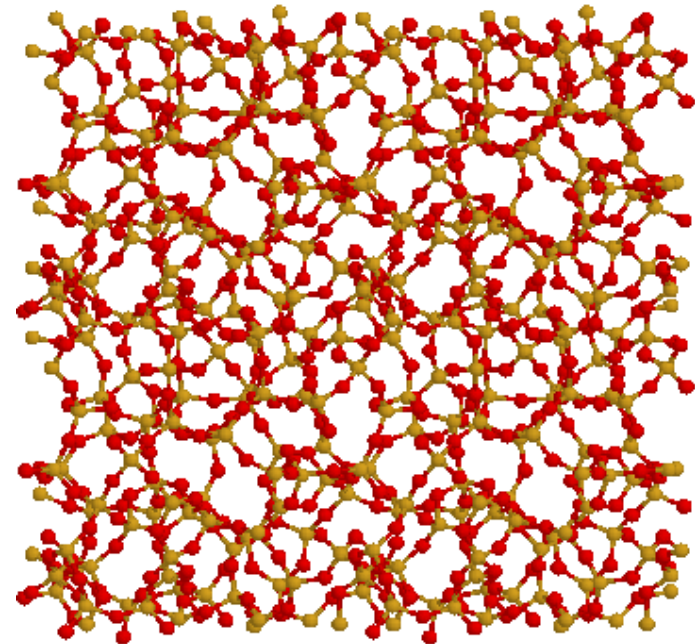


AMORPHOUS MATERIALS

- Disordered structure, no melting point
- glasses, polymers
- Short range order (few nm) (ex: Si^{4+} and O^{--} ions are situated in tetrahedra in silicate glasses)

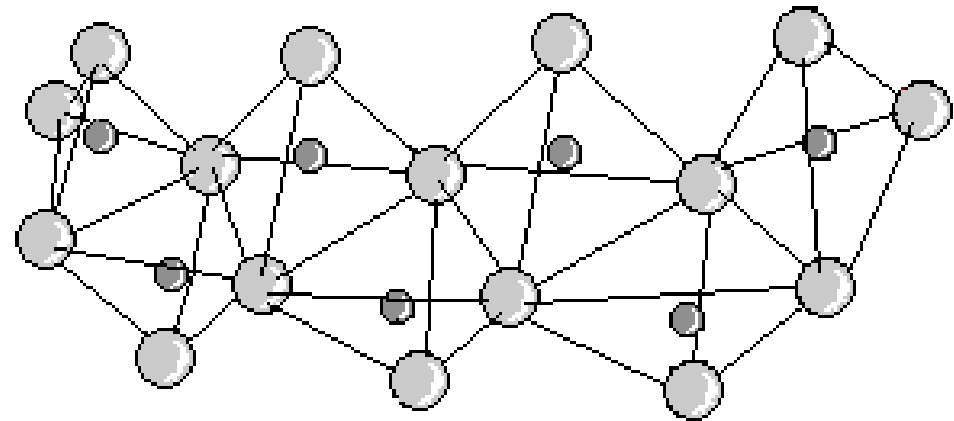
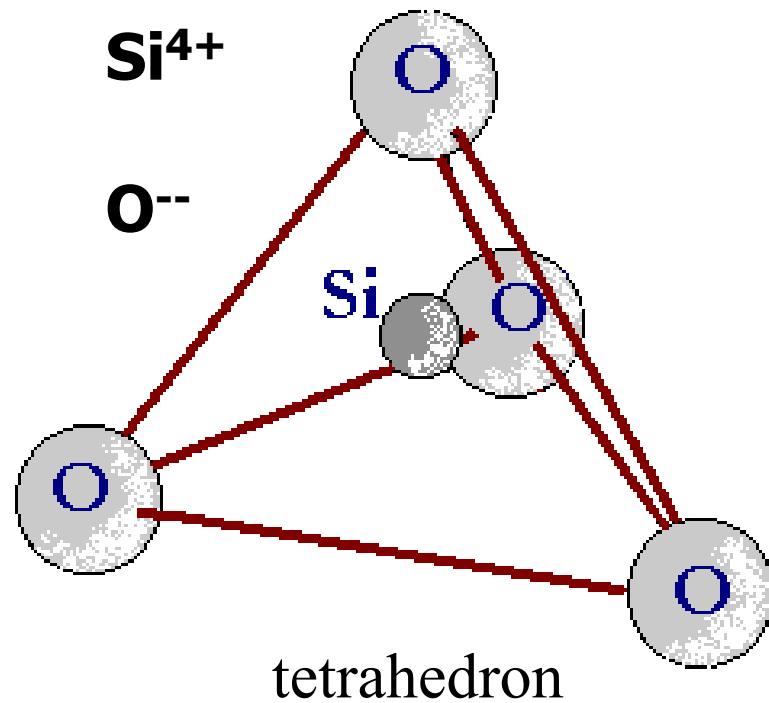
AMORPHOUS MATERIALS

- Local, short range order; bond angle and length are variable
- No order after few nanometers
- “Undercooled liquids”
- Quick solidification of a liquid
- “No time to obtain a crystalline structure” !



Amorphous silica

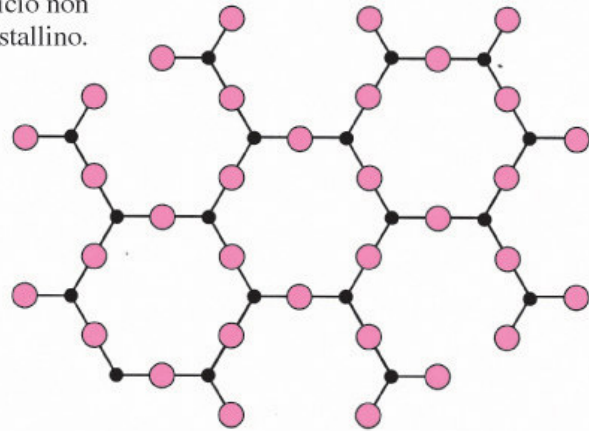
Amorphous materials: silica glasses



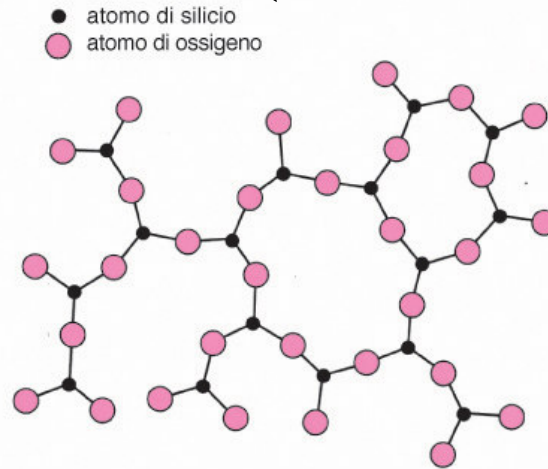
amorphous structure of silica:
tetrahedra chains

Crystalline vs. amorphous silica (SiO_2)

FIGURA 3.21
Schemi
bidimensionali della
struttura di: (a)
biossido di silicio
cristallino e (b)
biossido di silicio non
cristallino.



(a)



(b)

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AMORPHOUS MATERIALS

- Amorphous materials are not thermodynamically stable:
- They become crystalline if properly heated
- All liquids can be solidified quickly enough to obtain amorphous solids
- *All liquids can be solidified slowly enough to obtain poly- or mono-crystalline solids*

AMORPHOUS MATERIALS

- Ex: silica glass: bond angles, force and length differ in different tetrahedra
- No melting temperature
- Viscosity vs temperature
- *The opposite than crystalline materials!*

VISCOSITY

- **Viscosity is a measure of a fluid's resistance to flow.**
- It describes the internal friction of a moving fluid (e.g. tetrahedra chains in a silica glass).



VISCOSITY

- Viscosity Unit : Poise

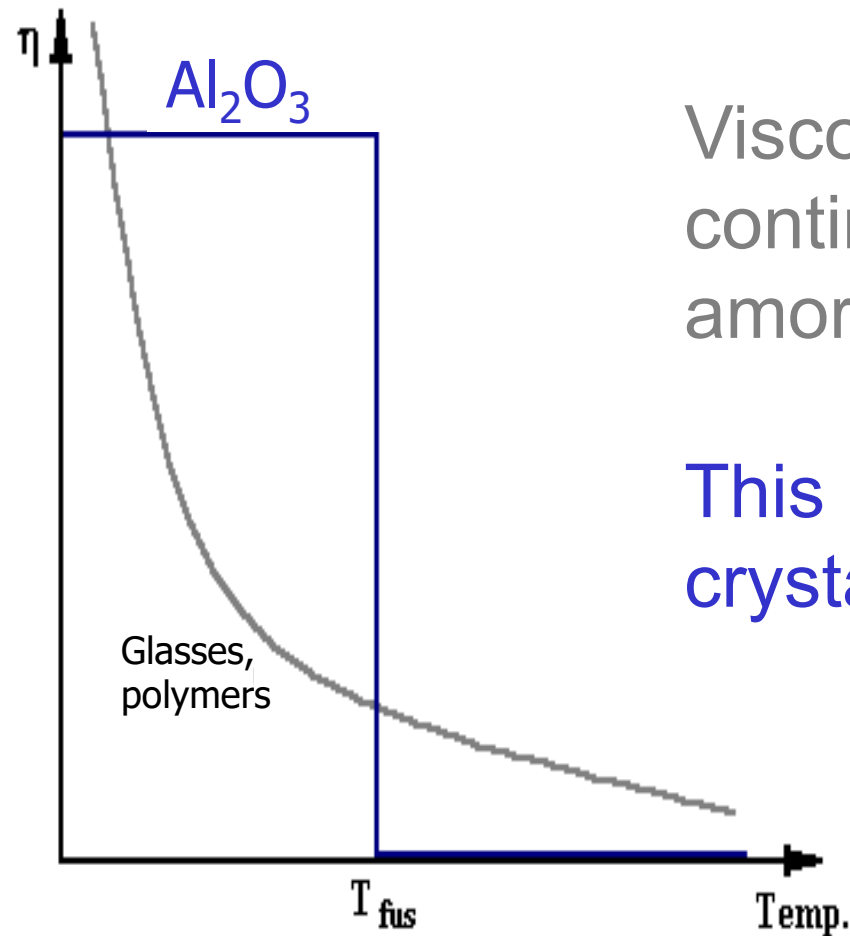
$$10 \text{ Poise} = 1 \text{ Pa}\cdot\text{s} = 1 (\text{N/m}^2)\cdot\text{s}$$

- Example of viscosity:

$$\text{H}_2\text{O (room T)} \quad \eta = 1 \times 10^{-3} \text{ Pa s}$$

$$\text{SiO}_2(\text{silica}) (1720^\circ) \quad \eta = 1 \times 10^6 \text{ Pa s}$$

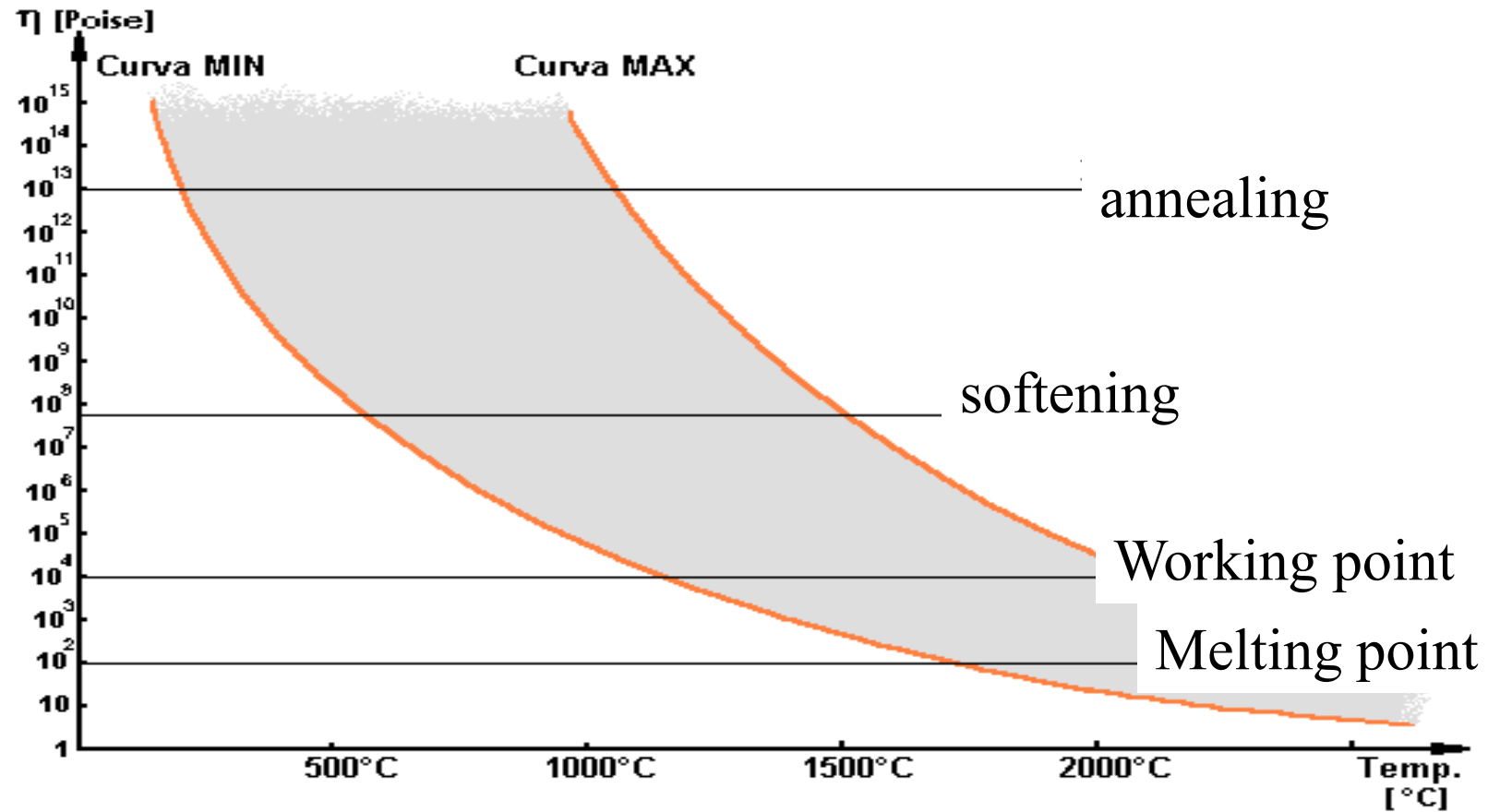
VISCOSITY: AMORPHOUS and CRYSTALLINE materials



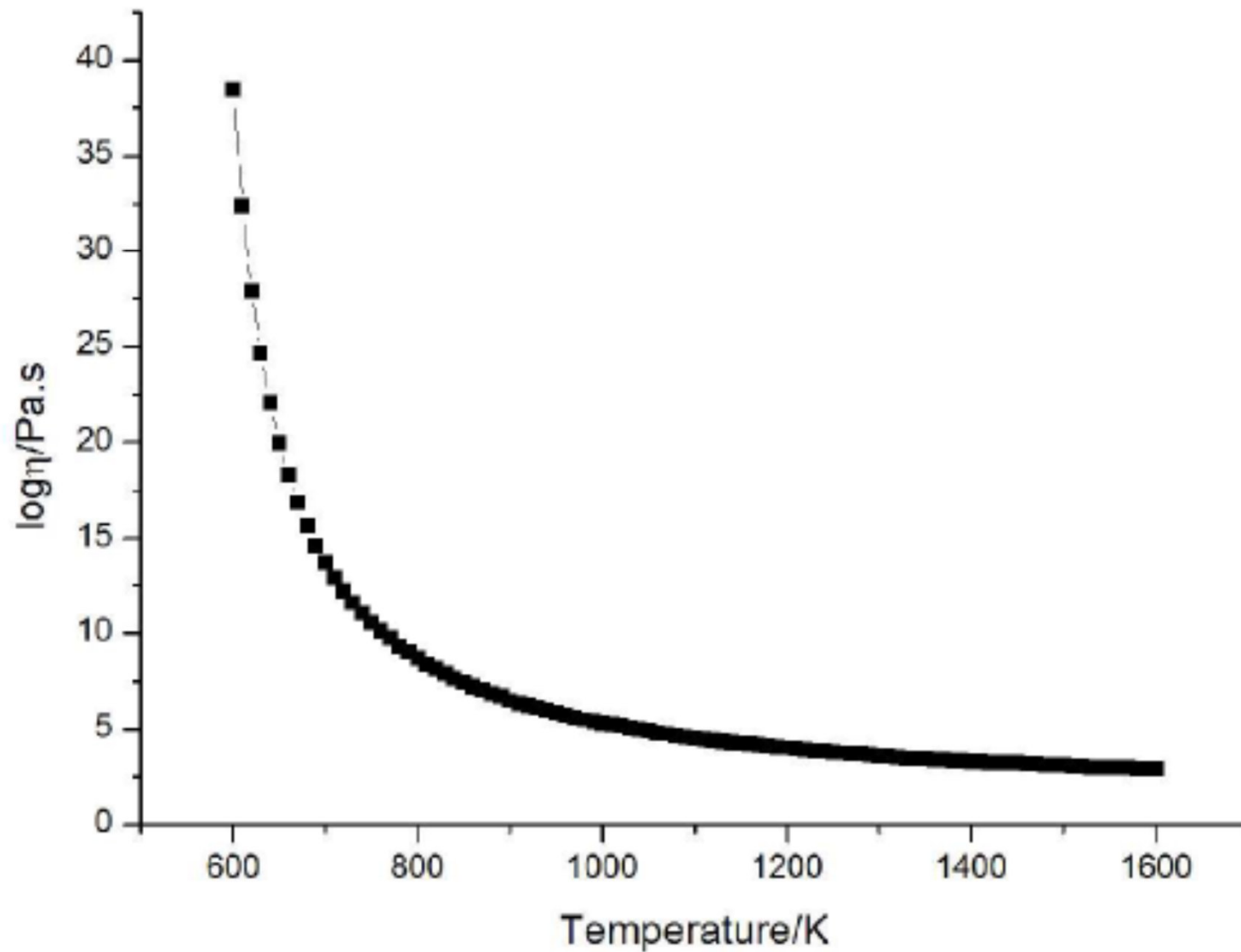
Viscosity decreases continuously for amorphous materials.

This is not the case for crystalline materials

VISCOSITY



Viscosity curve of a glass



Materion introduces zirconium-base **metallic glass** alloys

February 19, 2014

http://www.asminternational.org/news/industry/-/journal_content/56/10180/17353851/NEWS

Bulk metallic glasses are amorphous alloys derived from unique alloy combinations whose disparate atoms retain a random structure when they solidify.

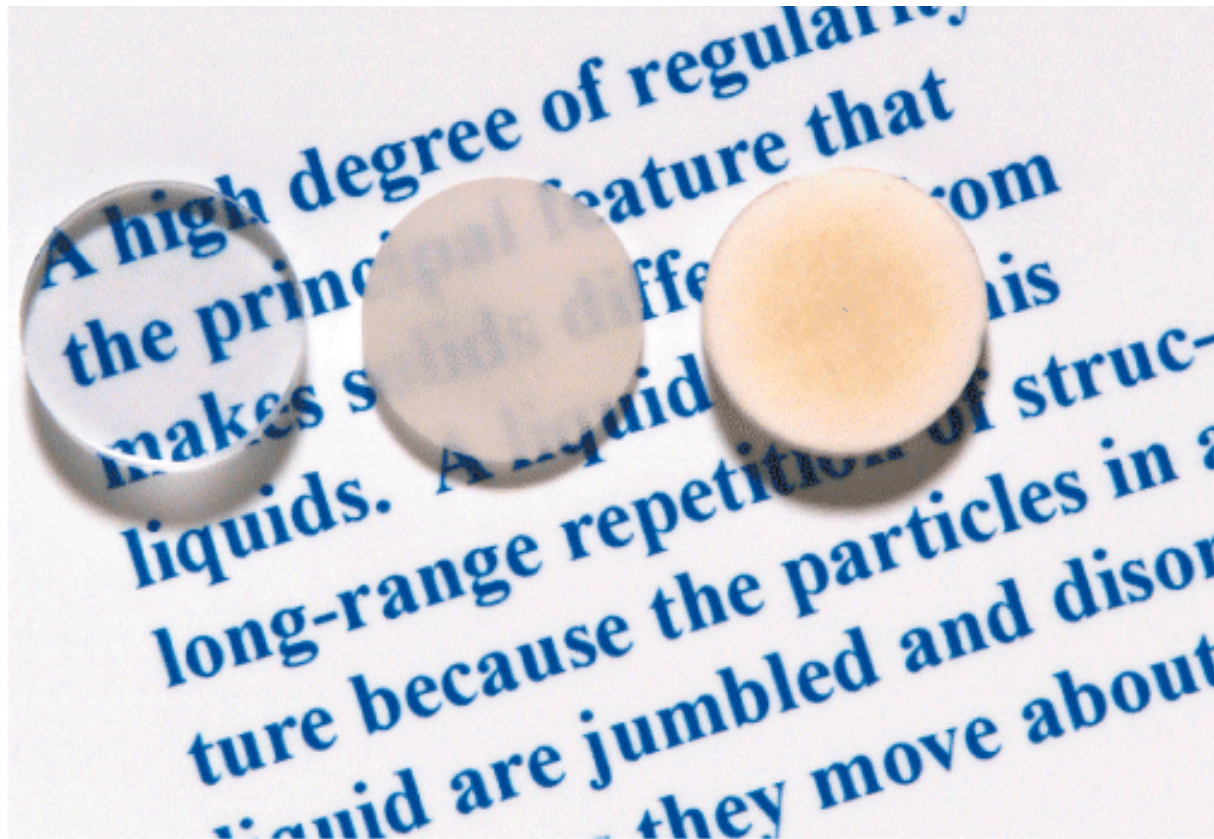
Amorphous alloys have unique properties, including very high specific strength (often twice that of titanium and stainless steel), relatively high hardness (> 50 Rc), high elastic range ($\sim 2\%$), **and exceptional corrosion resistance.**

Why?



A couple of notes on optical properties...

Why glass and diamond gem are transparent?



single-crystal
 Al_2O_3

polycrystalline
 Al_2O_3

polycrystalline Al_2O_3
with 5% porosity